TÉCNICO LISBOA

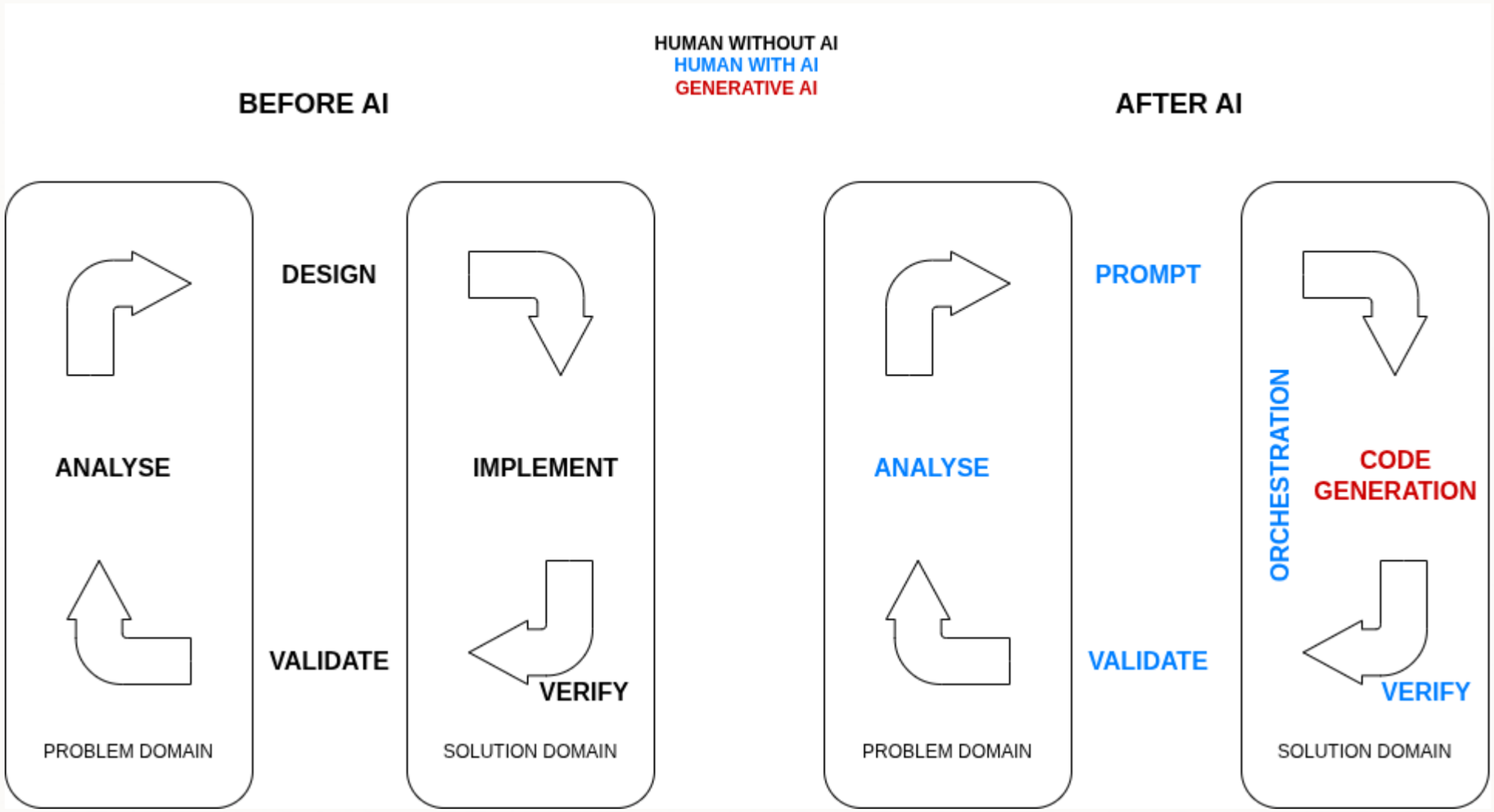
COMPUTER GRAPHICS FOR GAMES

P2 2026-27

Carlos Martinho


Epic Games
Student Showcase 2025

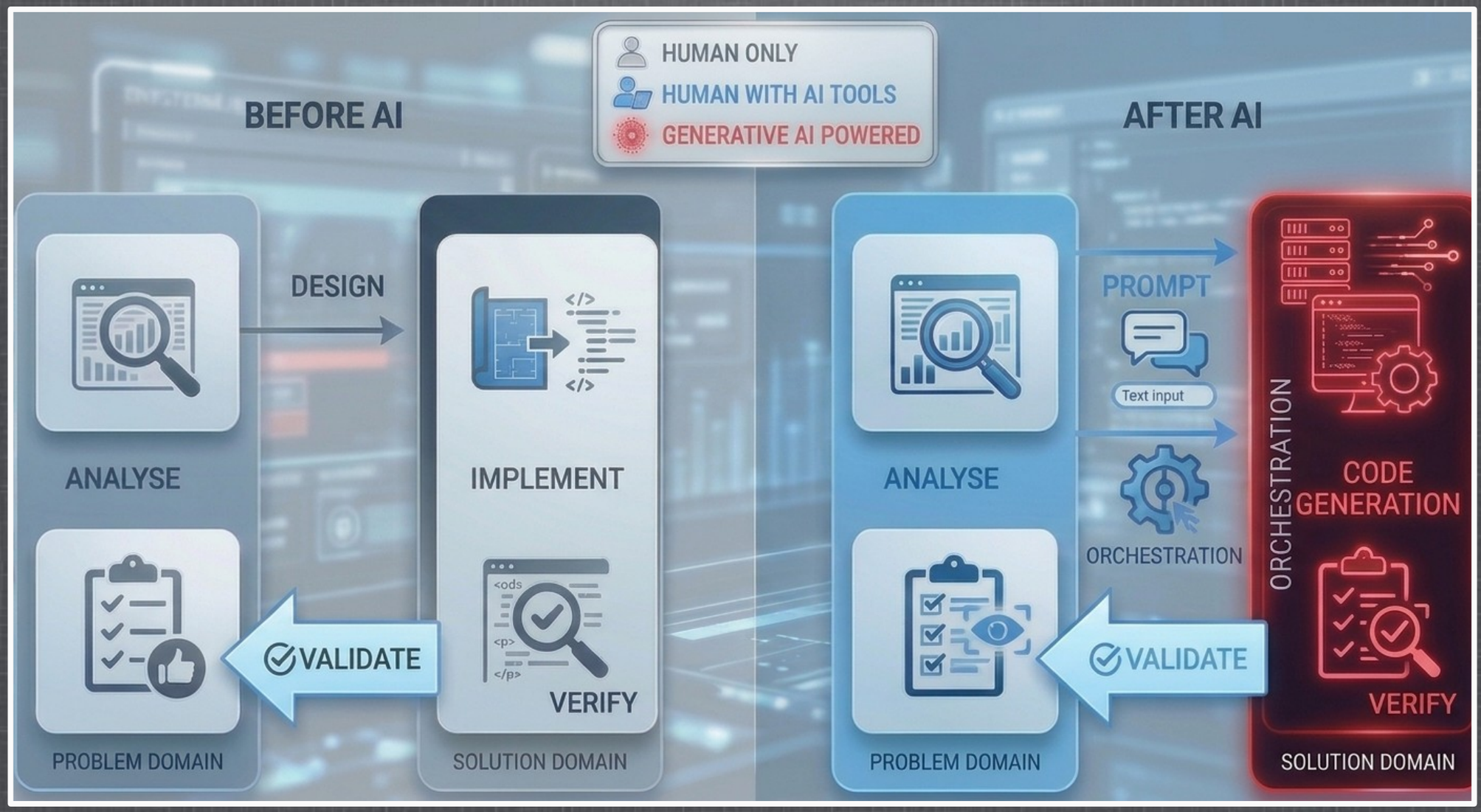
BEFORE AND AFTER GENAI



ONE SHOT VS POLISHING



WHAT DOES GOOD LOOKS LIKE?



MODERN “SLOP”

GAMEWATCH

News Update:

Steam NEXT FEST Is Being Flooded With AI-Generated Game Demos

Instagram: gamewatch.clips / Facebook: Gamewatch.clips

Steam Next Fest’s October 2026 edition has drawn thousands of demos, but players are raising concerns over AI-generated assets, low-effort clones and copycat games. With no built-in option to filter out AI-disclosed titles, calls for Valve to give players more control are growing.



GAME CLONING

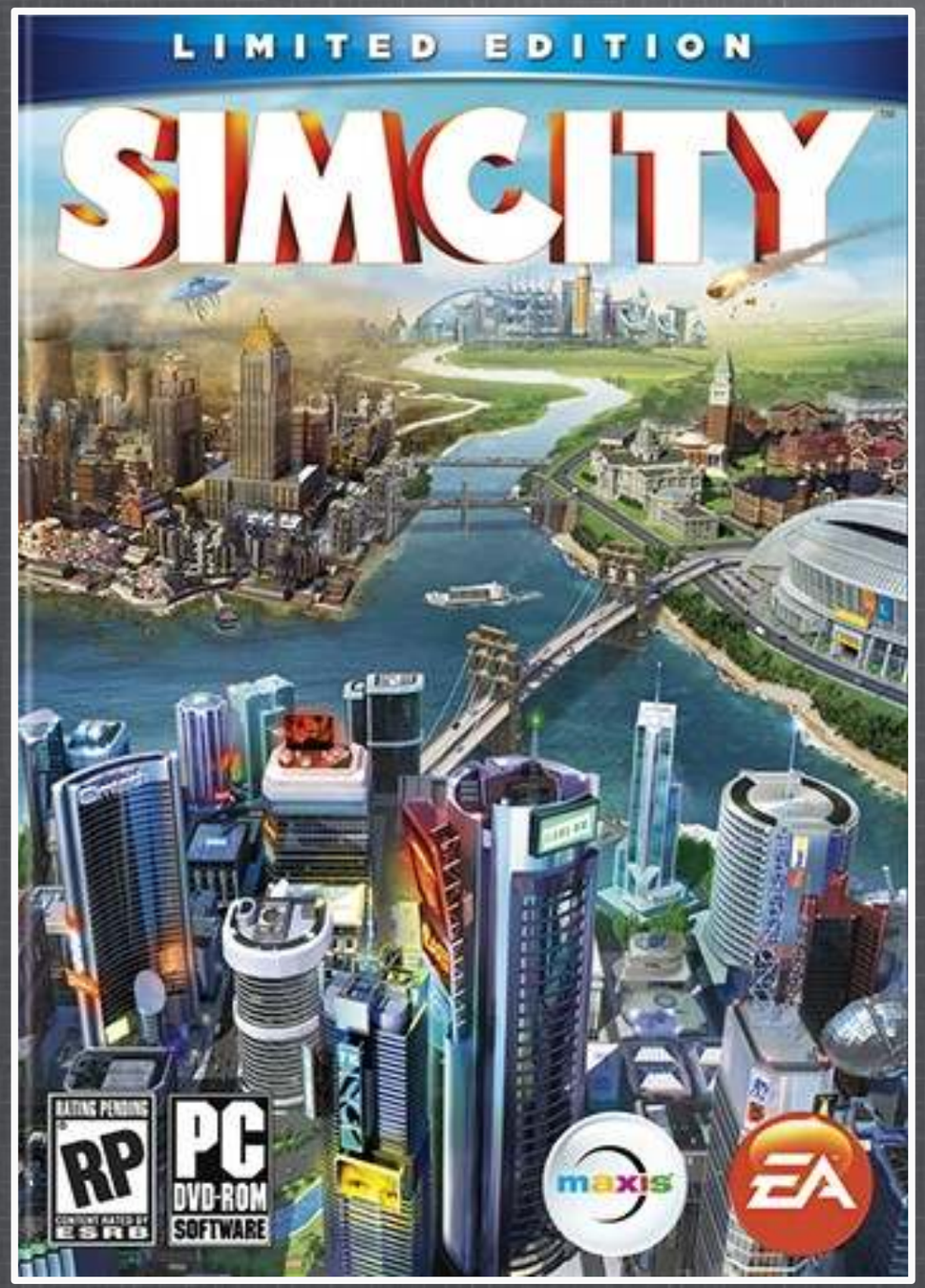


ATARI 2600 “SLOP”





STEAM RESPONSE TO PIRACY





STEAM RESPONSE TO AI SLOP




PLAY
TEST

CONSOLES RESPONSE TO AI SLOP







STEAMSTATION

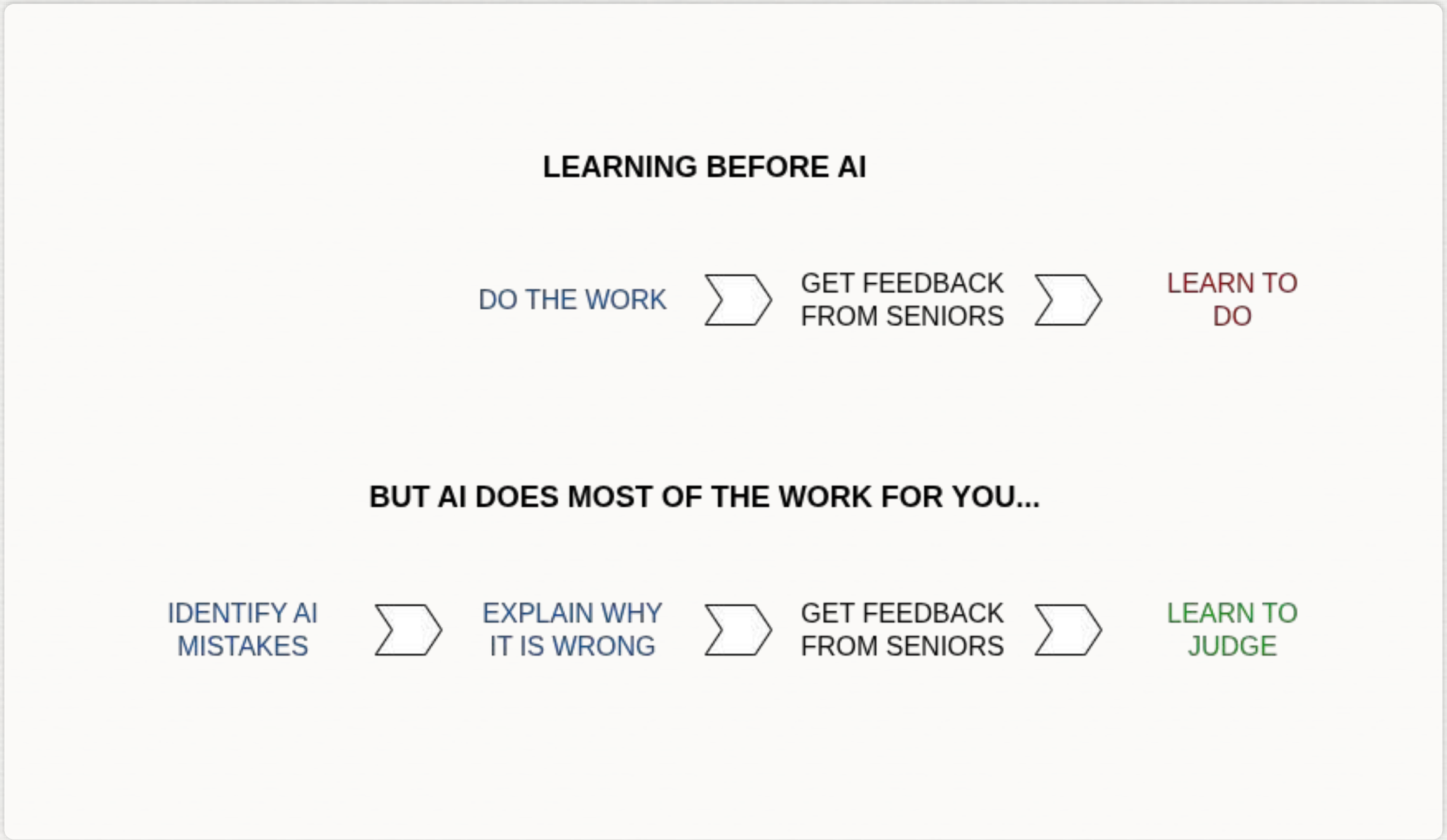




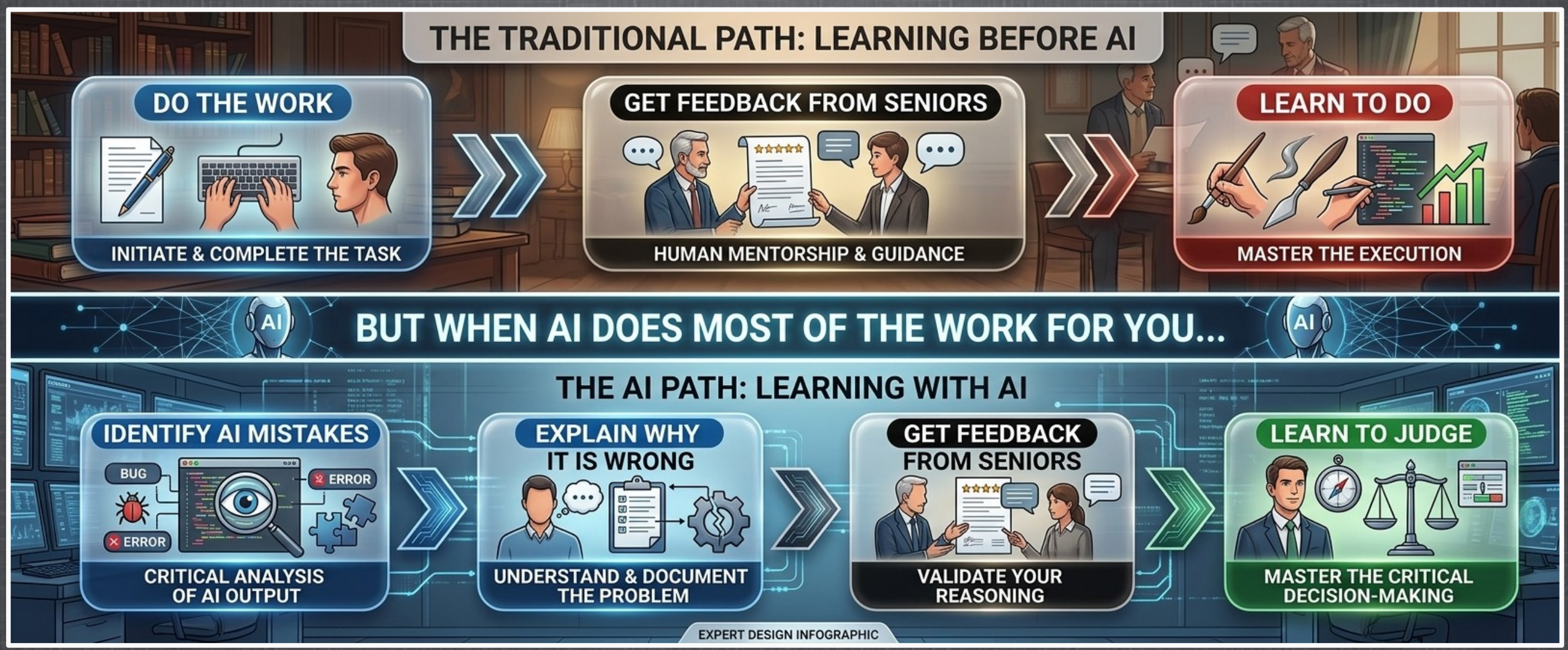
“WHAT I CANNOT CREATE, I DO NOT UNDERSTAND.”

Richard Feynman

LEARNING WITH GENAI

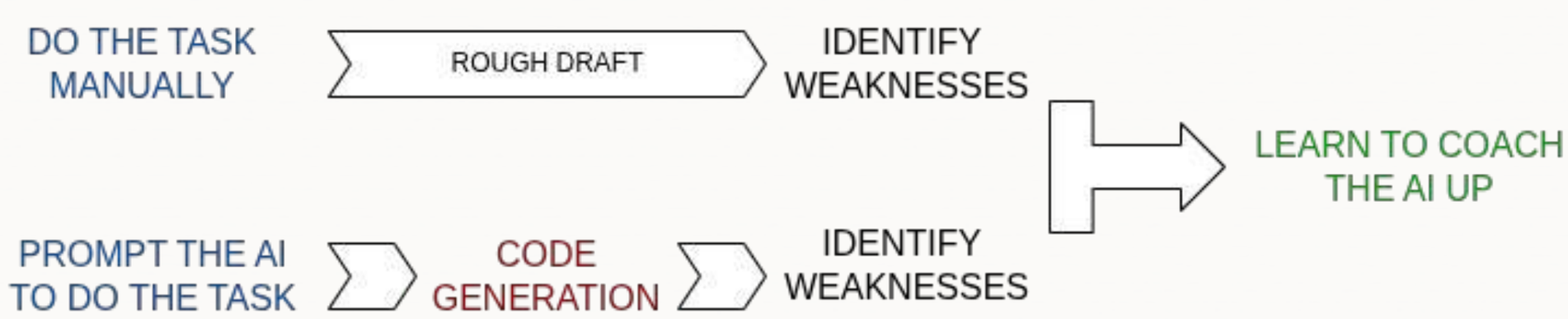


LEARNING WITH GENAI



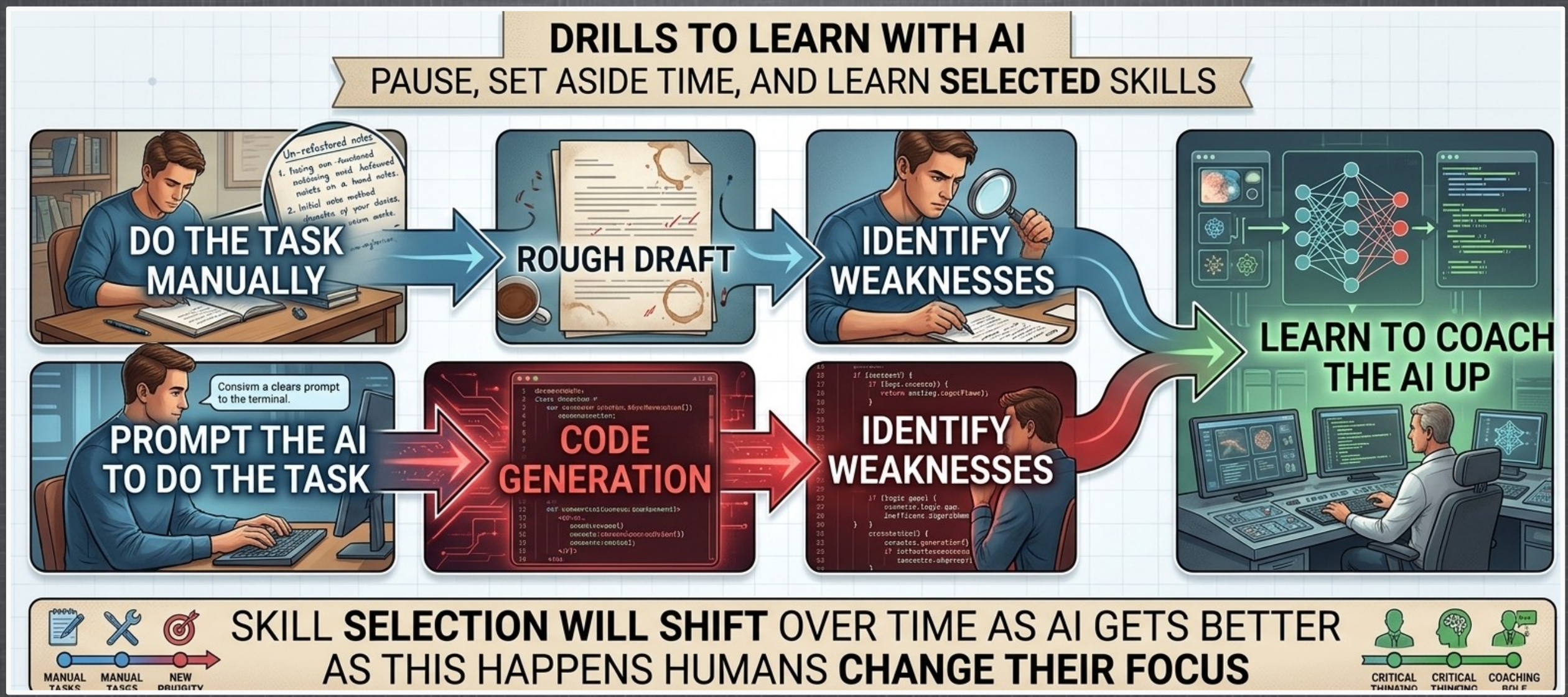
DRILLS WITH GENAI

DRILLS TO LEARN WITH AI
PAUSE, SET ASIDE TIME, AND LEARN **SELECTED** SKILLS



SKILL **SELECTION WILL SHIFT** OVER TIME AS AI GETS BETTER
AS THIS HAPPENS HUMANS **CHANGE THEIR FOCUS**

DRILLS WITH GENAI




Project
Based
Learning

OUTLINE

- Course introduction
- Importance of Computer Graphics
- Brief history of Computer Graphics
- Quarter planning

GOALS

GOALS

- Develop a real-time graphics engine ‘from scratch’ using modern 3D graphics programming with GPU.
(*technical challenge*)

- Understand the theory and the algorithms underlying modern 3D graphics systems.
(*intellectual challenge*)

- Development will be in Modern C++ / OpenGL / GLSL.

- Help creating a strong portfolio.

PREREQUISITES

PREREQUISITES

- Linear algebra (review in the context of Computer Graphics and addenda next lectures).
- Proficient object programming skills. Good knowledge of C++ and STL will help.
- Knowledge of Computer Graphics (e.g. image formats, raster algorithms, etc.) may help.

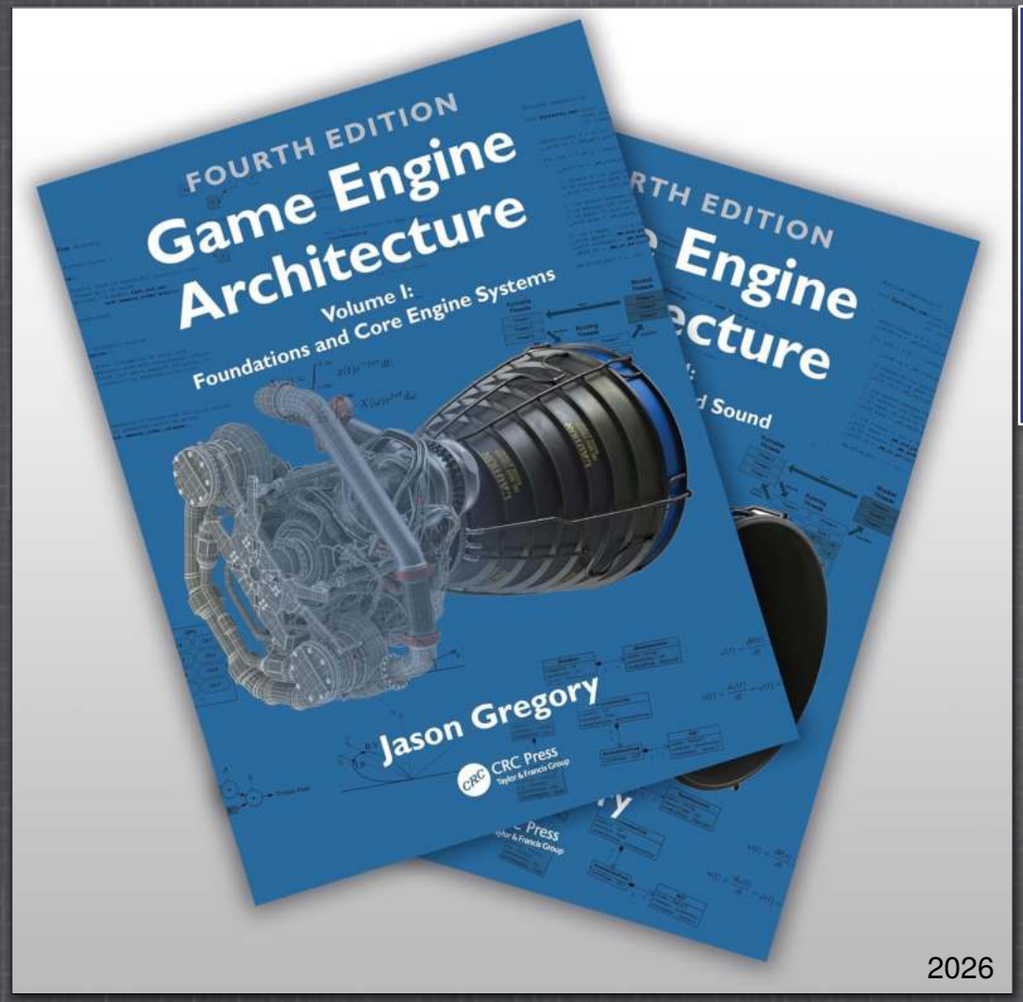
GRADING

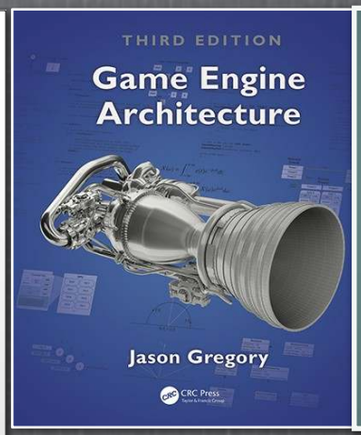
BASE PROJECT
(T2, 40%, >9.5)

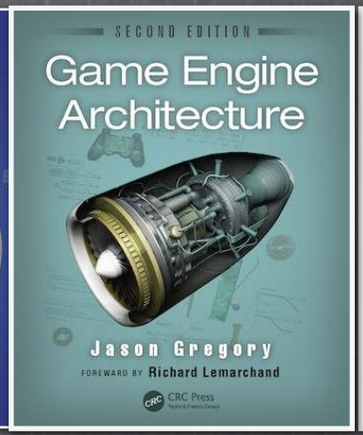
+

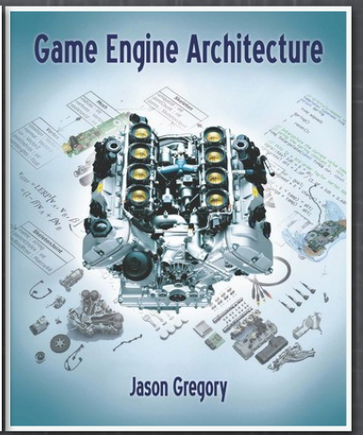
SHADER PROJECT
(T1, 60%, >9.5)

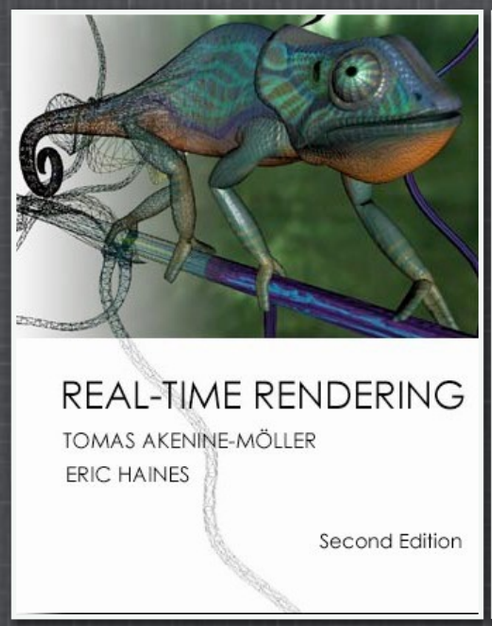
BIBLIOGRAPHY

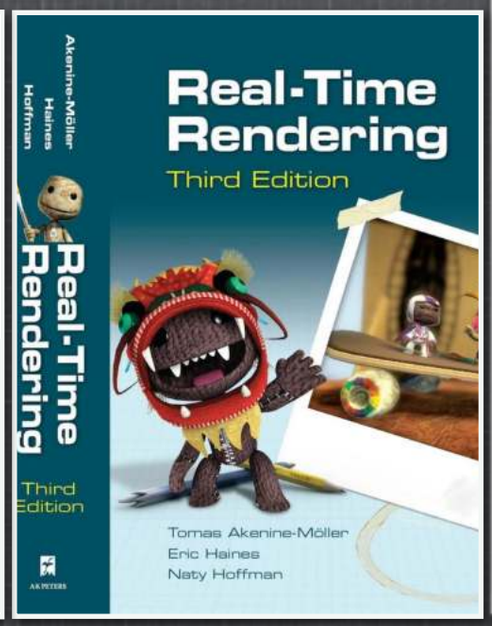

FOURTH EDITION
Game Engine Architecture
Volume I:
Foundations and Core Engine Systems
Jason Gregory
CRC Press
Taylor & Francis Group
2026

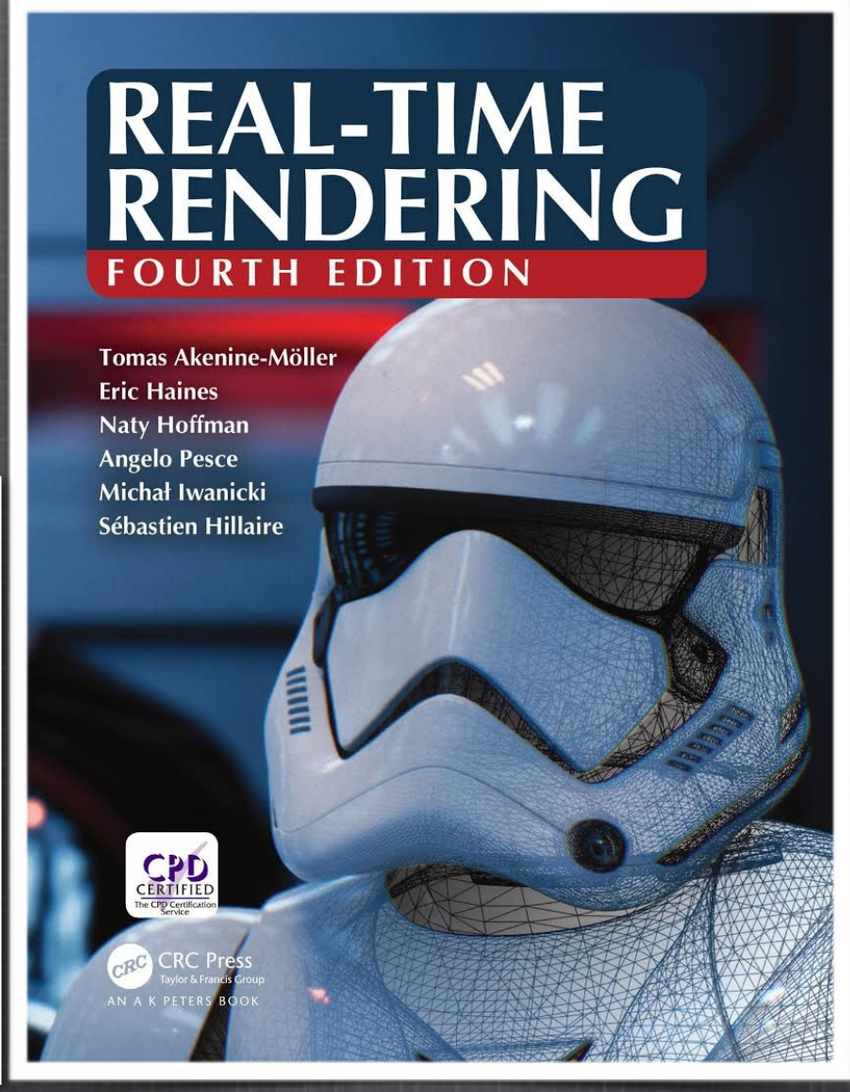

THIRD EDITION
Game Engine Architecture
Jason Gregory
CRC Press

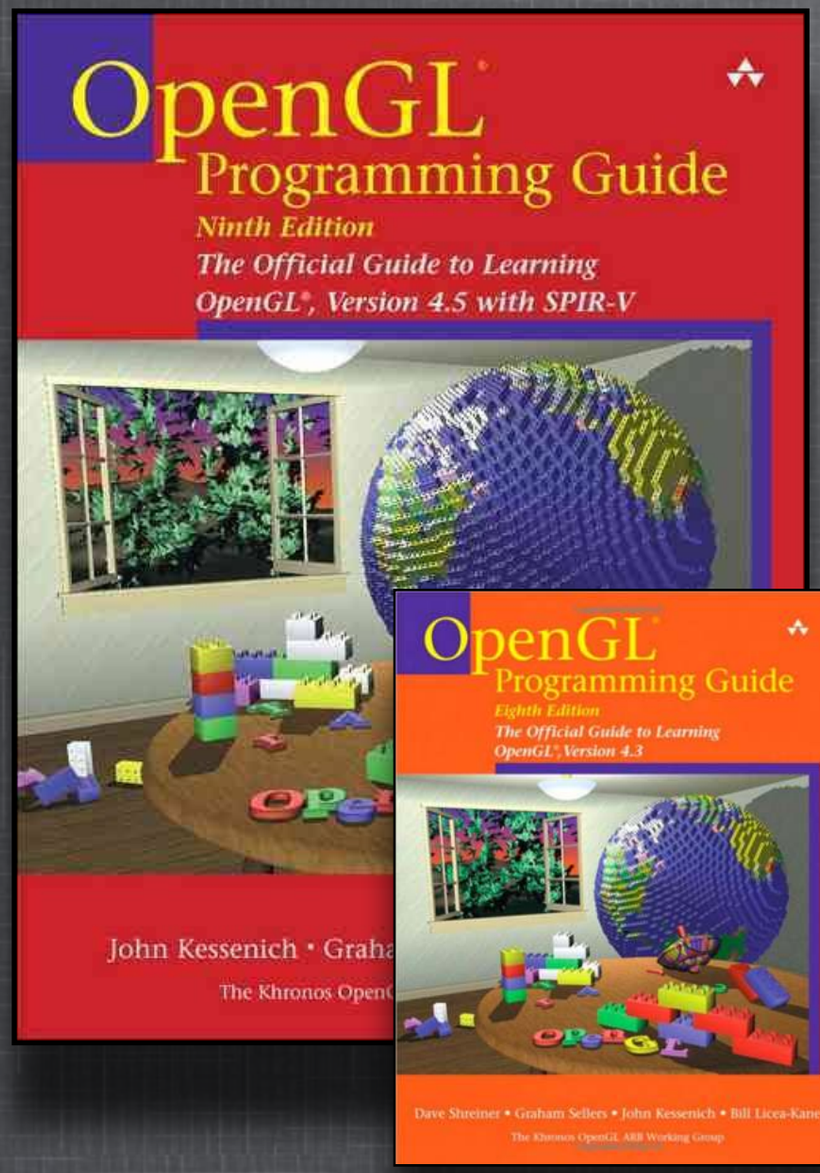

SECOND EDITION
Game Engine Architecture
Jason Gregory
FOREWARD BY Richard Lemarchand
CRC Press

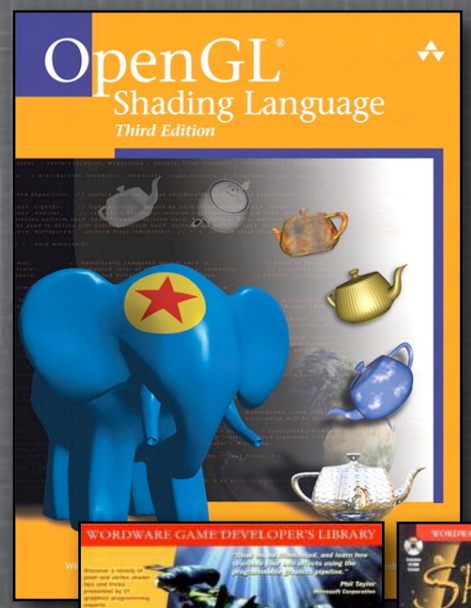

Game Engine Architecture
Jason Gregory


REAL-TIME RENDERING
TOMAS AKENINE-MÖLLER
ERIC HAINES
Second Edition


Akenine-Möller
Haines
Hoffman
Real-Time Rendering
Third Edition
Real-Time Rendering
Third Edition
Tomas Akenine-Möller
Eric Haines
Naty Hoffman

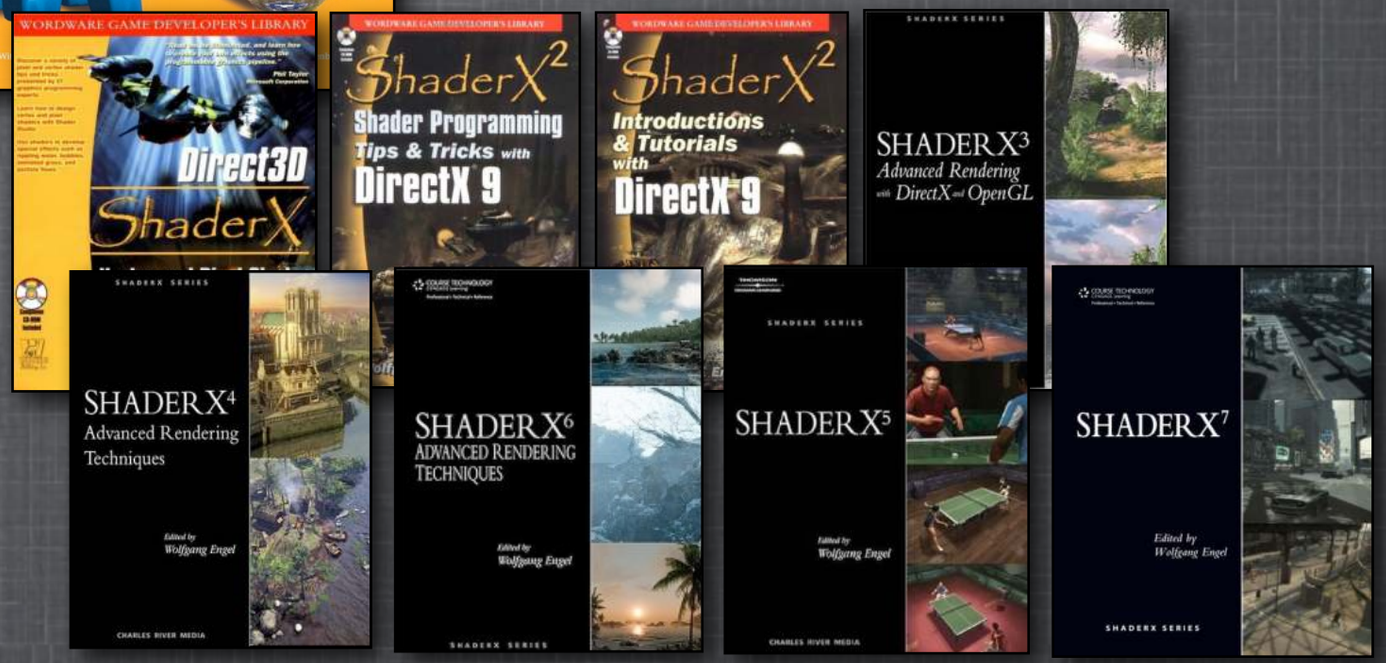

REAL-TIME RENDERING
FOURTH EDITION
Tomas Akenine-Möller
Eric Haines
Naty Hoffman
Angelo Pesce
Michał Iwanicki
Sébastien Hillaire
CPD CERTIFIED
The CPD Certification Service
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Taylor & Francis Group
AN A K PETERS BOOK



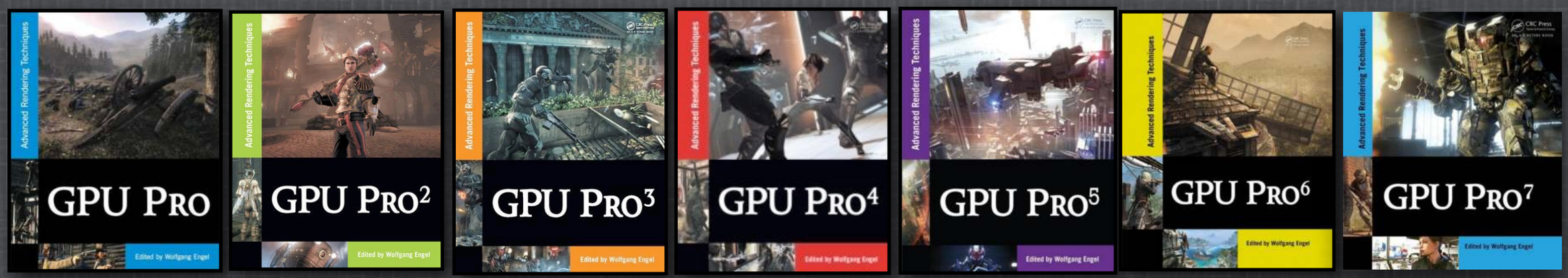




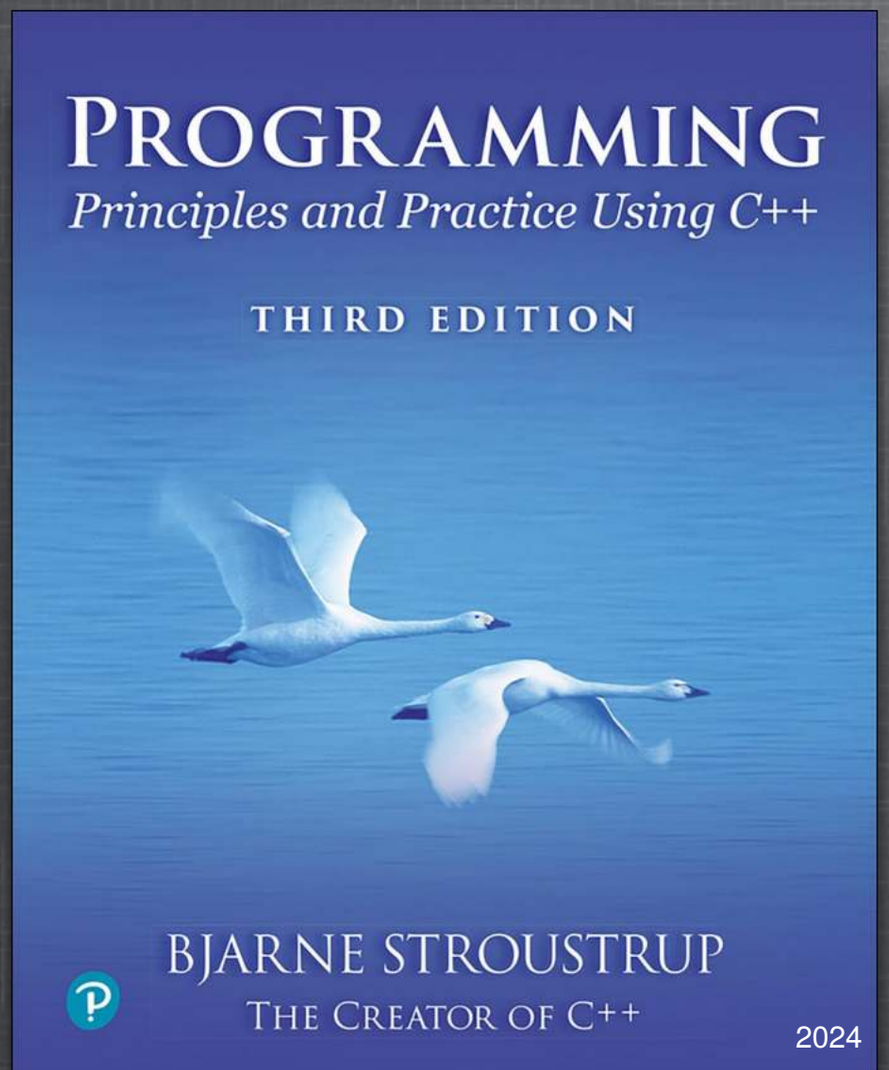
2004-2007

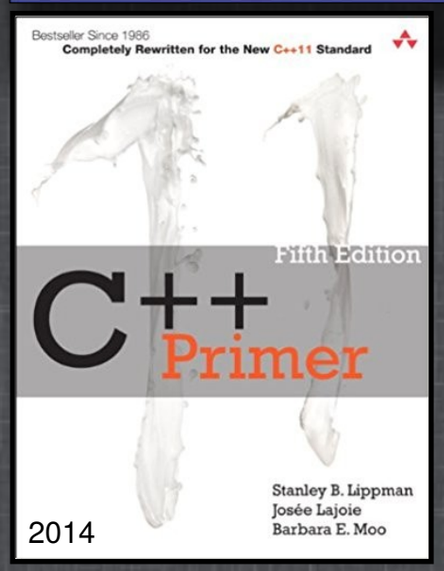


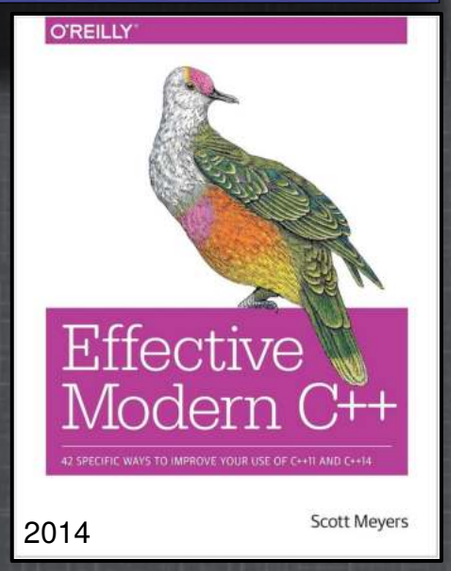
2002-2009



2010-2016

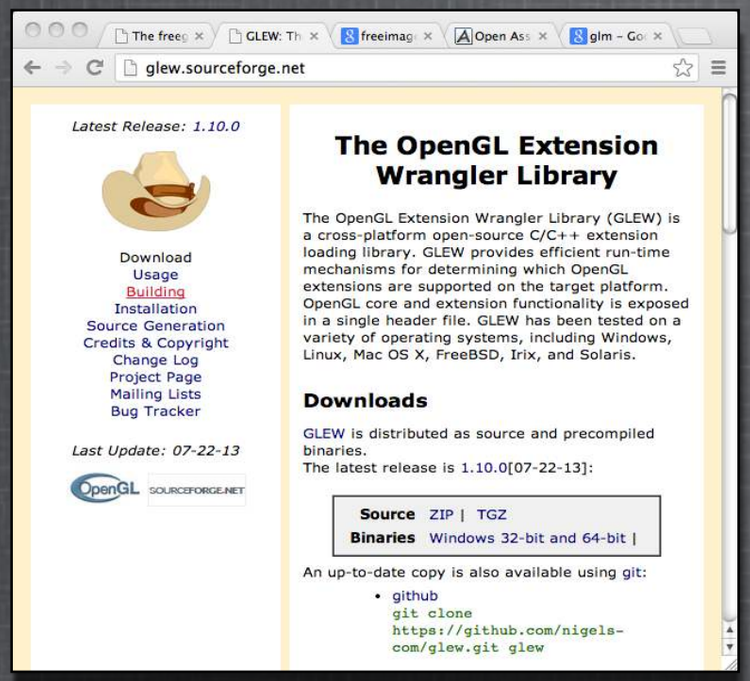

PROGRAMMING
Principles and Practice Using C++
THIRD EDITION
BJARNE STROUSTRUP
THE CREATOR OF C++
2024

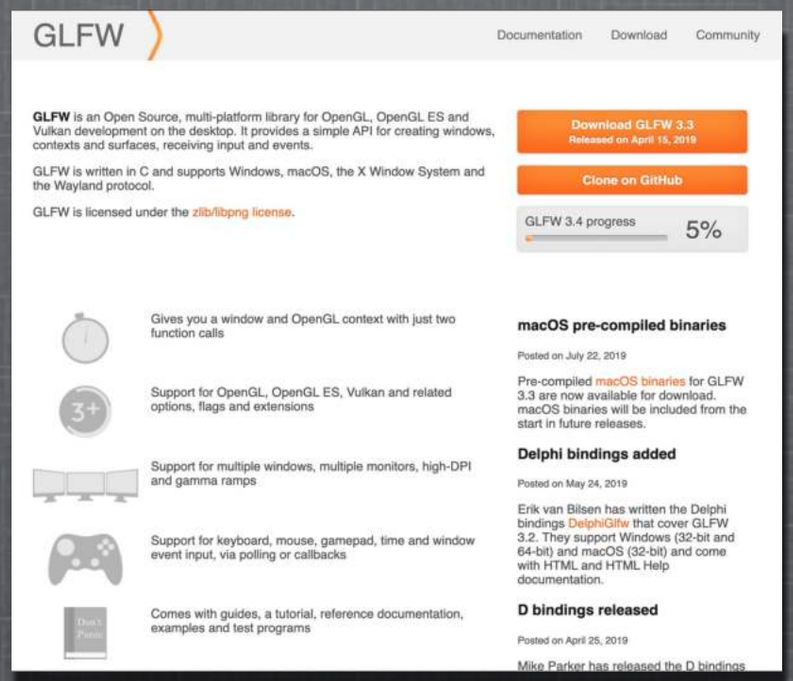

Bestseller Since 1986
Completely Rewritten for the New C++11 Standard
Fifth Edition
C++ Primer
Stanley B. Lippman
Josée Lajoie
Barbara E. Moo
2014

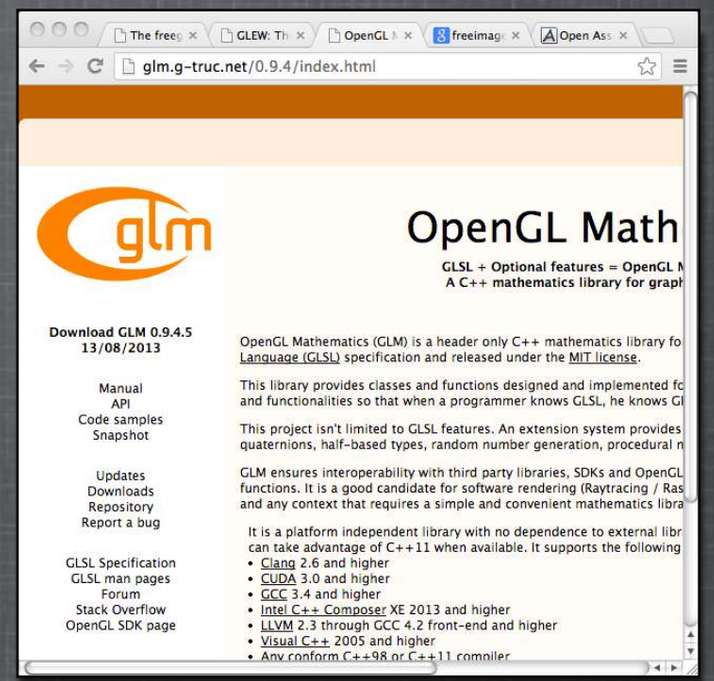

O'REILLY
Effective Modern C++
42 SPECIFIC WAYS TO IMPROVE YOUR USE OF C++11 AND C++14
Scott Meyers
2014

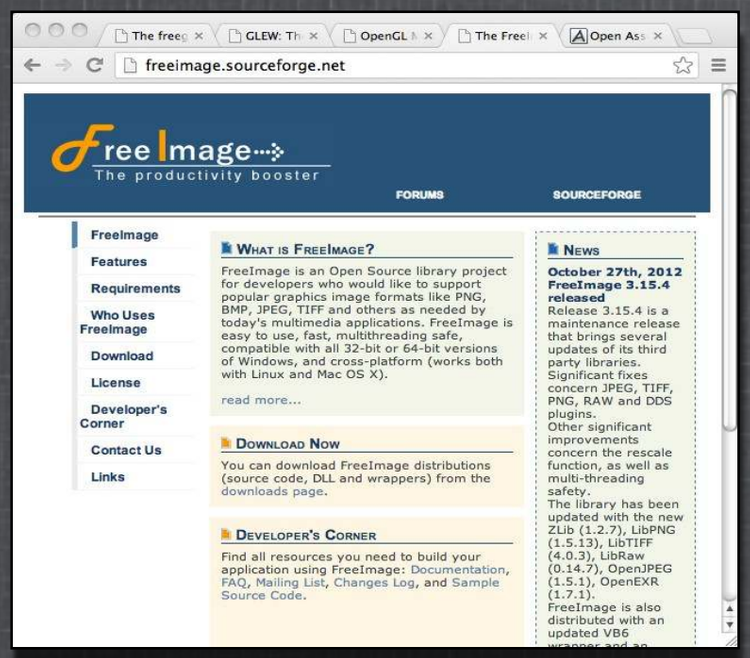

cppreference.com
Create account
Page
Discussion
Standard revision: Diff
C++ reference
C++11, C++14, C++17, C++20, C++23, C++26 | Compiler support C++11, C++14, C++17, C++20,
Language
Keywords – Preprocessor
ASCII chart
Basic concepts
Comments
Names (lookup)
Types (fundamental types)
The main function
Expressions
Value categories
Evaluation order
Operators (precedence)
Conversions – Literals
Statements
if – switch
for – range-for (C++11)
while – do-while
Declarations – Initialization
Functions – Overloading
Classes (unions)
Templates – Exceptions
Freestanding implementations
Standard library (headers)
Named requirements
Feature test macros (C++20)
Language – Standard library
Language support library
Program utilities
Signals – Non-local jumps
Basic memory management
Variadic functions
source_location (C++20)
Coroutine support (C++20)
Comparison utilities (C++20)
Type support – type_info
numeric_limits – exception
initializer_list (C++11)
Concepts library (C++20)
Diagnostics library
Assertions – System error (C++11)
Exception types – Error numbers
basic_stacktrace (C++23)
Debugging support (C++26)
Memory management library
Allocators – Smart pointers
Memory resources (C++17)
Metaprogramming library (C++11)
Type traits – ratio
integer_sequence (C++14)
General utilities library
Function objects – hash (C++11)
Swap – Type operations (C++11)
Integer comparison (C++20)
pair – tuple (C++11)
optional (C++17)
expected (C++23)
variant (C++17) – any (C++17)
bitset – Bit manipulation (C++20)
Containers library
vector – deque – array (C++11)
list – forward_list (C++11)
map – multimap – set – multiset
unordered_map (C++11)
unordered_multimap (C++11)
unordered_set (C++11)
unordered_multiset (C++11)
Container adaptors
span (C++20) – mdspan (C++23)
Iterators library
Ranges library (C++20)
Range factories – Range adaptors
generator (C++23)
Algorithms library
Numeric algorithms
Execution policies (C++17)
Constrained algorithms (C++20)
Strings
Text pro
Numeric
Date an
Input/ou
Concurr
Executio
Technical specifications
Standard library extensions (library fundamentals TS)
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Standard library extensions v2 (library fundamentals TS v2)
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observer_ptr – Detection idiom
Standard library extensions v3 (library fundamentals TS v3)
scope_exit – scope_fail – scope_success – unique_resource
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External Links – Non-ANSI/ISO Libraries – Index – std Symbol Index

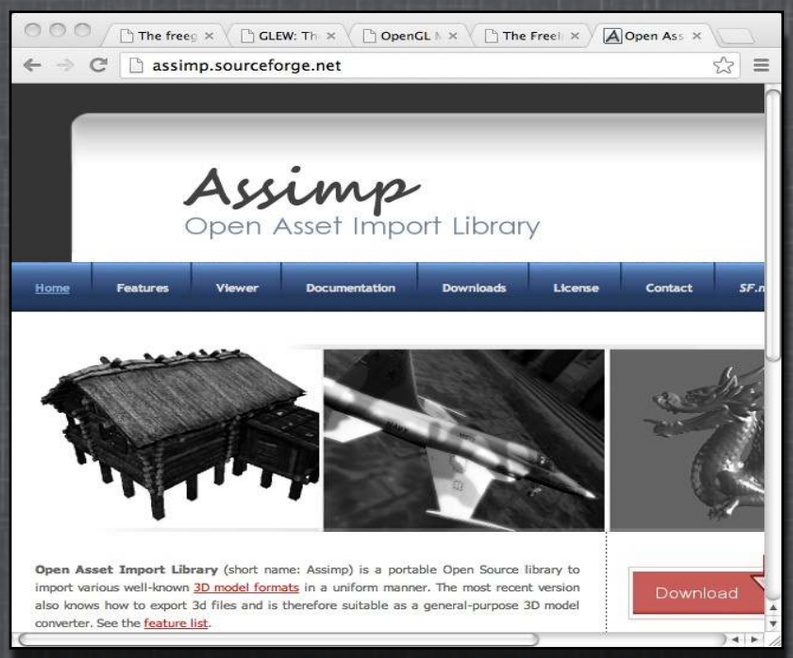
ONLINE LIBRARIES

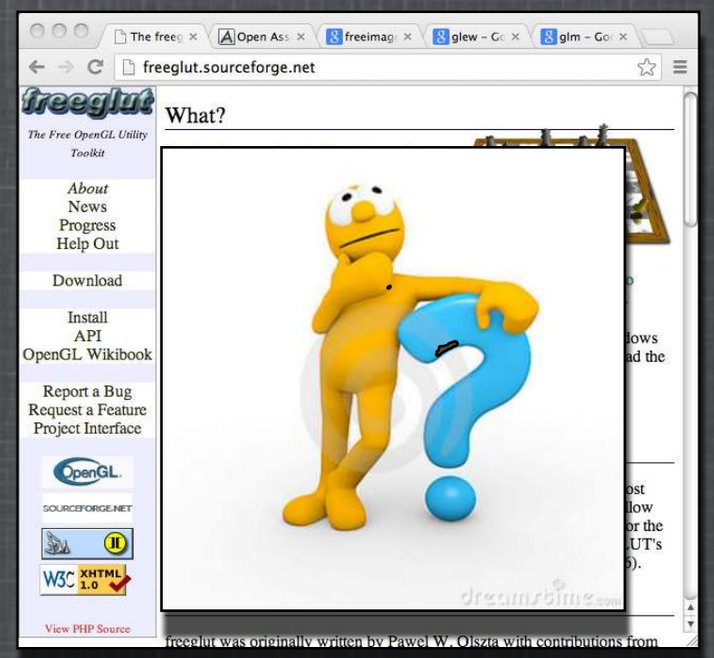




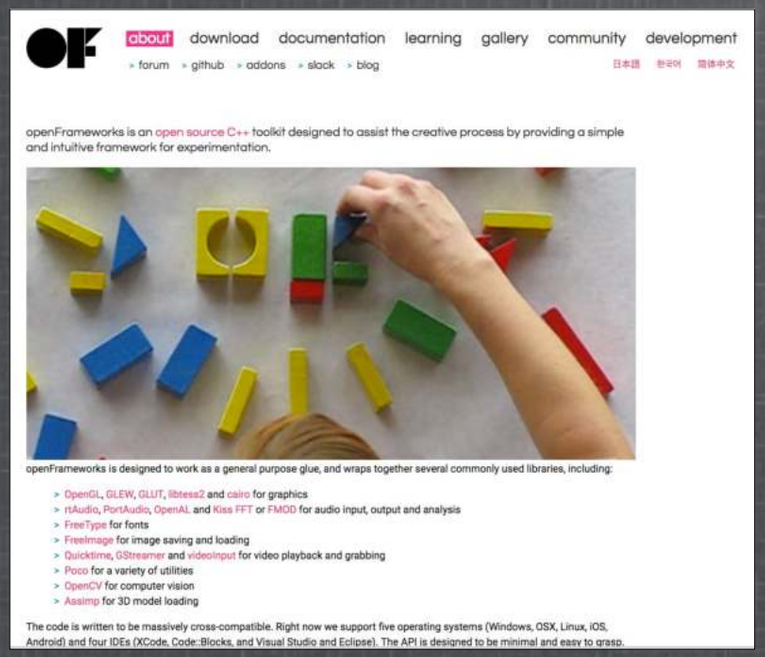




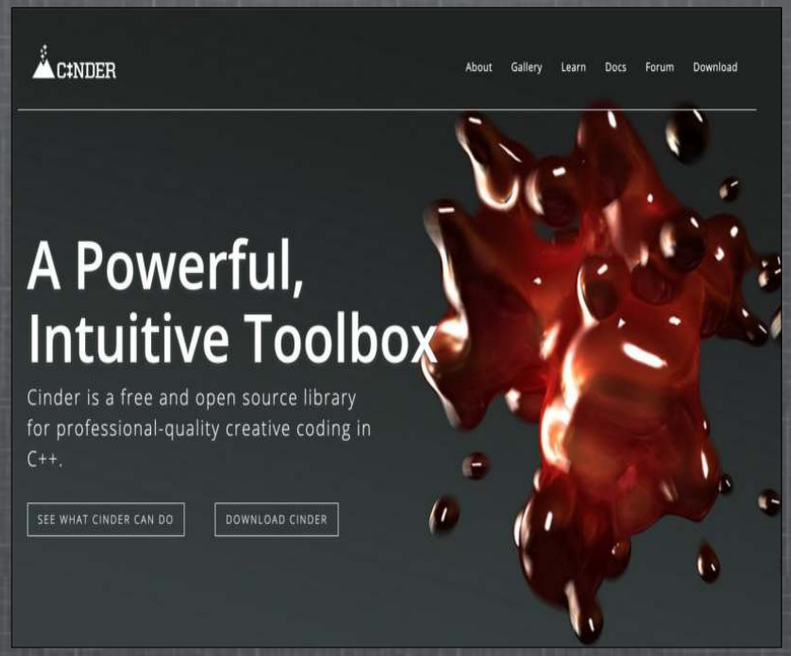




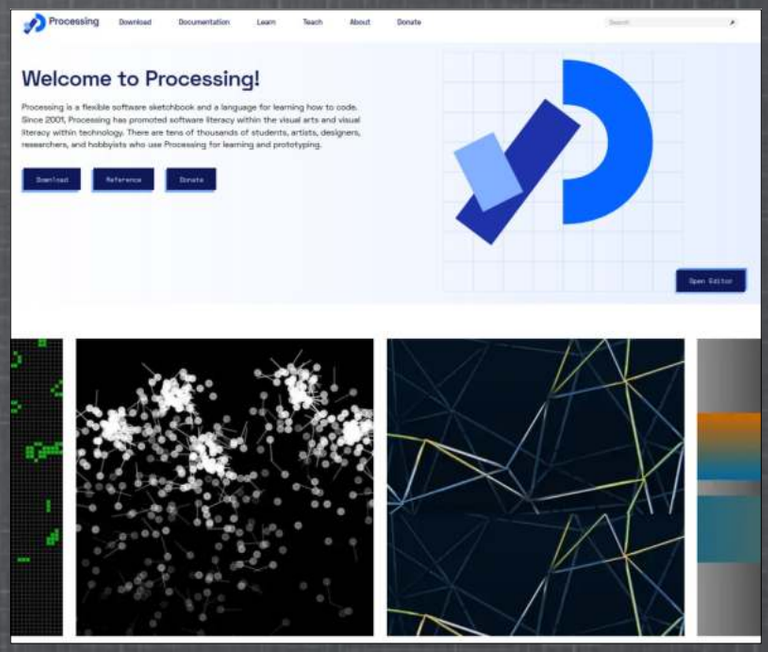
CREATIVE CODING



OPEN FRAMEWORKS

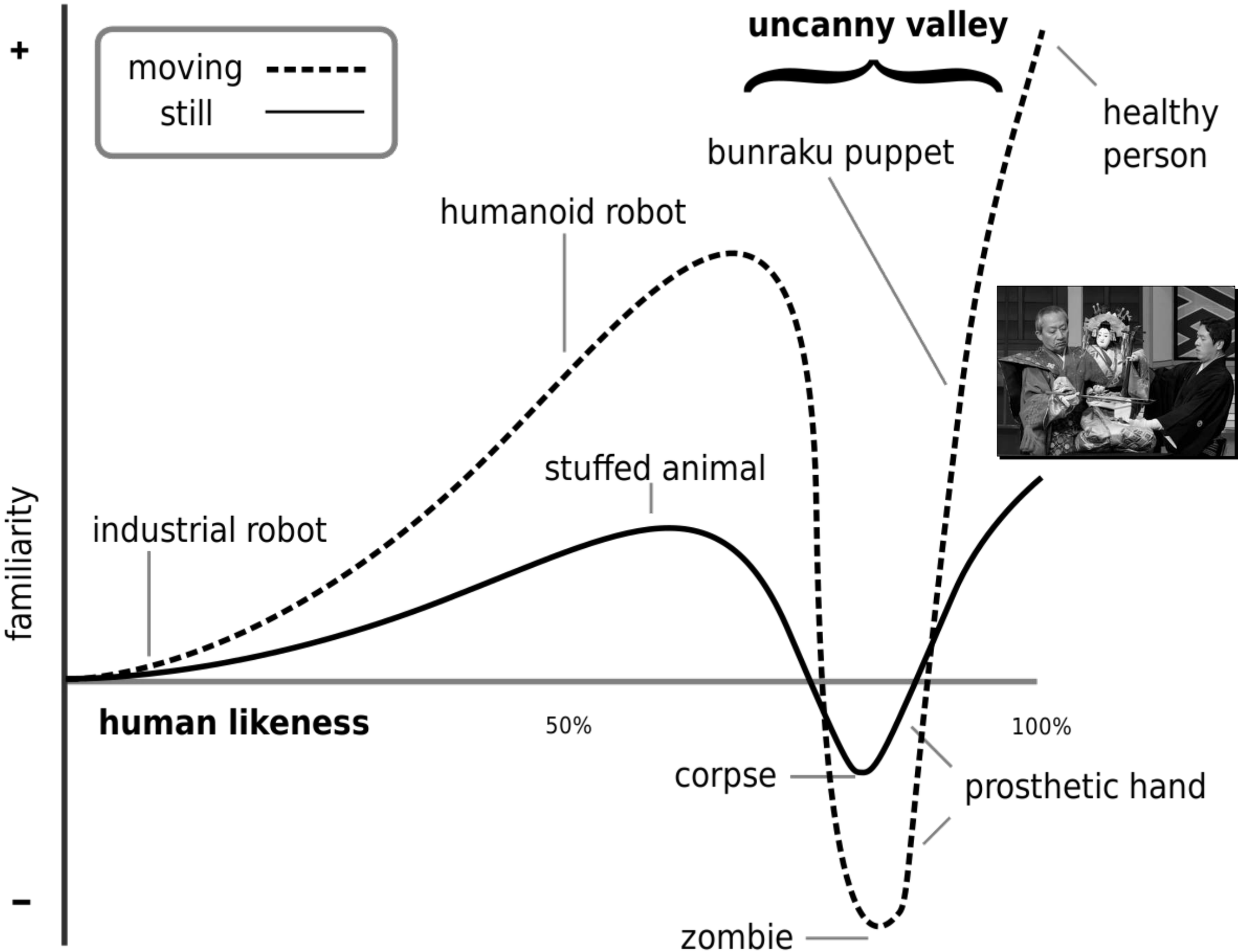


CINDER



PROCESSING

IMPORTANCE OF COMPUTER GRAPHICS


+
familiarity
−
moving
still
uncanny valley
healthy person
bunraku puppet
humanoid robot
stuffed animal
industrial robot
human likeness
50%
100%
corpse
prosthetic hand
zombie

COMPUTER GRAPHICS IN MOVIES



(1982)



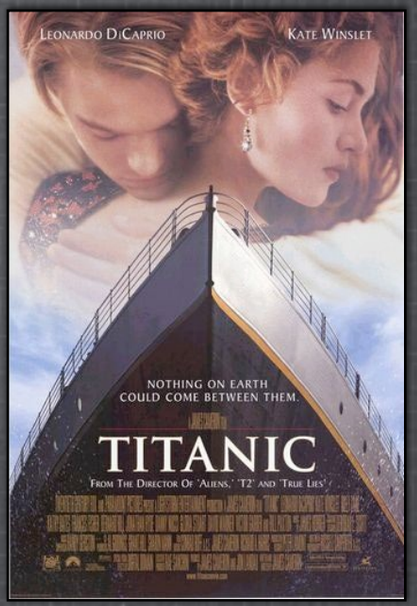
(1991)



(1993)



(1995)



(1997)



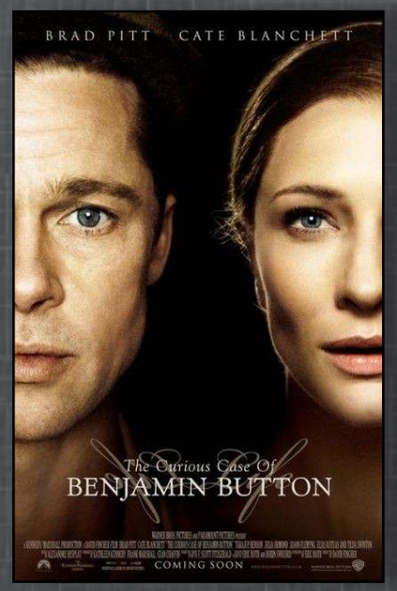
(1999,2002,2005)



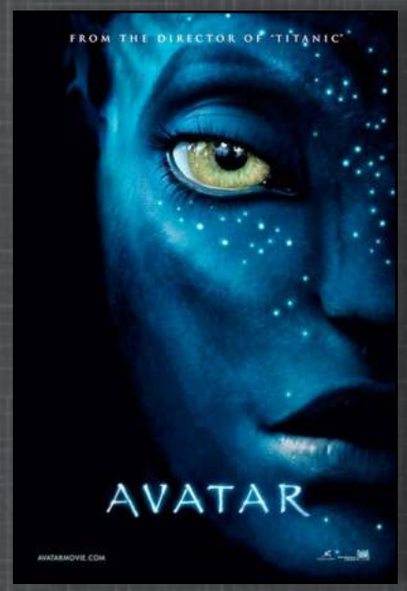
(1999)



(2001, 2002, 2003)



(2008)



(2009)

COMPUTER GRAPHICS IN GAMES



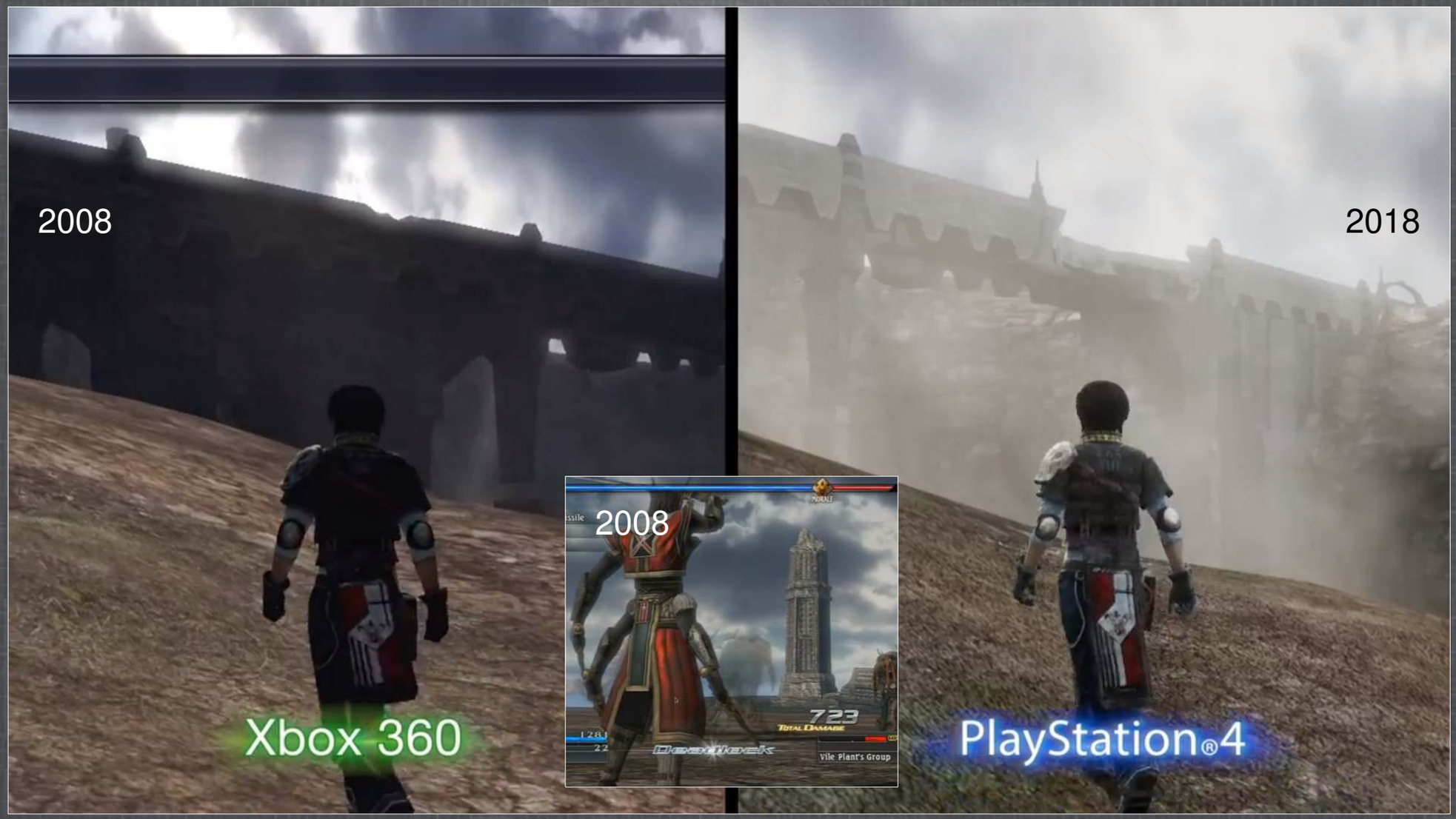
DIGITALFOUNDRY 2017
(KINGDOM HEARTS III, SQUARE ENIX, 2019)

COMPUTER GRAPHICS IN GAMES



BIOSHOCK 2 (2K BOSTON, 2010)

COMPUTER GRAPHICS IN GAMES



LAST REMNANT, 2018 REMASTER

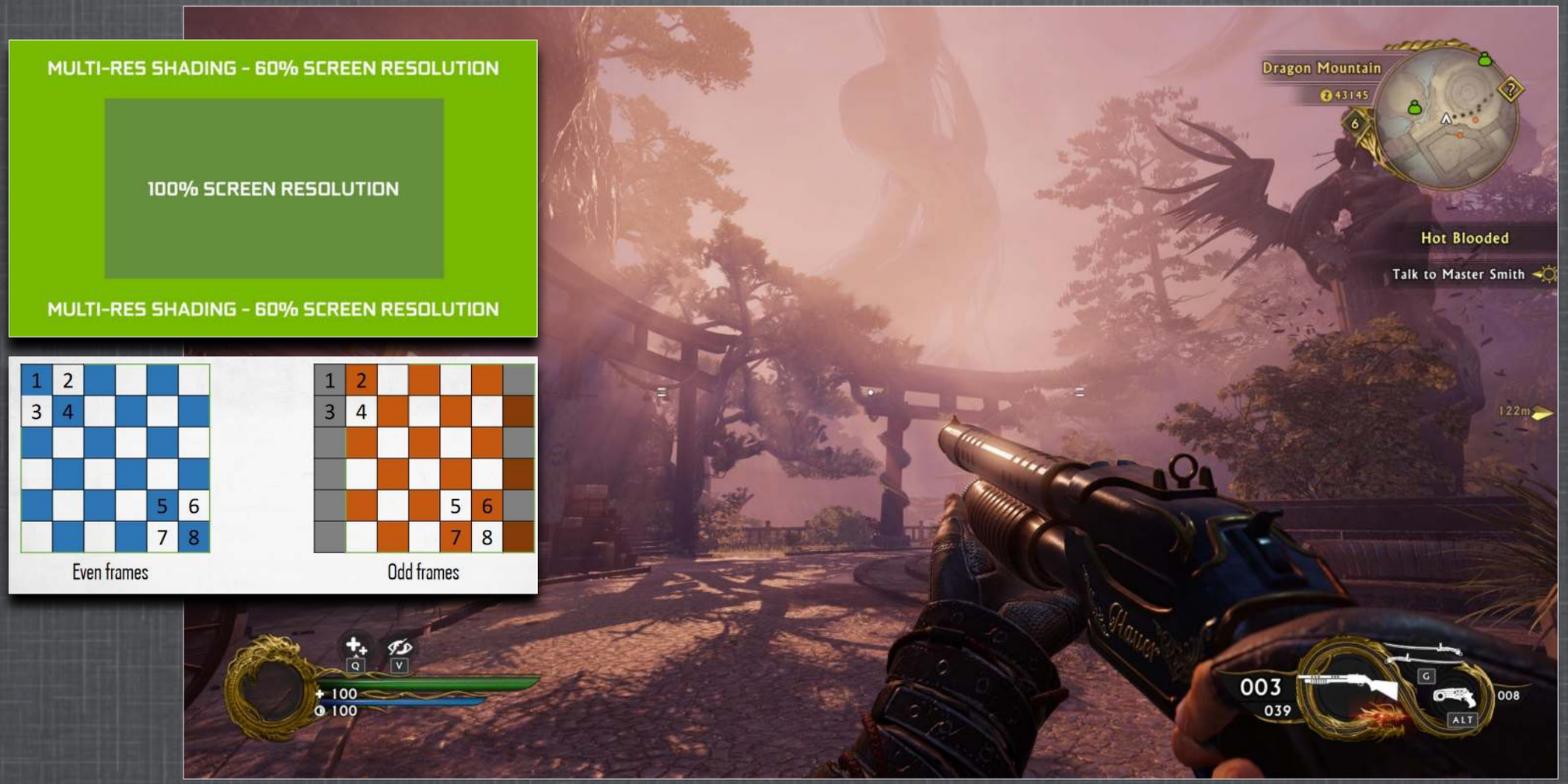
COMPUTER GRAPHICS IN GAMES





DIGITAL FOUNDRY, 2019
BORDERLANDS 2 AND 3

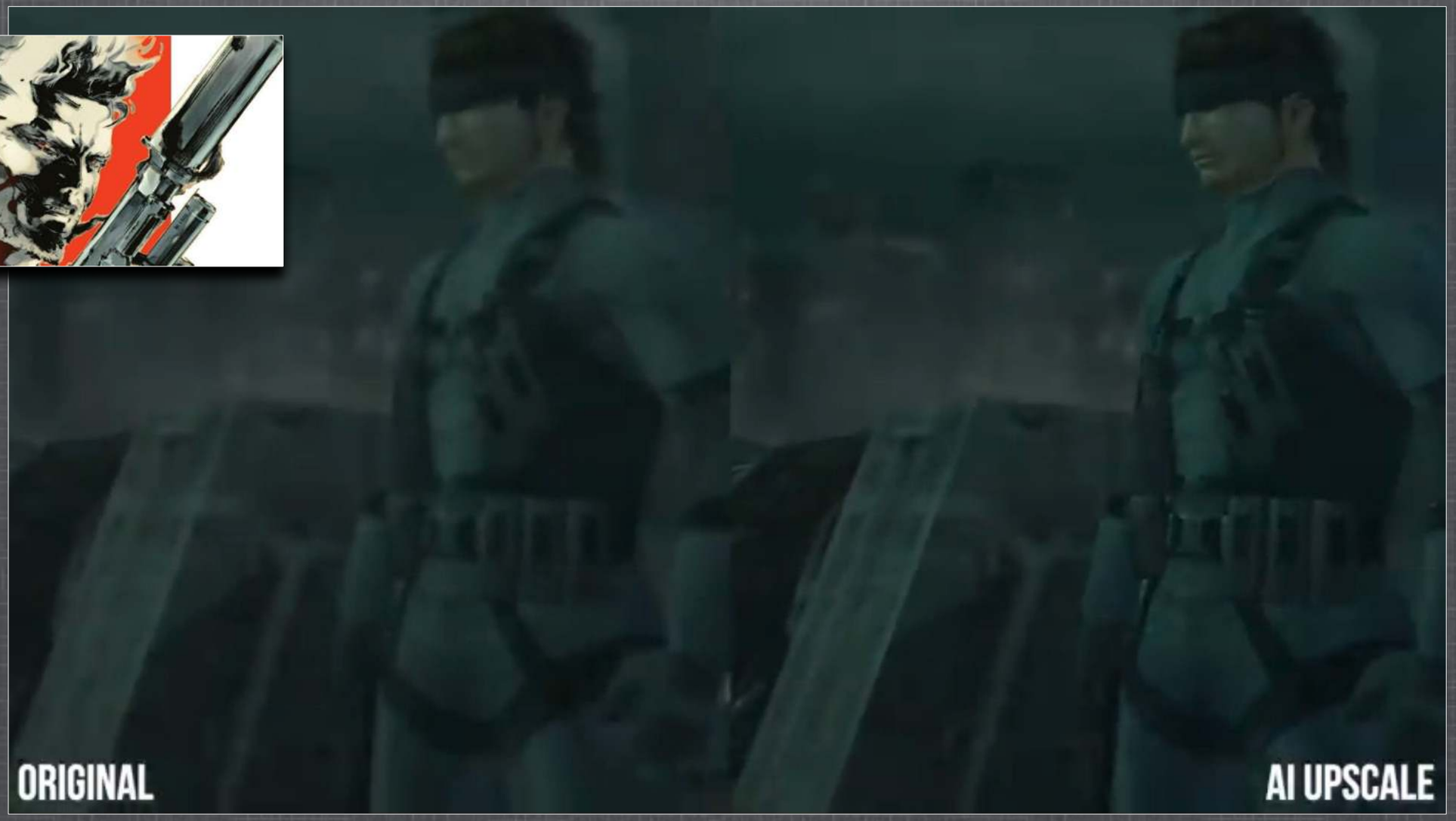
COMPUTER GRAPHICS IN GAMES



SHADOW WARRIOR 2
MULTI-RESOLUTION MULTI-FRAME SHADING

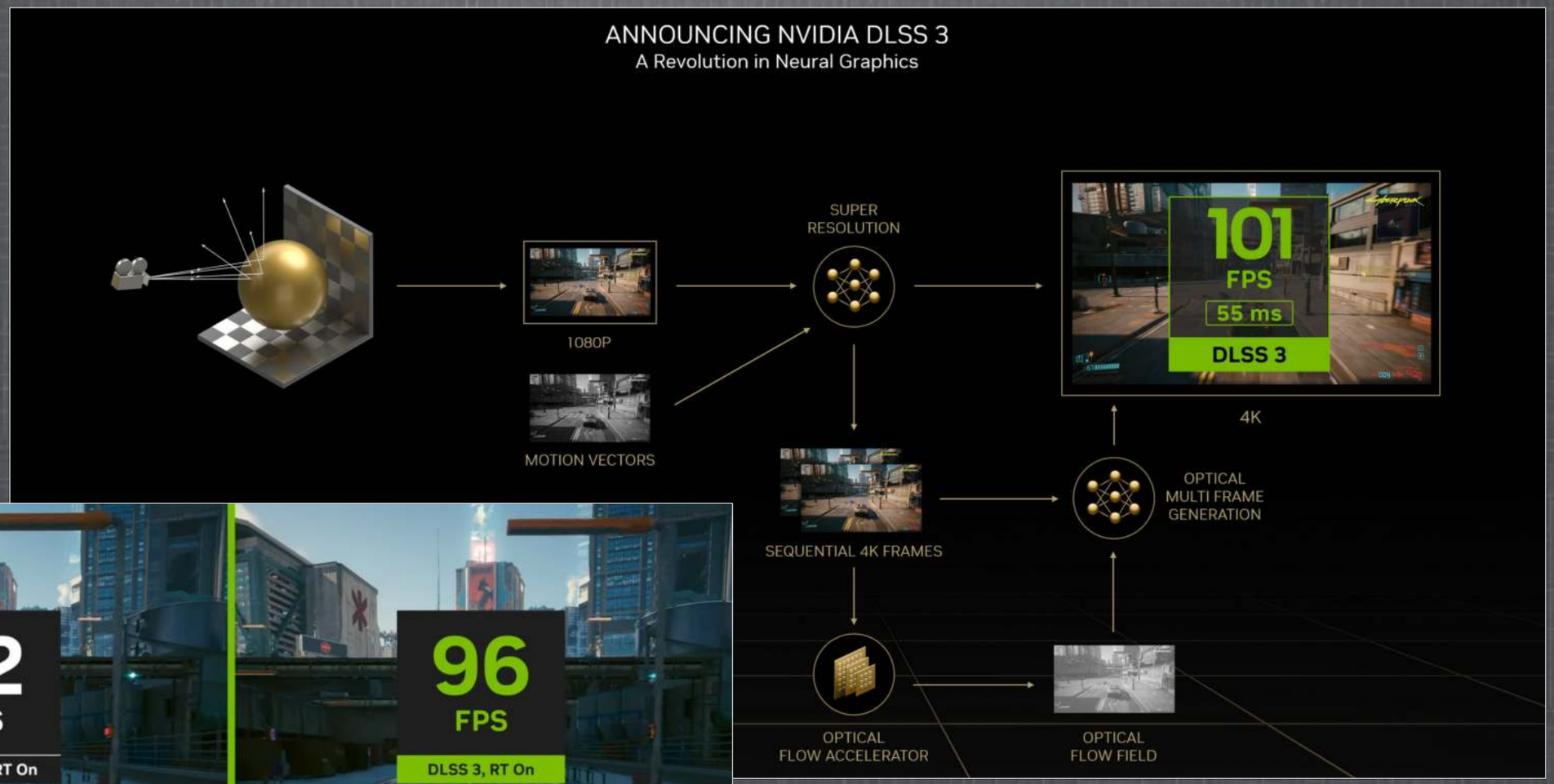
COMPUTER GRAPHICS IN GAMES





METAL GEAR SOLID 2 – 2021 UPSCALED TRAILER

COMPUTER GRAPHICS IN GAMES





CYBERPUNK 2077 @ GTC 2022

COMPUTER GRAPHICS IN GAMES

tom's HARDWARE

US Edition

Best Picks CPUs GPUs PC Components News Laptops Desktops Softw

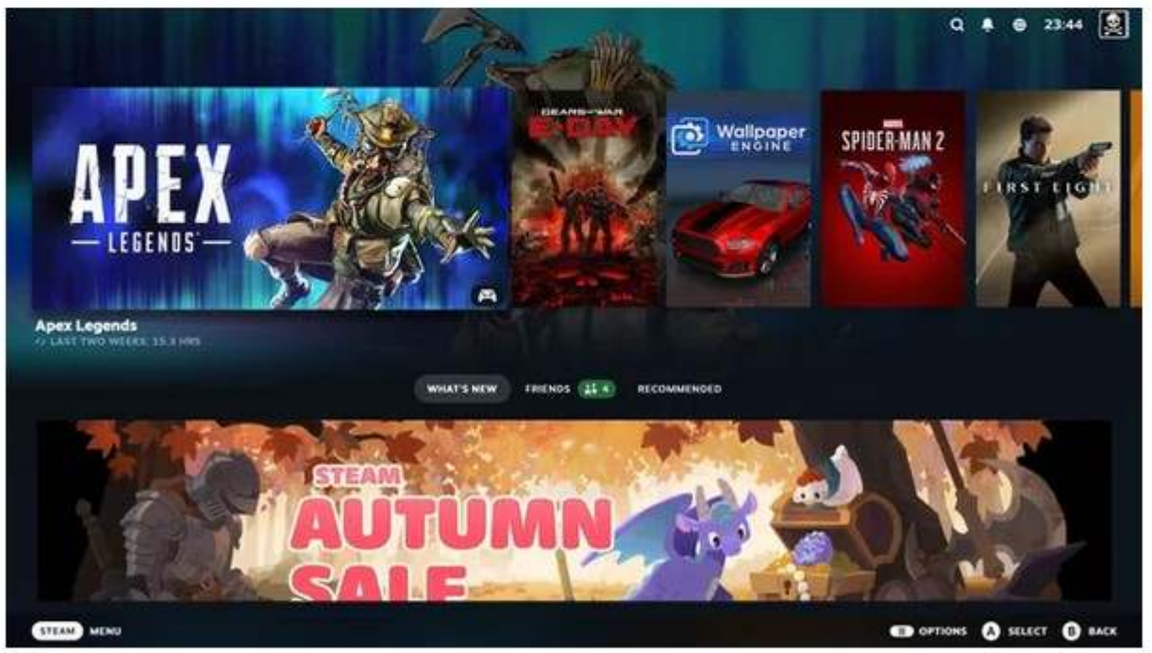
TRENDING RTX Spark Deals Nova Lake RAM Shortage AI Data

Video Games > PC Gaming

Steam hits record $1.7 billion in September 2026 despite soaring PC hardware costs — platform approaches 21,000 new releases in 2026, on track to top $20 billion annually

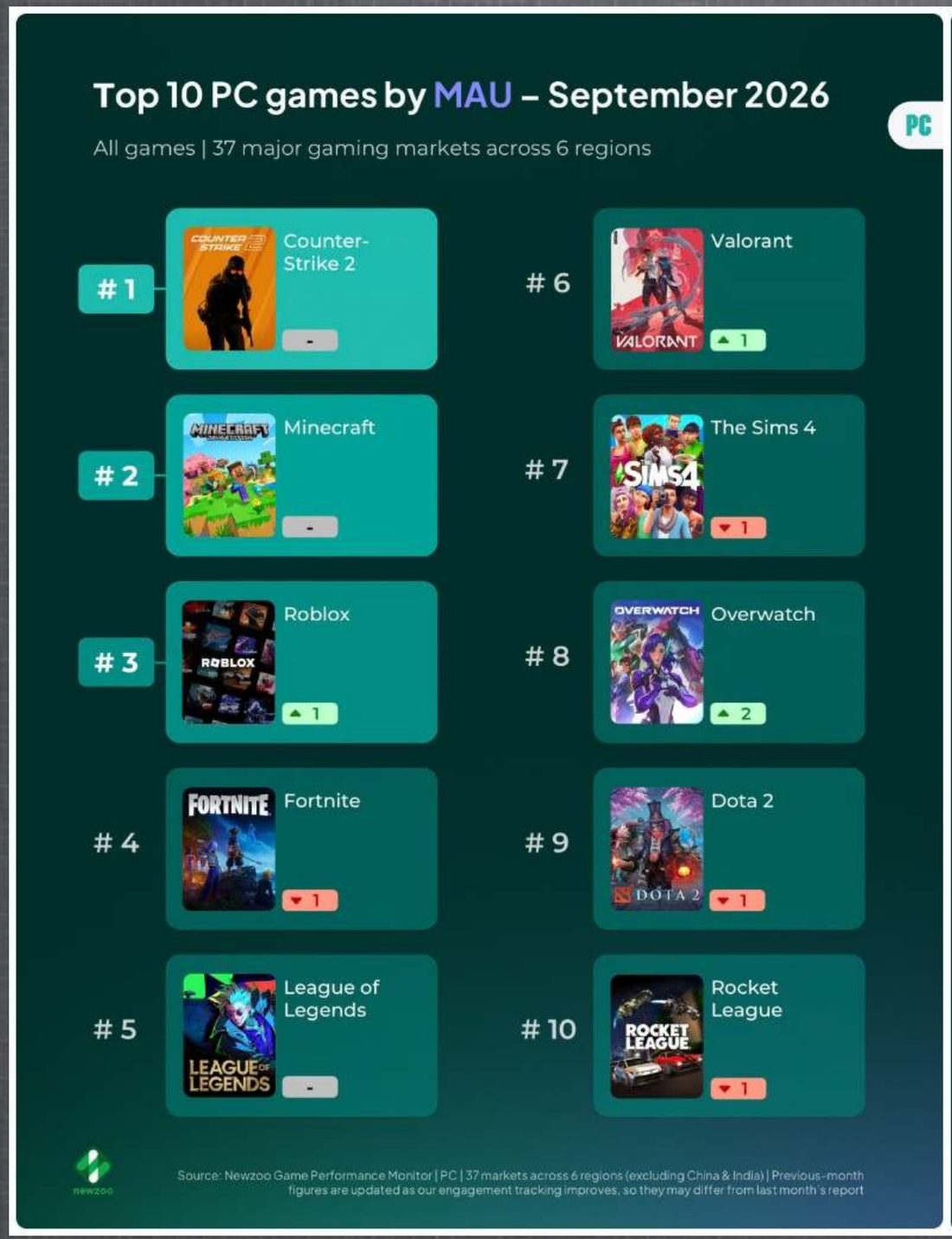
News By Kunal Khullar | Published October 3, 2026

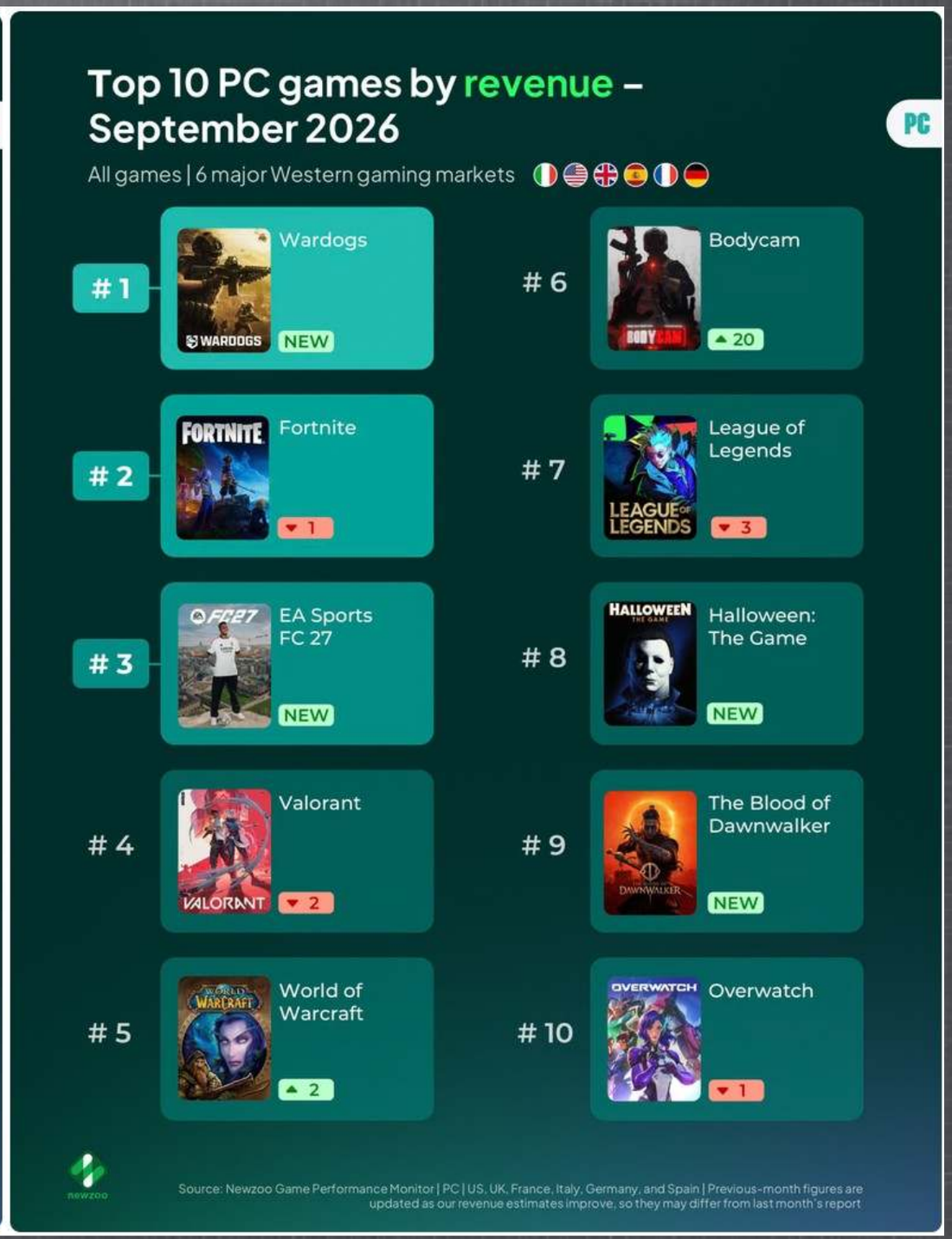
PC gaming stays strong.



(Image credit: Future)

COMPUTER GRAPHICS IN GAMES



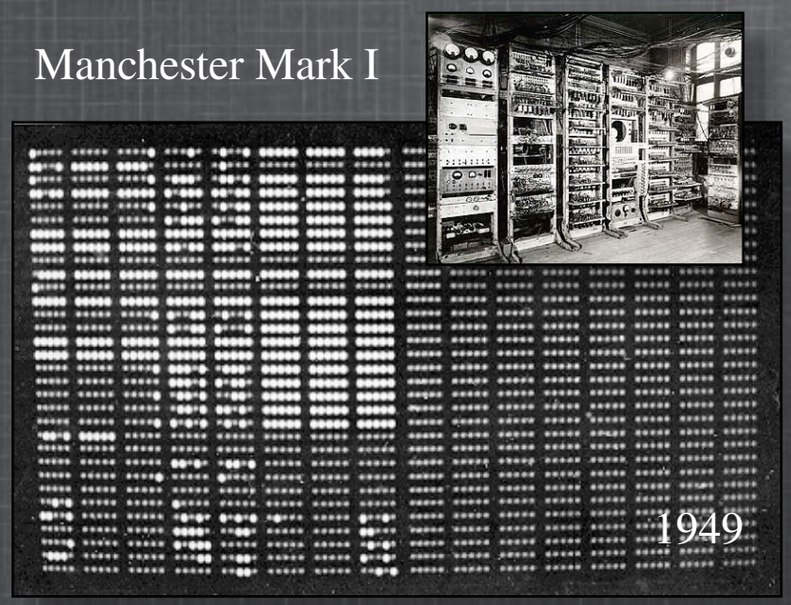


DIGITAL MEDIA

DIGITAL MEDIA

From *text* to *hypertext*, *image* and *video*

Manchester Mark I



Fiction (1975-77)



Reality (1999)



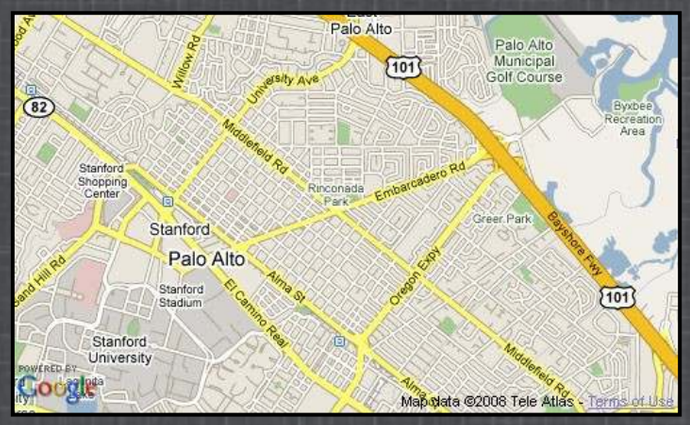
flickr · facebook · YouTube · twitch

DIGITAL MEDIA

Digital Twins

Google Earth and Google Maps



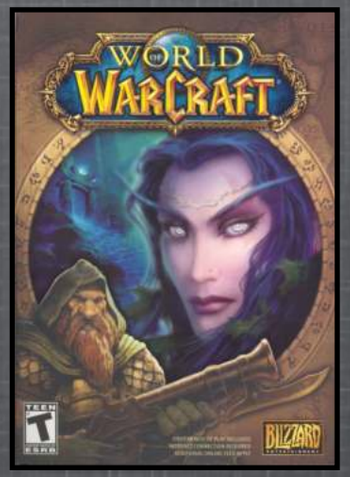


Home Simulation





Social Spaces



Second Life



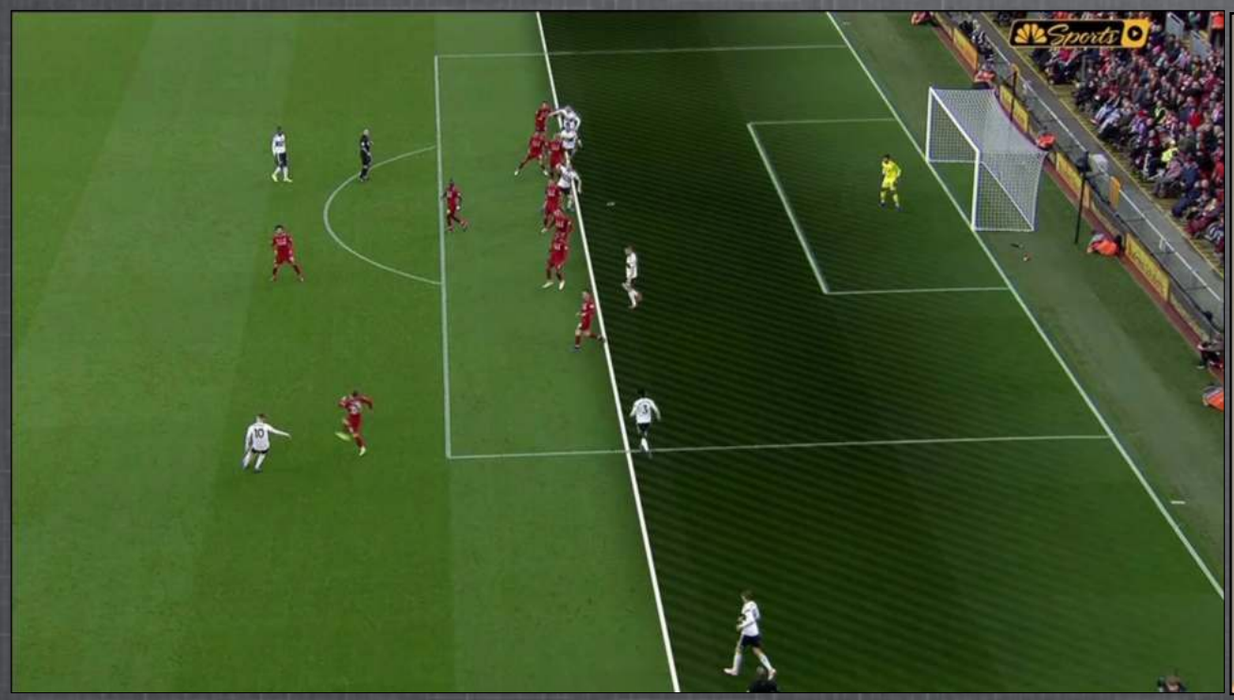
Electronic Publishing



DIGITAL MEDIA

Processing Feelings

Information overlay



Which one is the original? Are they real?



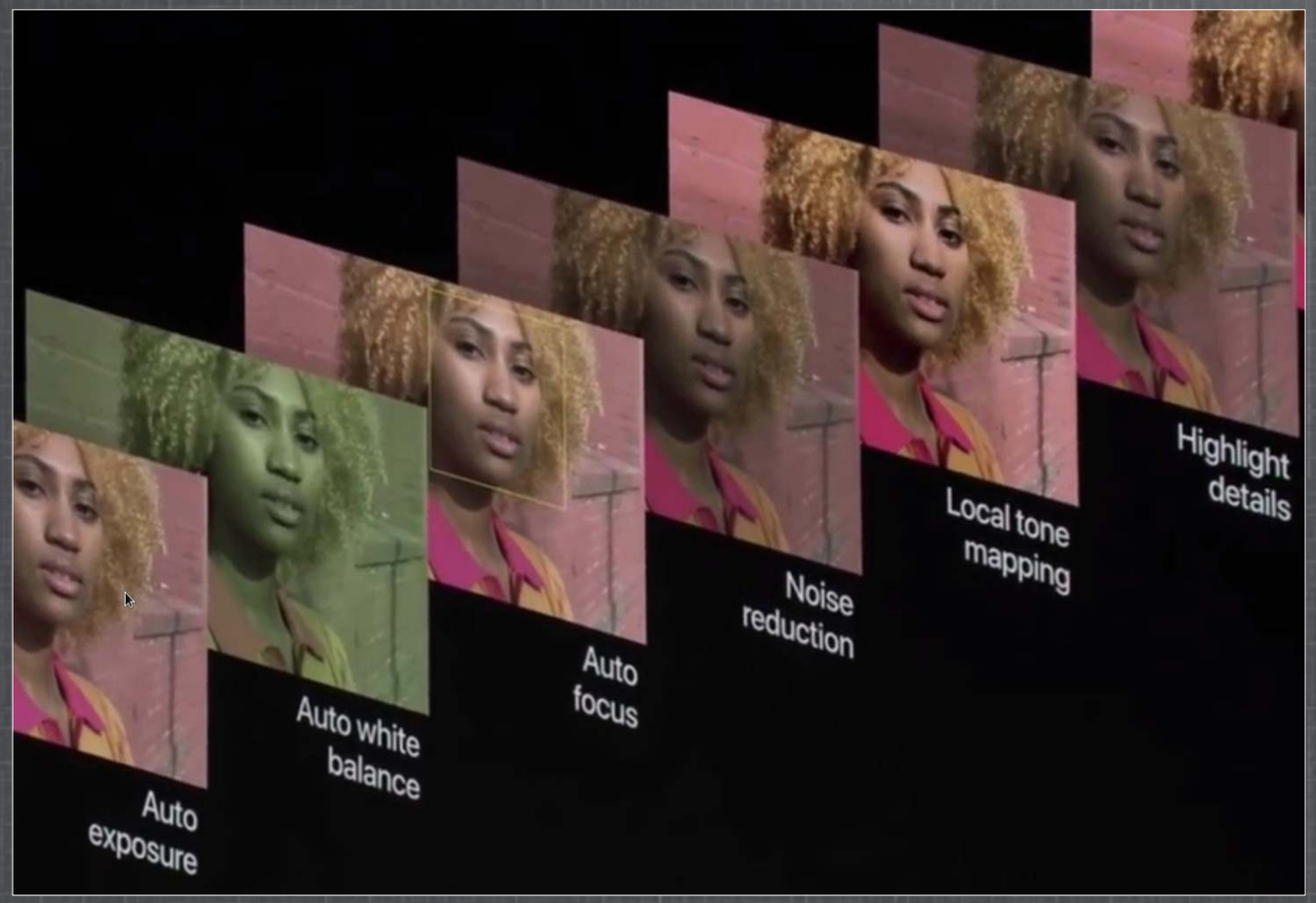




Composition Enhancement

DIGITAL MEDIA

How real is our memory of a capture?



Apple X s MAX launch (sep 2018)



DIGITAL MEDIA

How real is our memory of a capture?





Lens Flare

DIGITAL MEDIA

Future of Media



COMPUTER AIDED DESIGN AND VISUALISATION



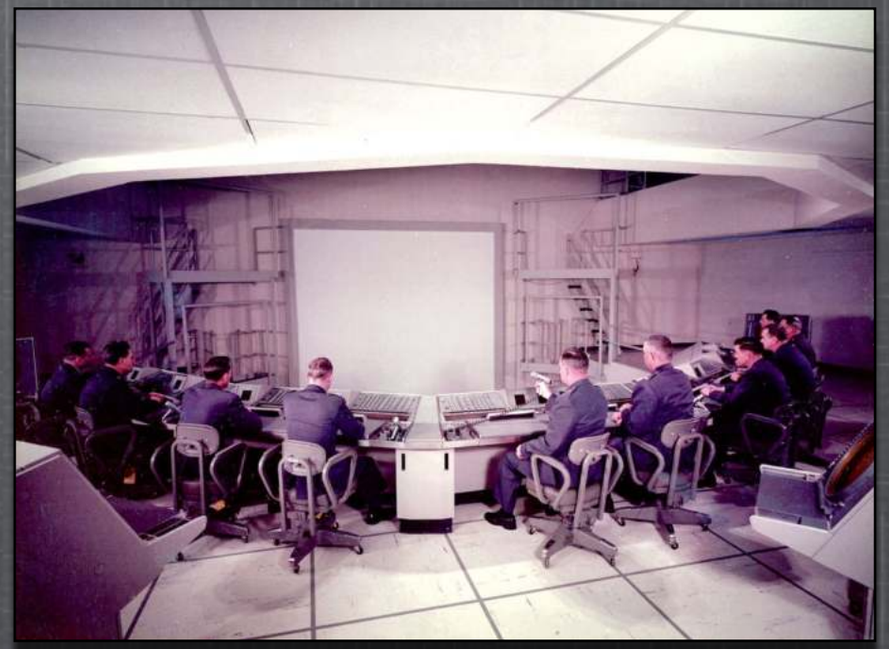
Unreal Engine 4 - Unreal Paris (January 2015)

BRIEF HISTORY OF COMPUTER GRAPHICS

COMPUTER GRAPHICS

TERM COINED BY WILLIAM FETTER (BOEING) IN 1960



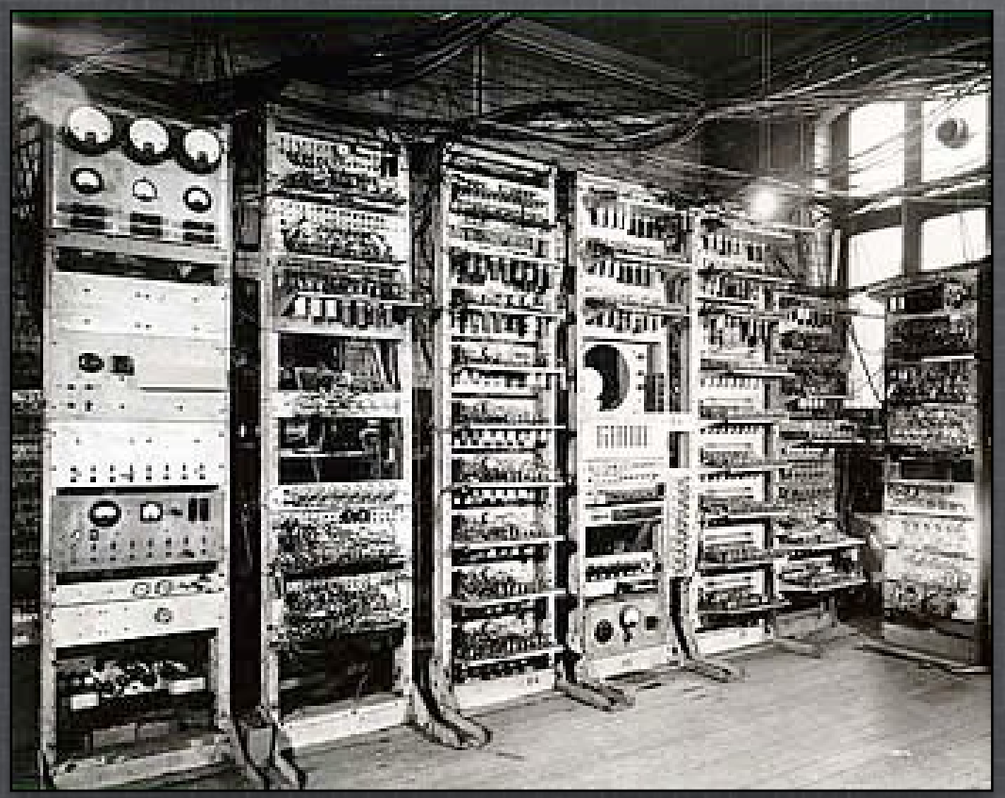


1ST GRAPHICS SYSTEM

USAF SAGE RADAR DATA (MIT) IN MID 1950'S

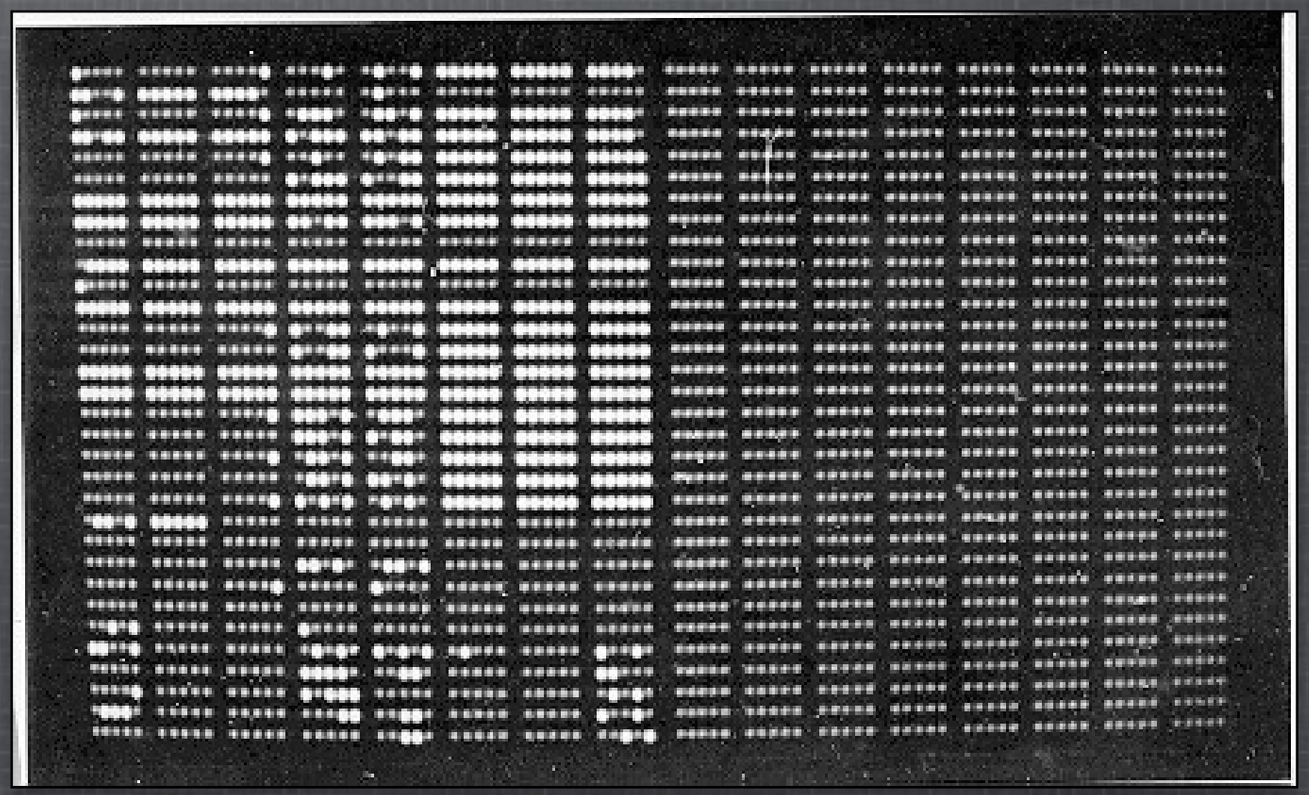
2D COMPUTER GRAPHICS

1ST TEXT SYSTEMS

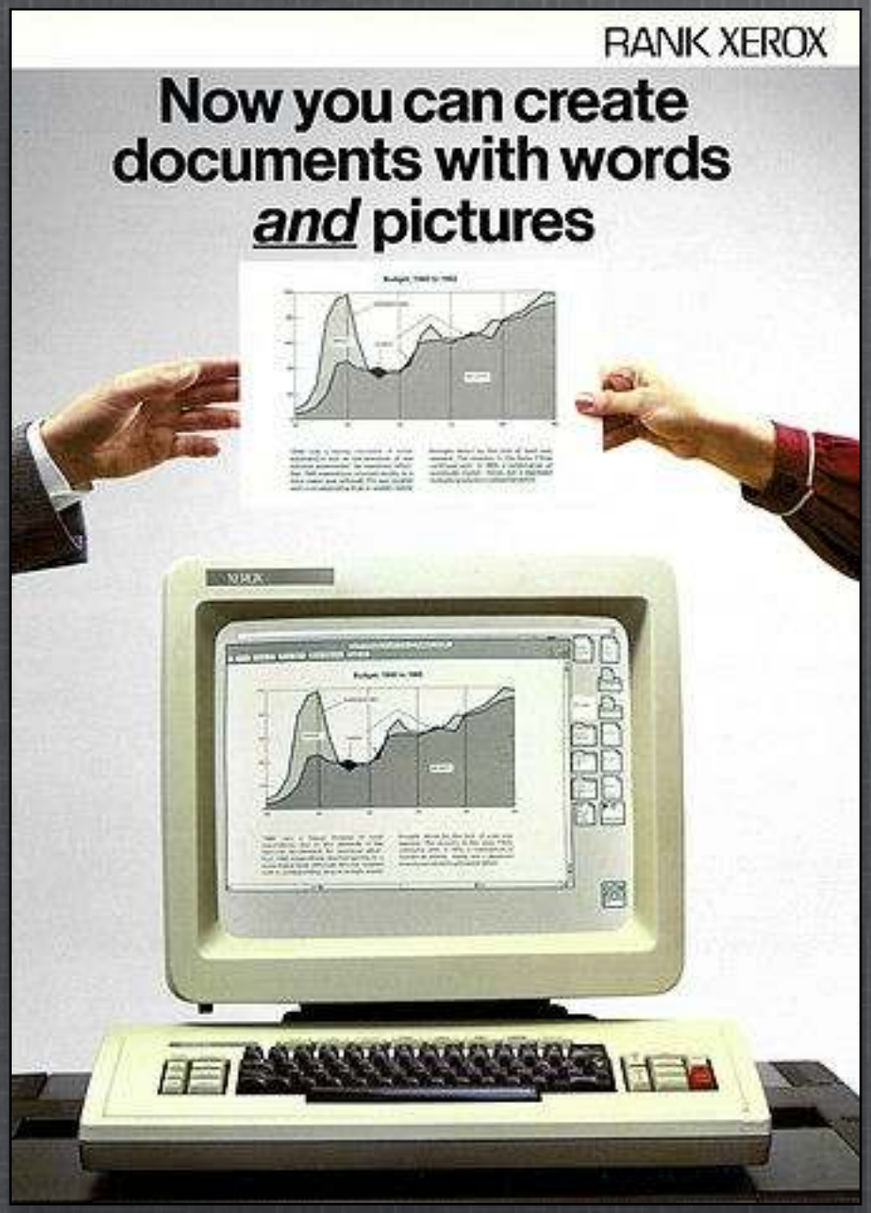


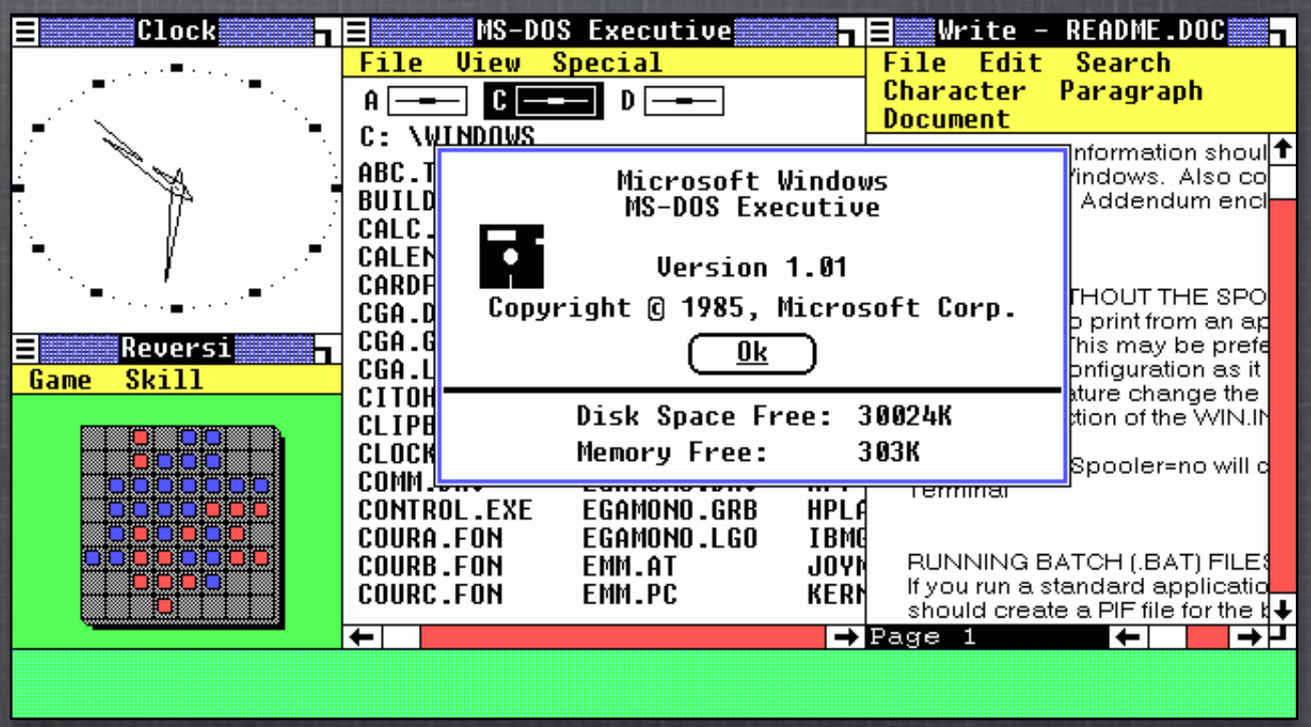
Manchester Mark I

Display

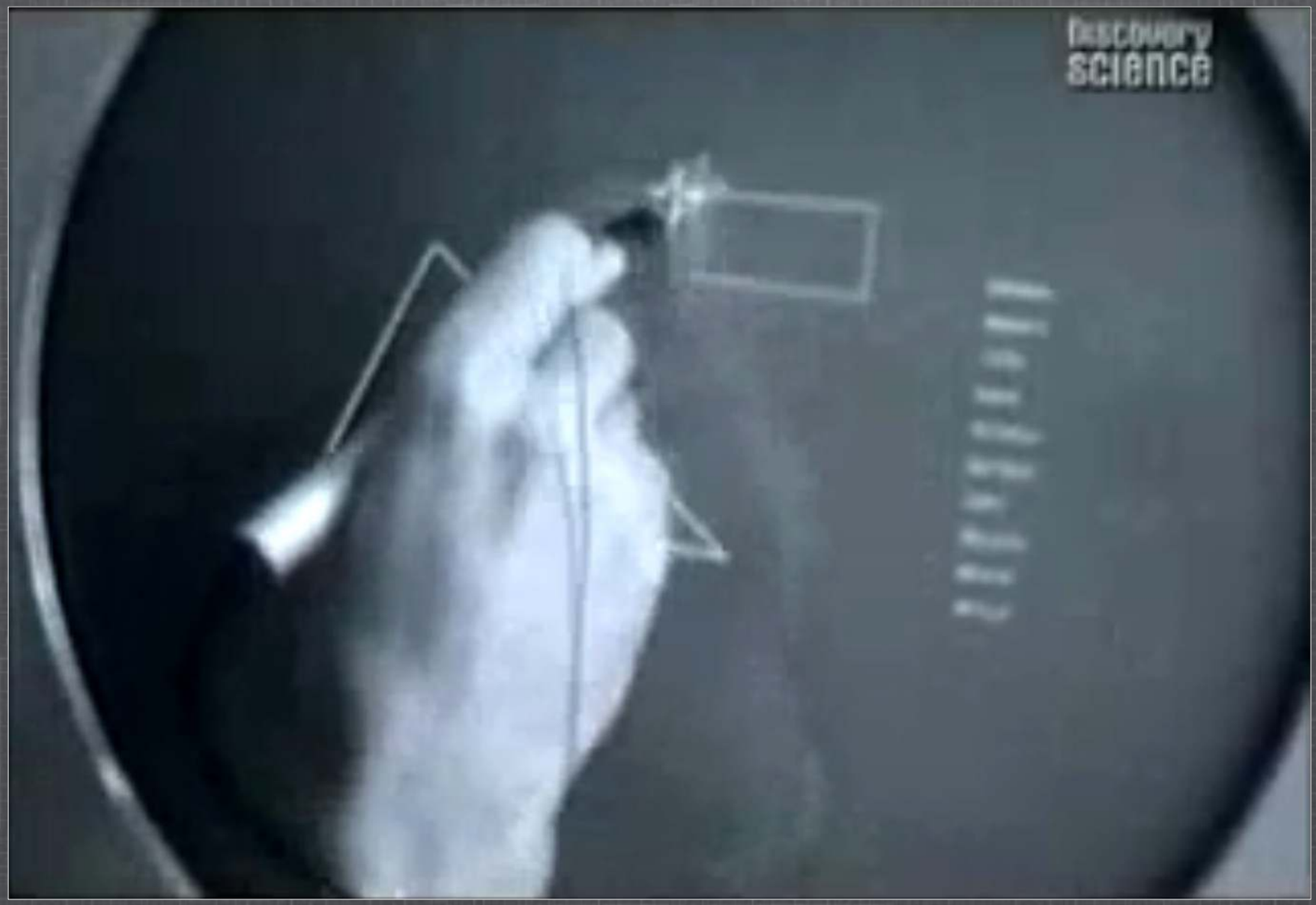


1ST GRAPHICAL USER INTERFACES





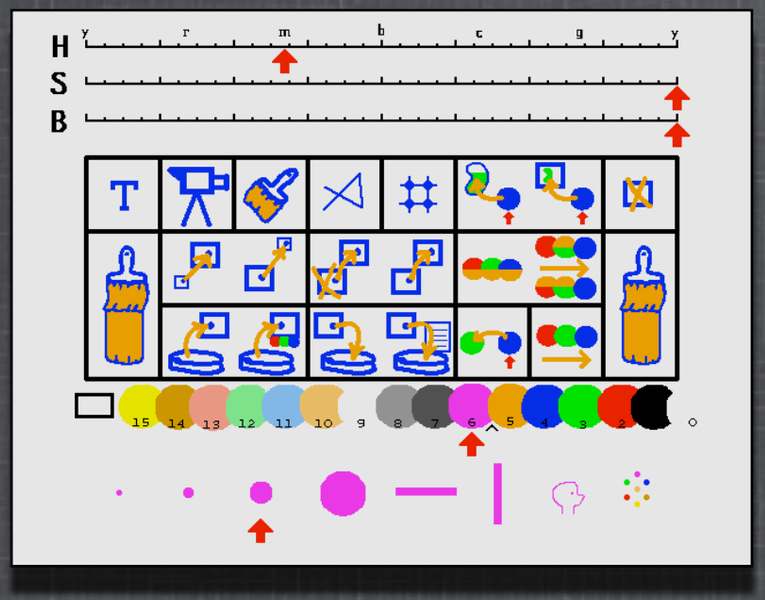
1ST DRAWING SYSTEMS

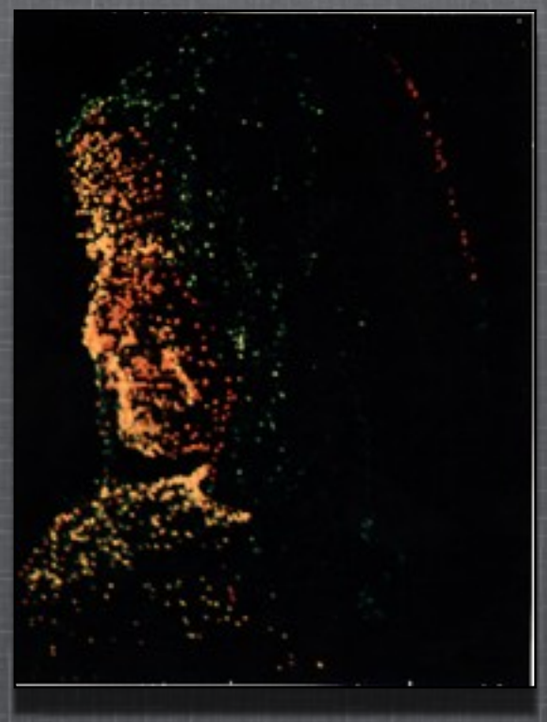


Sketchpad (Sutherland, MIT 1963)



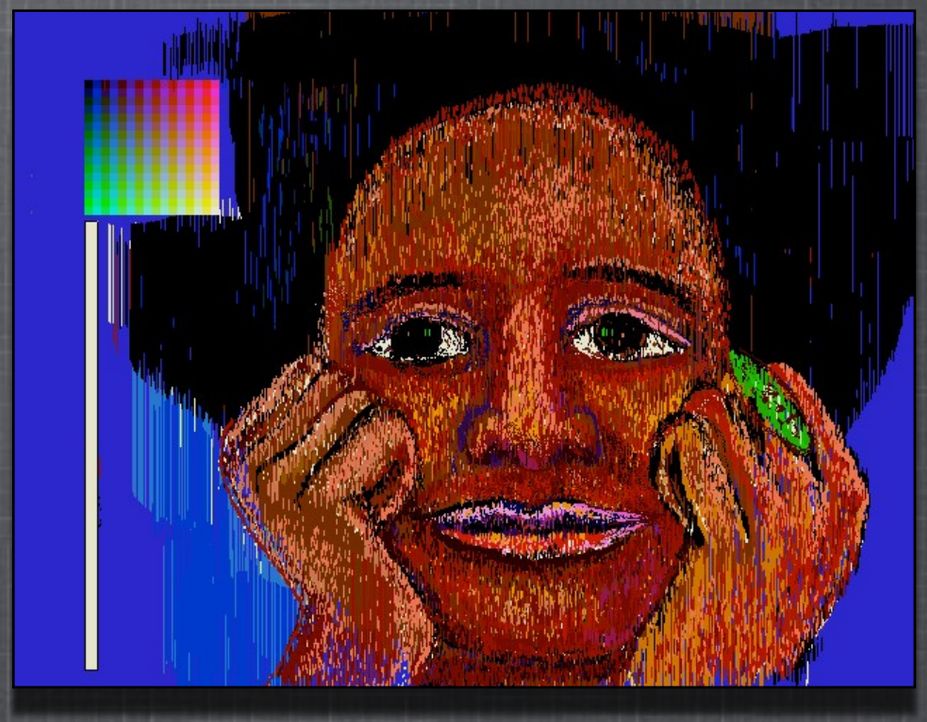
1ST PAINT SYSTEMS





Janet, by Bob Flegal

Black Girl, by Fritz Fisher

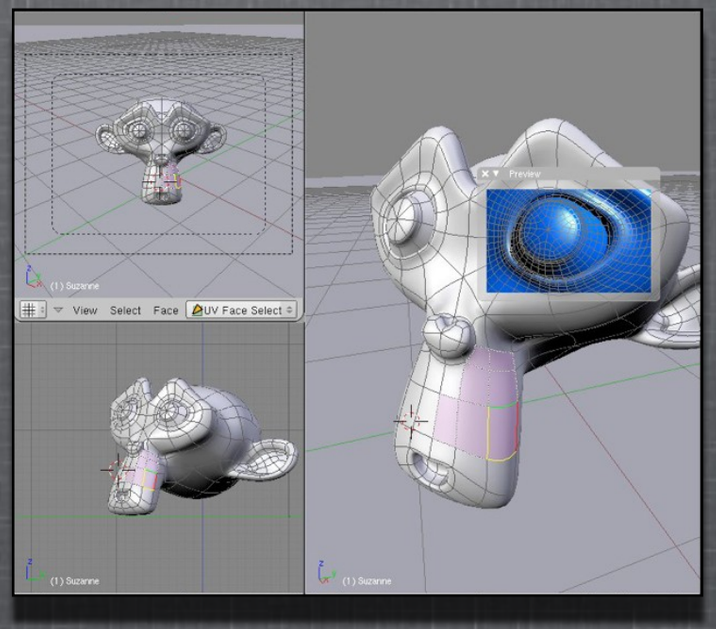


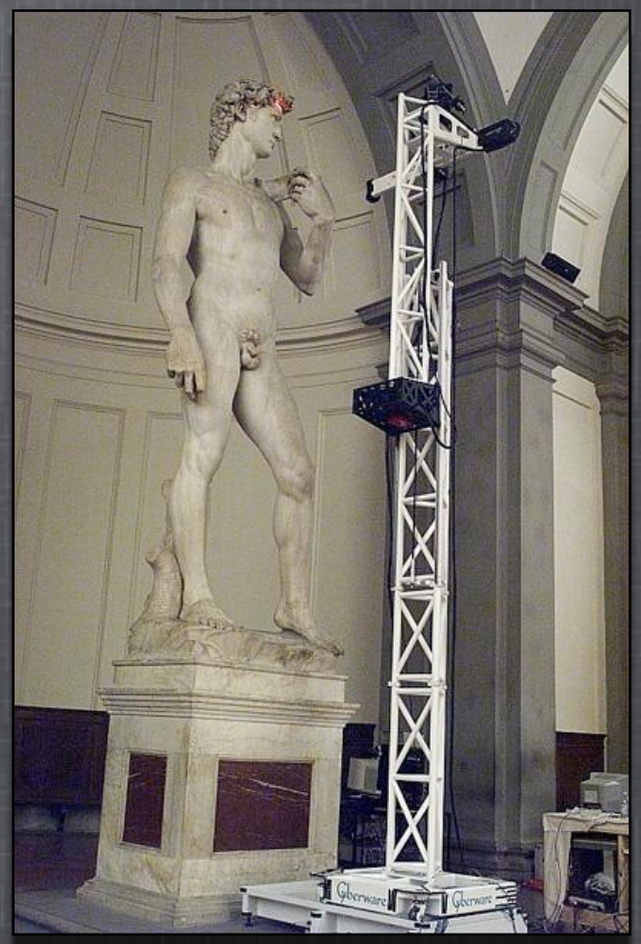
SuperPaint Framebuffer System (Shoup and Smith, 1973-1979)

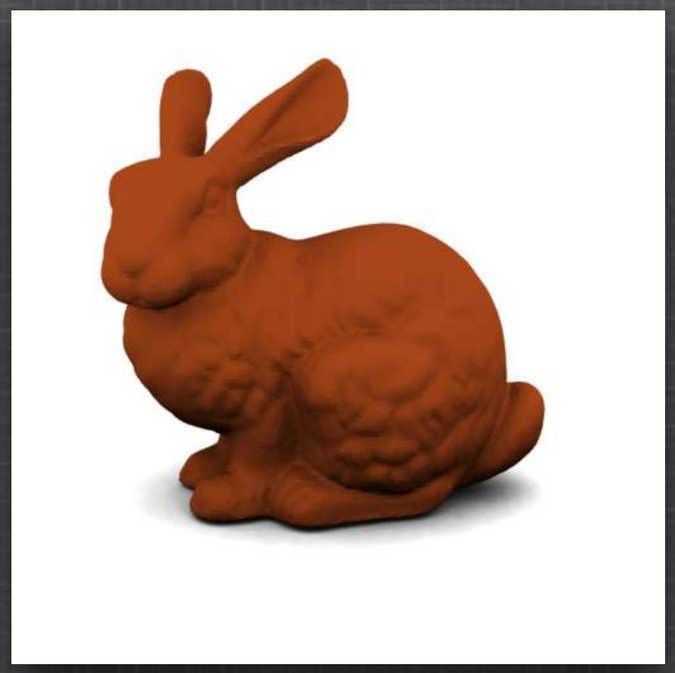
3D COMPUTER GRAPHICS

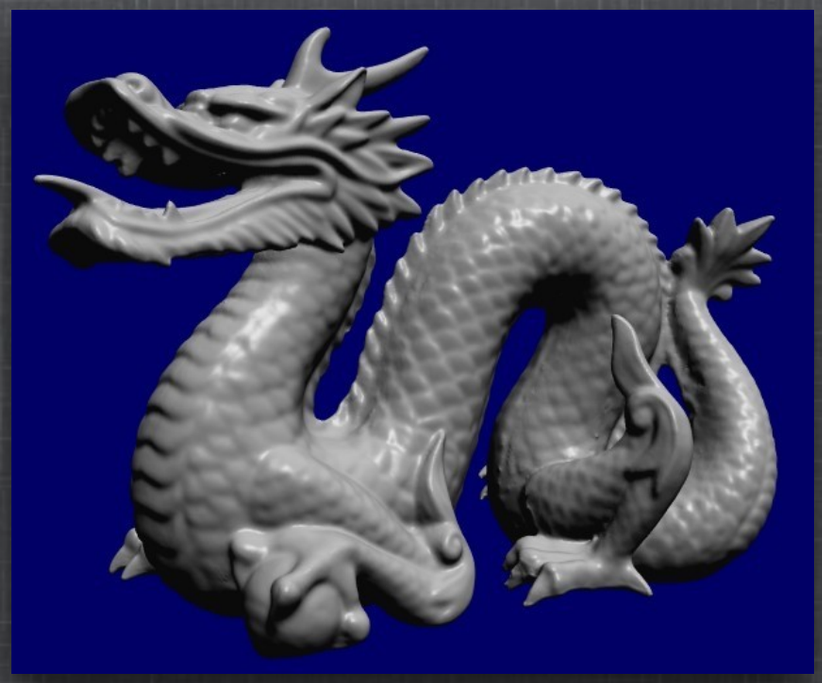
MODELING

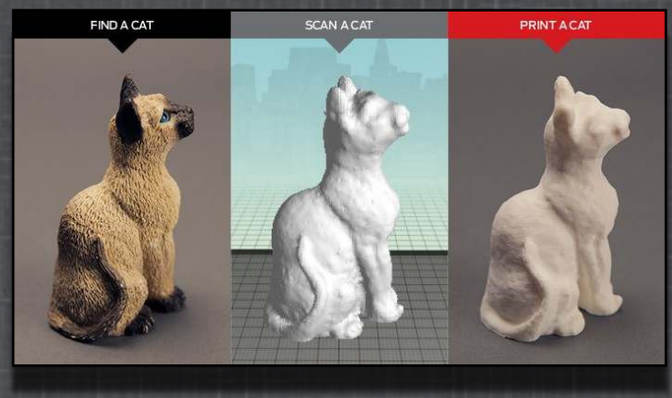



(1) Suzanne
View
Select
Face
UV Face Select
Preview
(1) Suzanne
(1) Suzanne

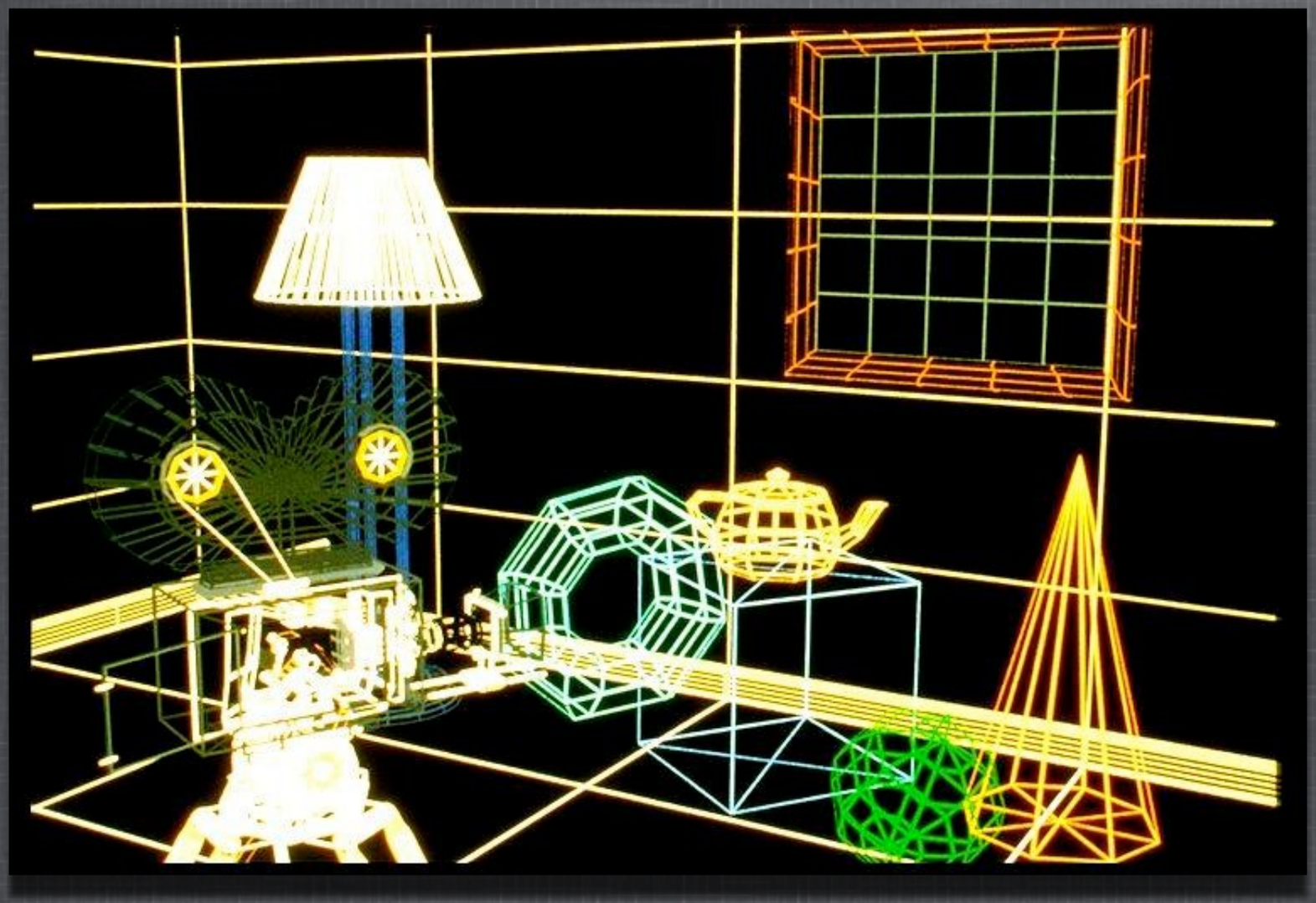






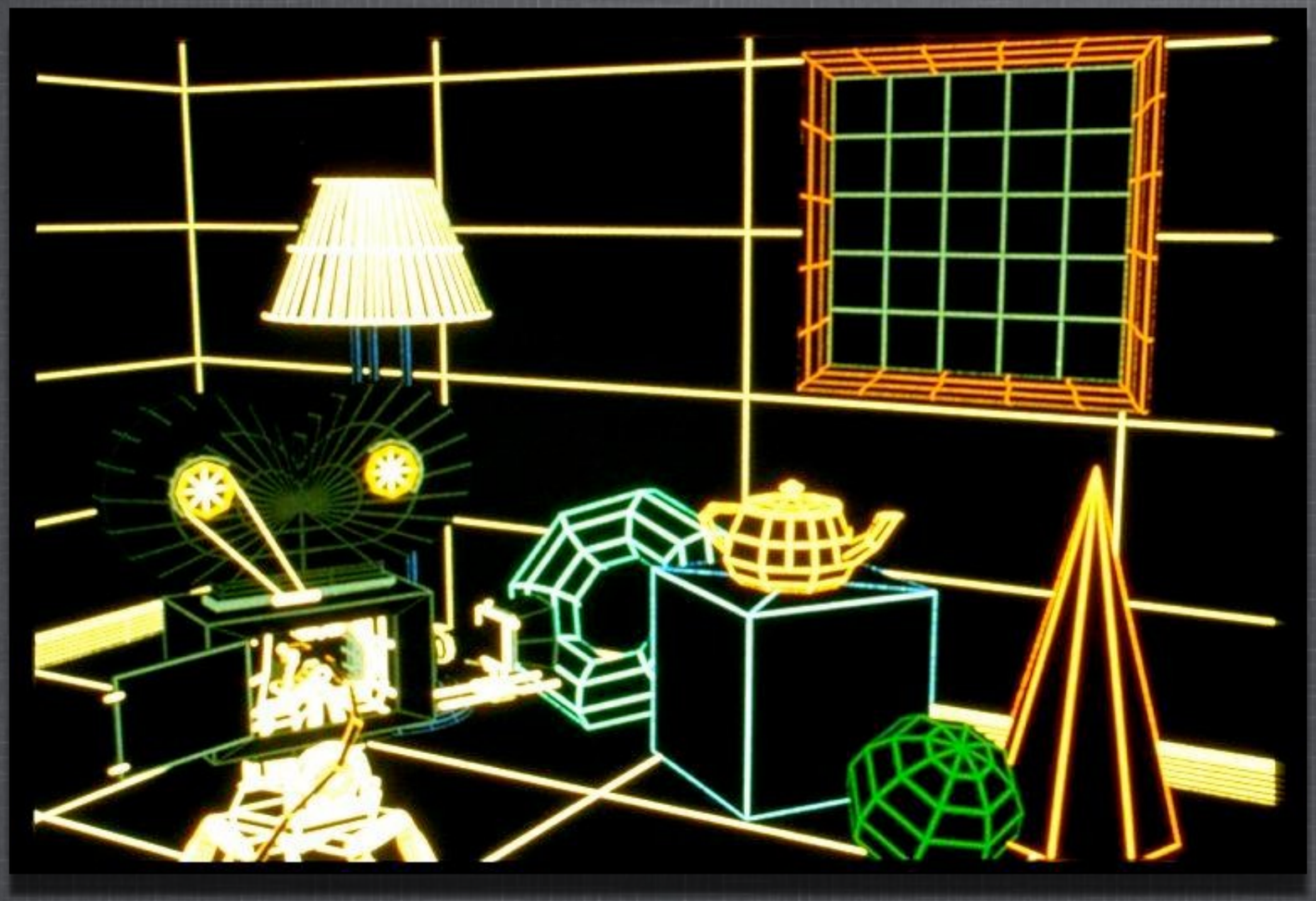

FIND A CAT
SCAN A CAT
PRINT A CAT

RENDERING



RENDERING

Visibility (1960's)



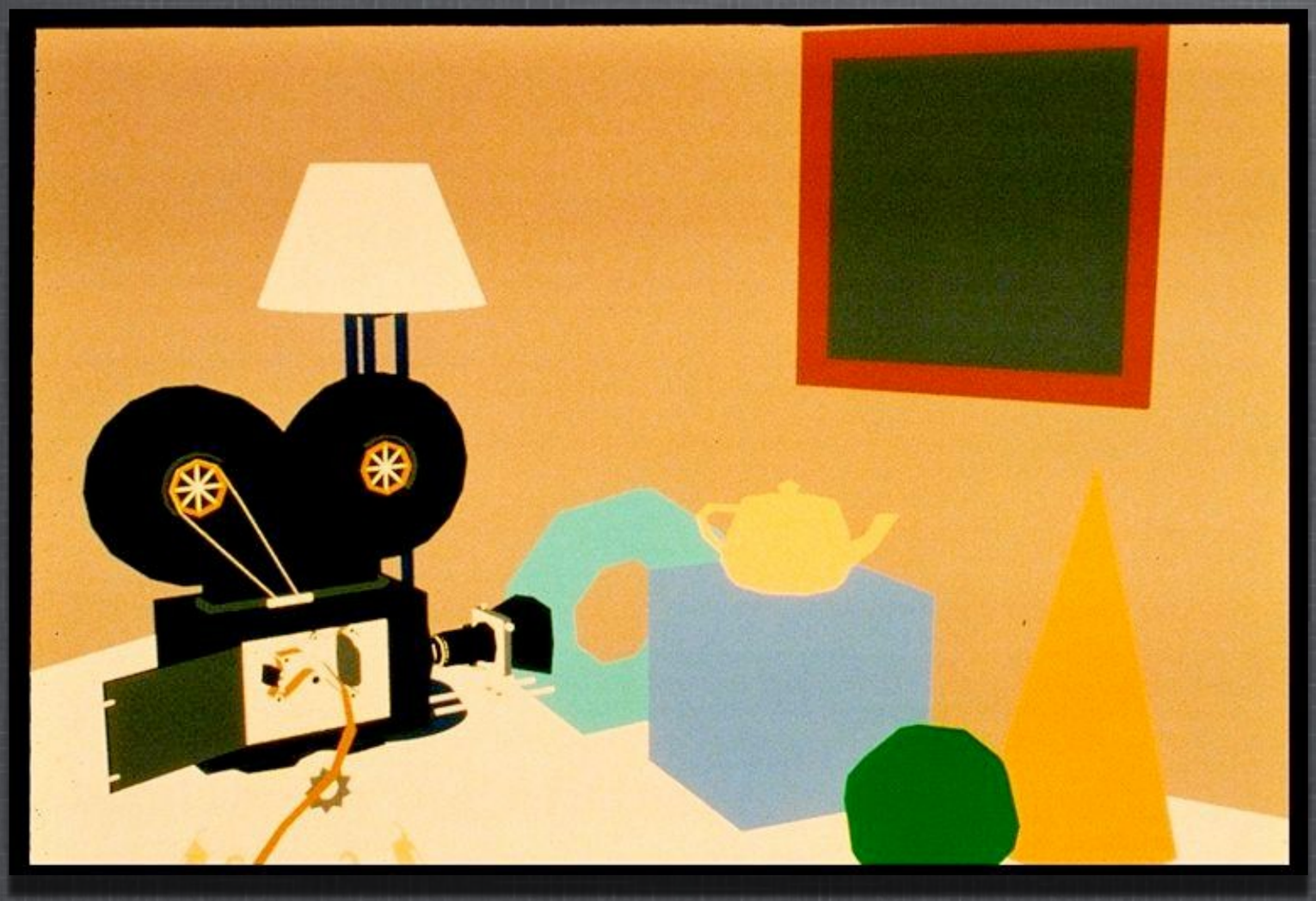
Hidden Line Algorithms:
Roberts (63), Appel (67)

Hidden Surface Algorithms:
Warnock (69), Watkins (70)

Visibility and Sorting:
Sutherland (74)

RENDERING

Visibility and Lighting (1970's)



Hidden Line Algorithms:
Roberts (63), Appel (67)

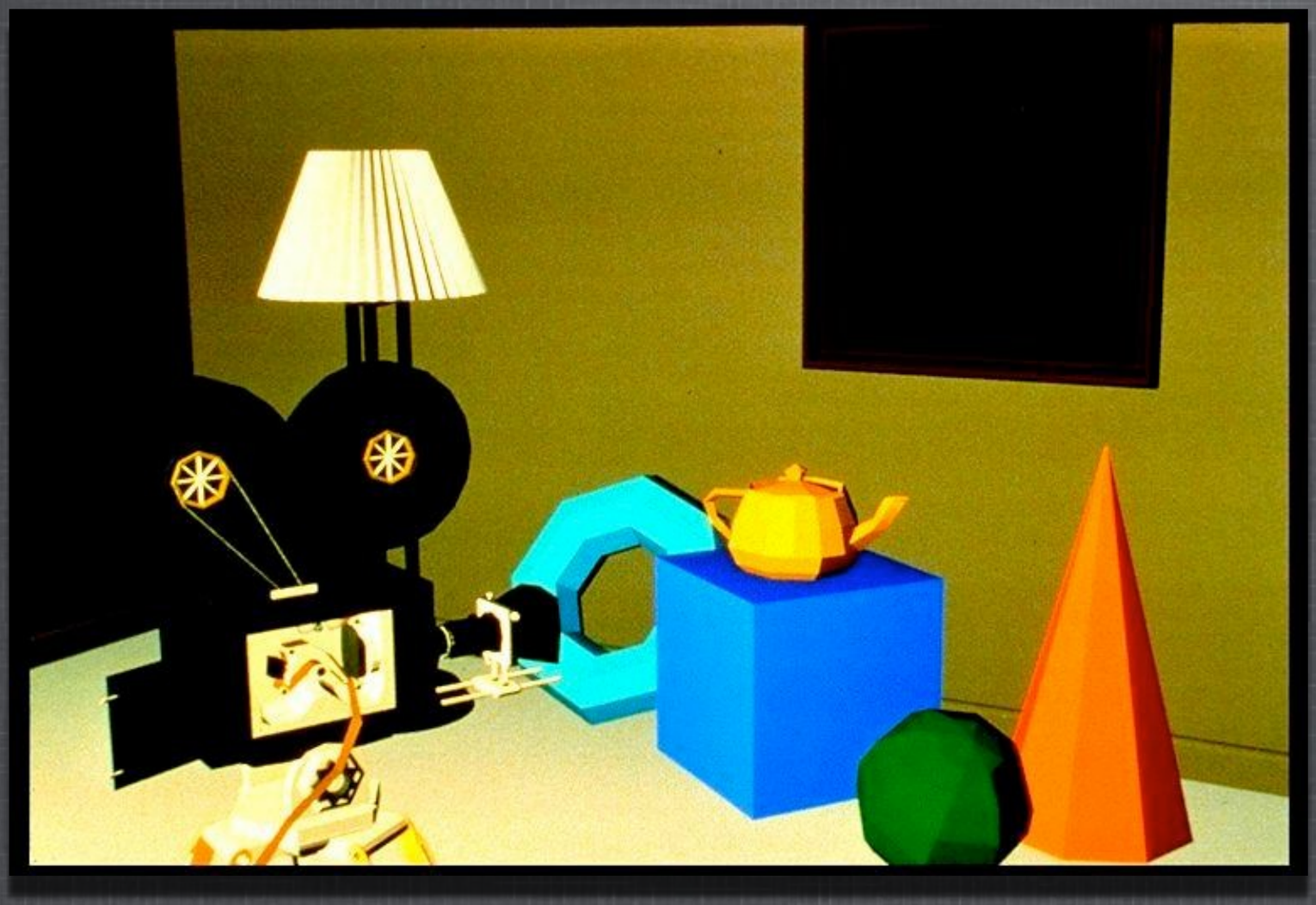
Hidden Surface Algorithms:
Warnock (69), Watkins (70)

Visibility and Sorting:
Sutherland (74)

Constant shading

RENDERING

Lighting (1970's)



Flat shading

RENDERING

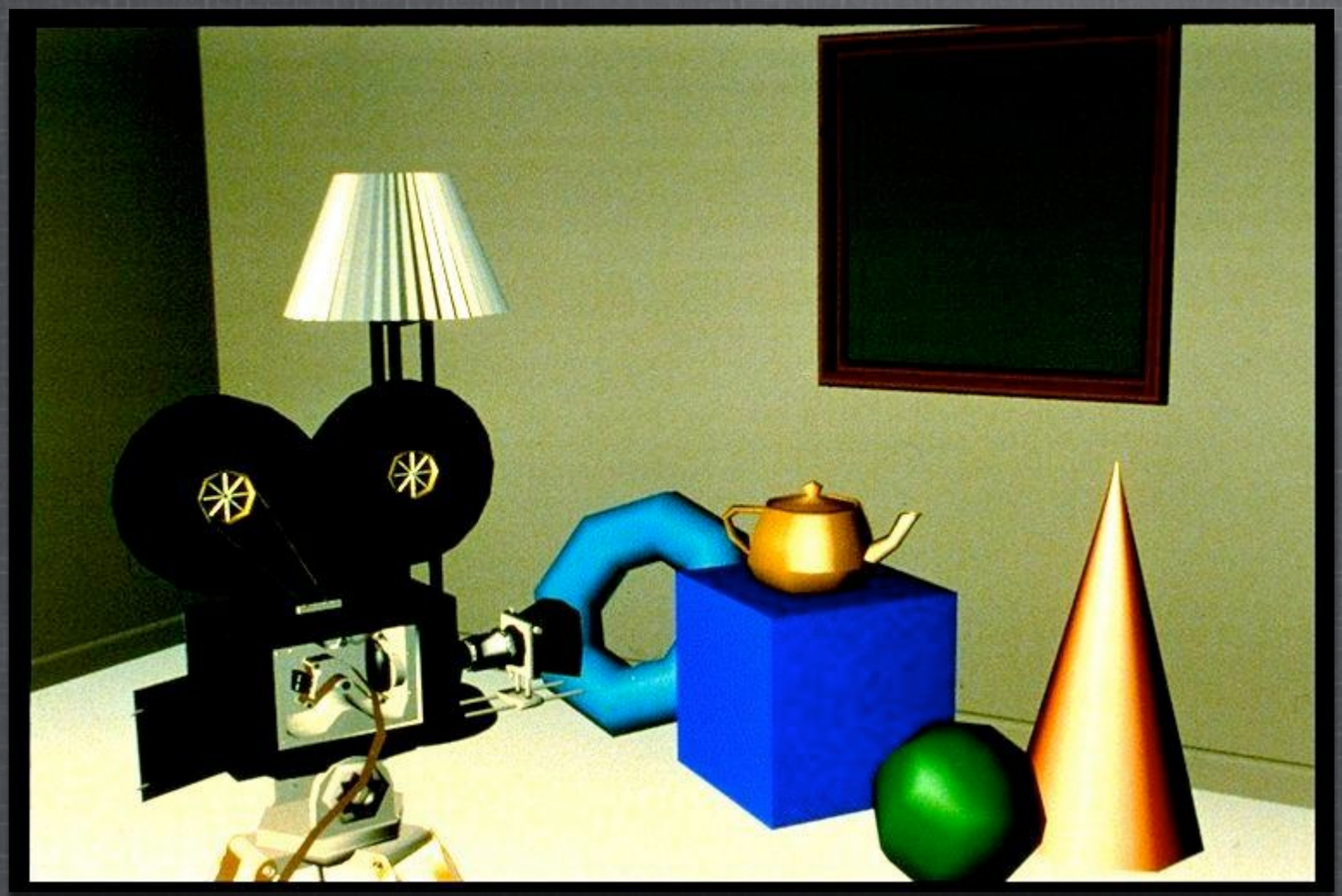
Lighting (1970's)



Diffuse lighting
(Gouraud 1971)

RENDERING

Lighting (1970's)



Diffuse lighting
(Gouraud 1971)

Specular lighting
(Phong 1974)

RENDERING

Lighting (1970's)



Diffuse lighting
(Gouraud 1971)

Specular lighting
(Phong 1974)

RENDERING

Lighting (1970's)



Diffuse lighting
(Gouraud 1971)

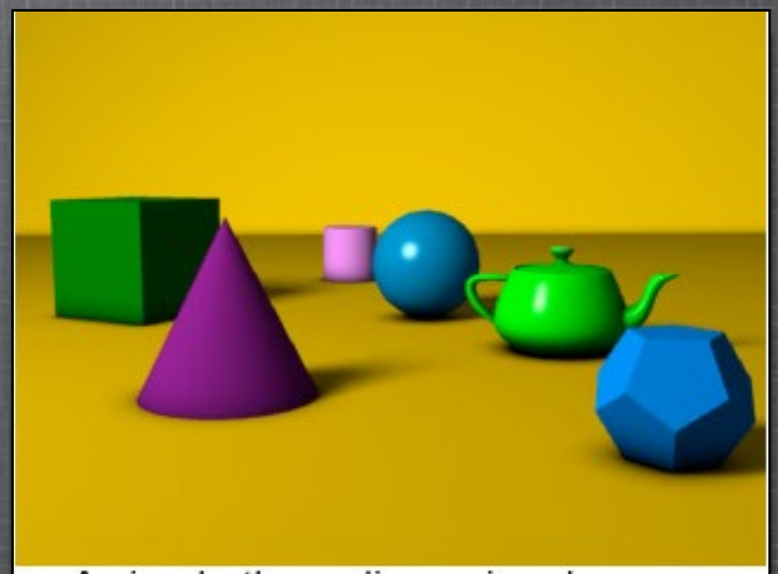
Specular lighting
(Phong 1974)

Curved Surfaces, Texture
(Blinn, 1974)



RENDERING

Z-buffer Hidden Surface
(Catmull, 1974)



A simple three dimensional scene



Z-buffer representation

RENDERING

Lighting (1970's)



Diffuse lighting
(Gouraud 1971)

Specular lighting
(Phong 1974)

Curved Surfaces, Texture
(Blinn, 1974)

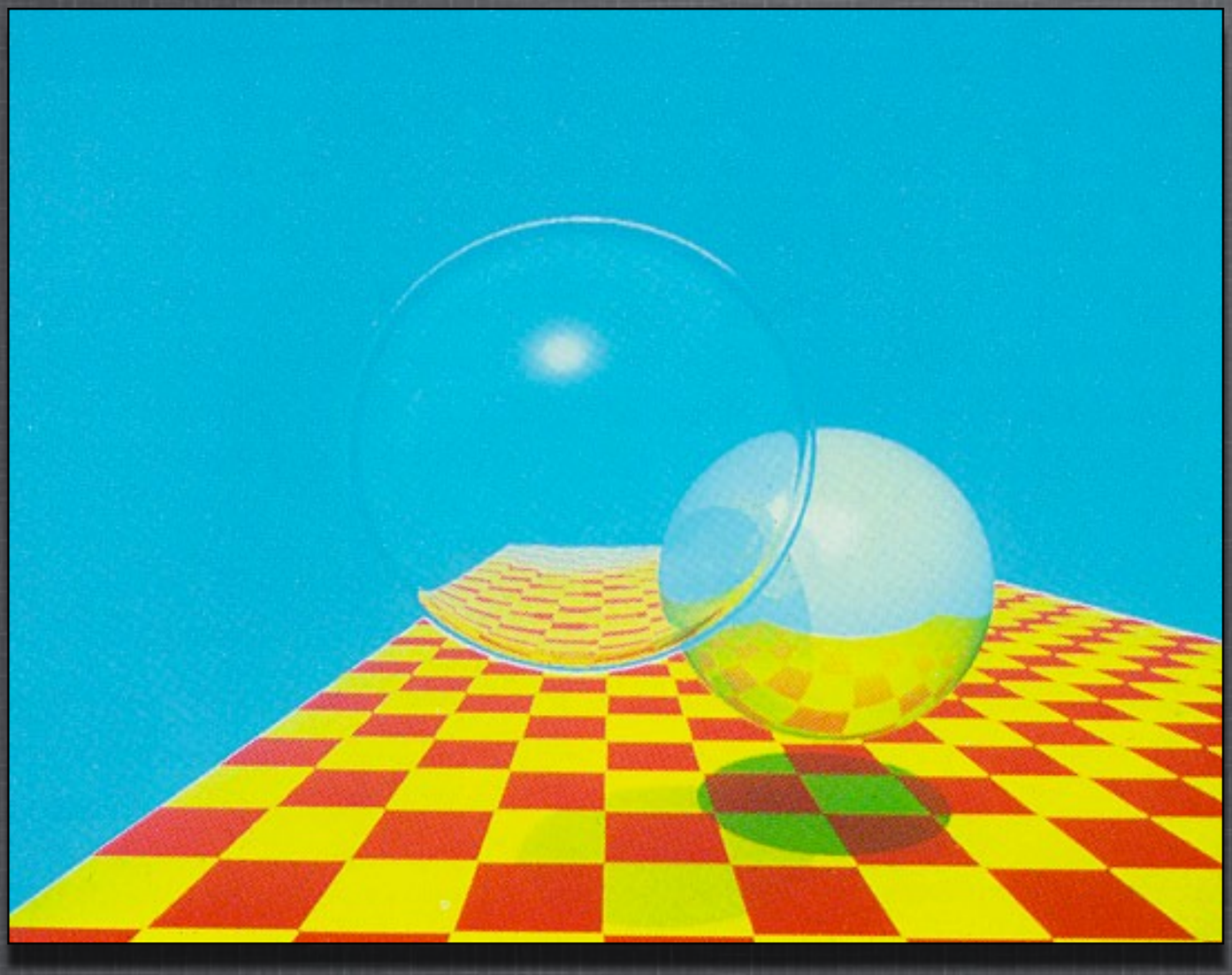
Antialias and Shadows
Crow (1977)

GLOBAL ILLUMINATION

(1980's and 1990's)

RENDERING

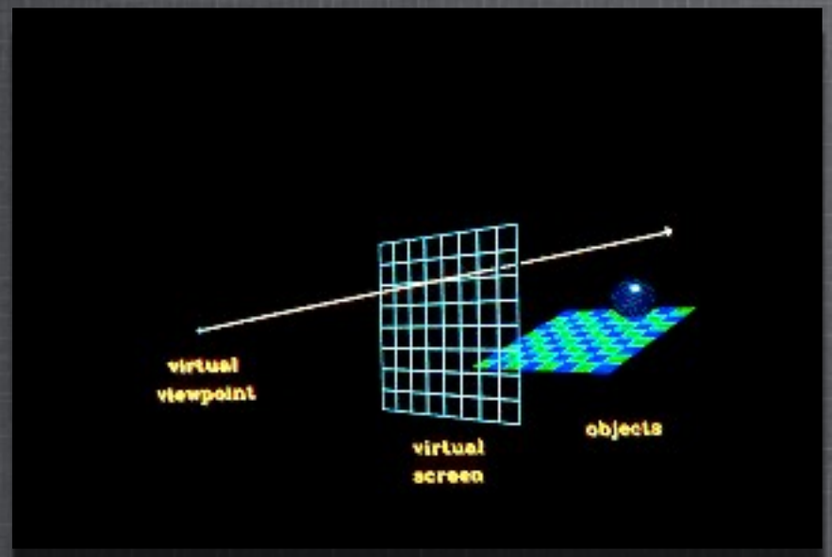
Global Illumination (1980's and 1990's)

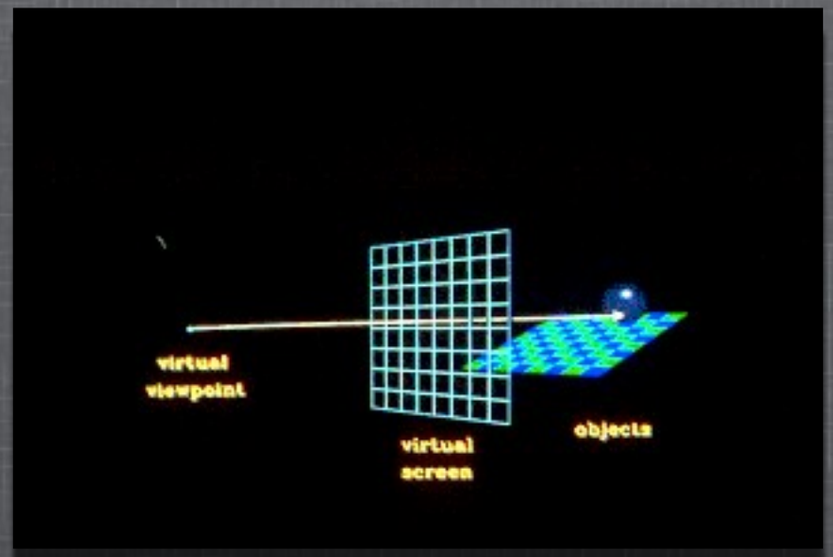


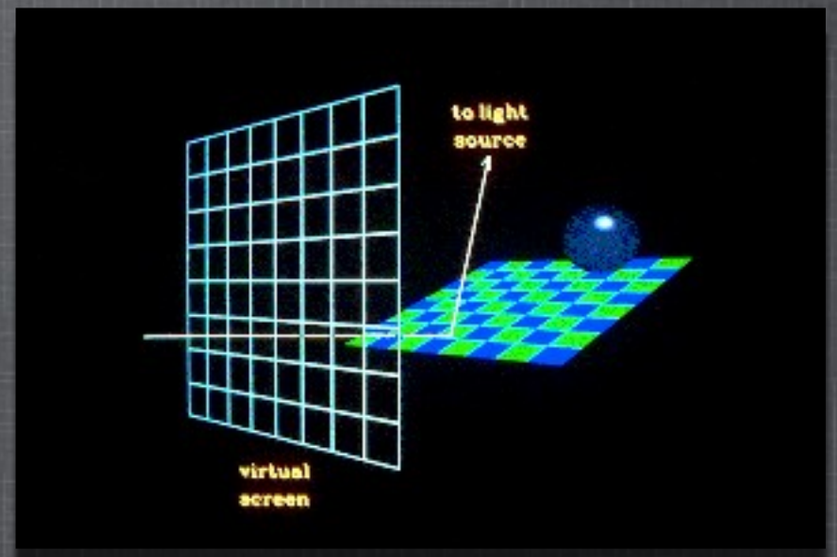
Raytracing (recursive)
(Whitted 1980)

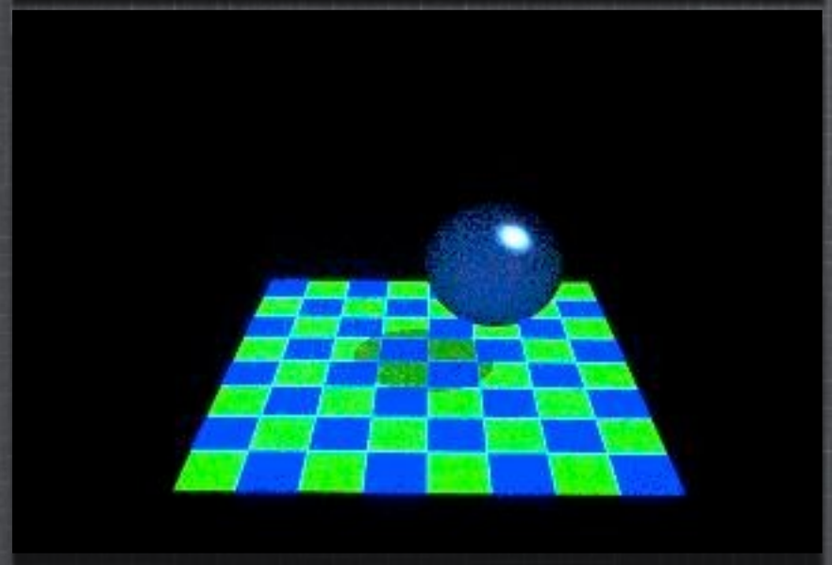
RENDERING

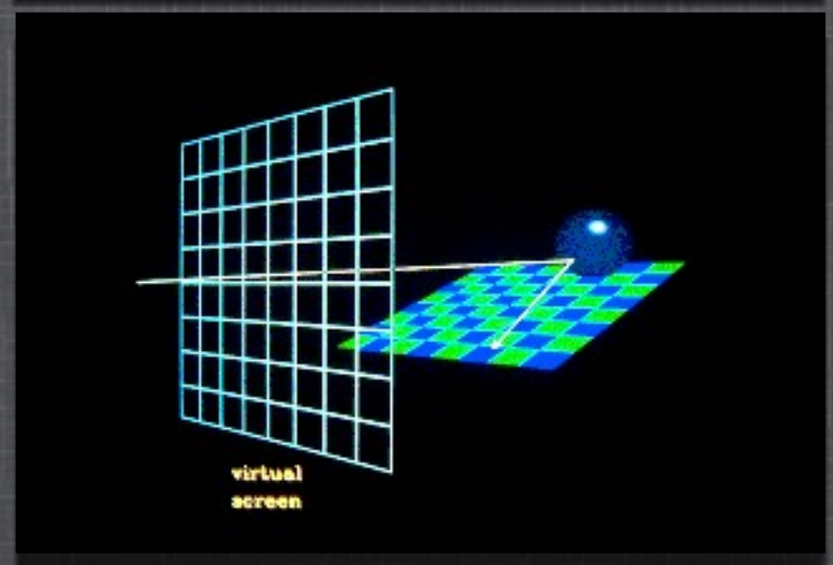
Raytracing (Whitted 1980)

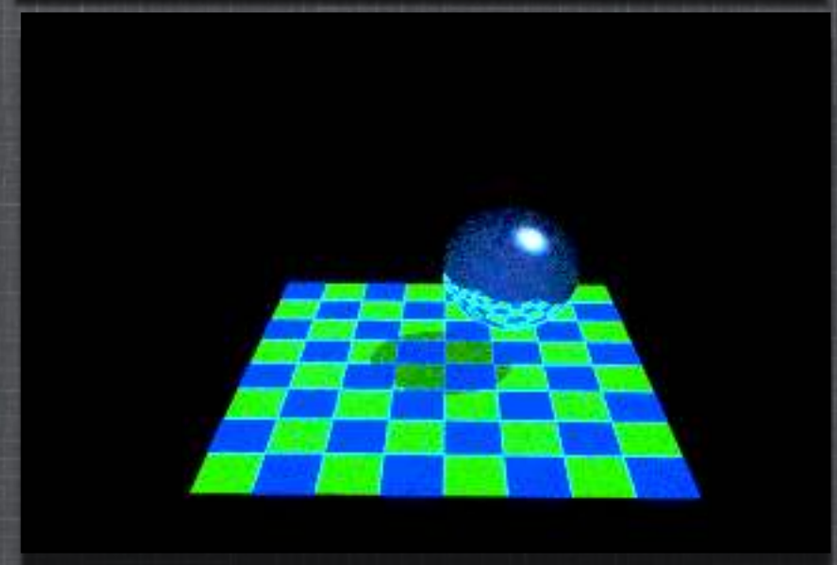






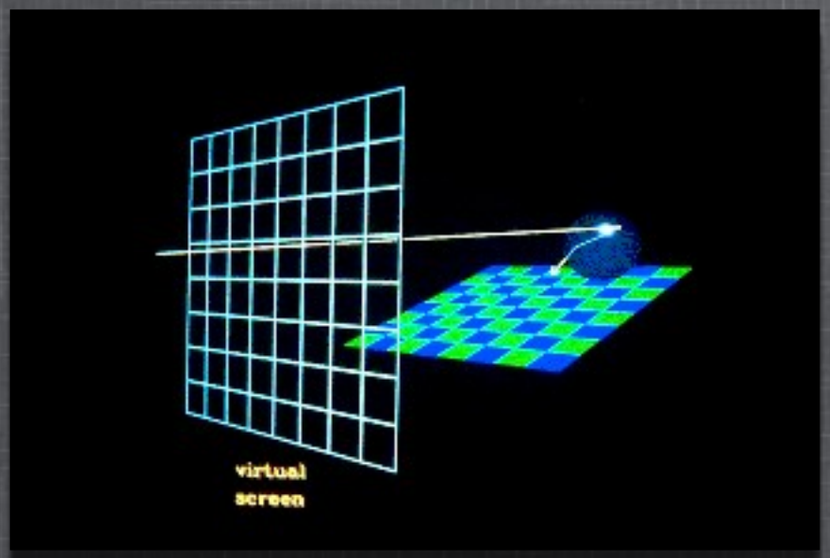


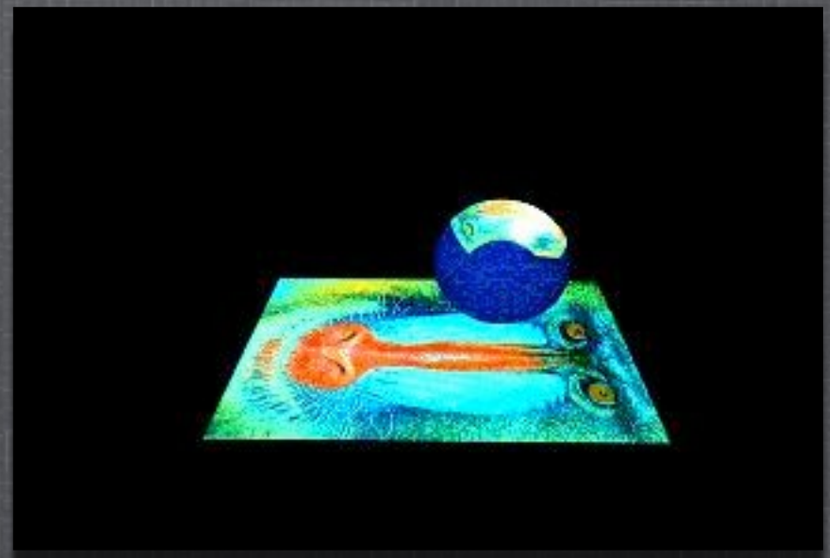




RENDERING

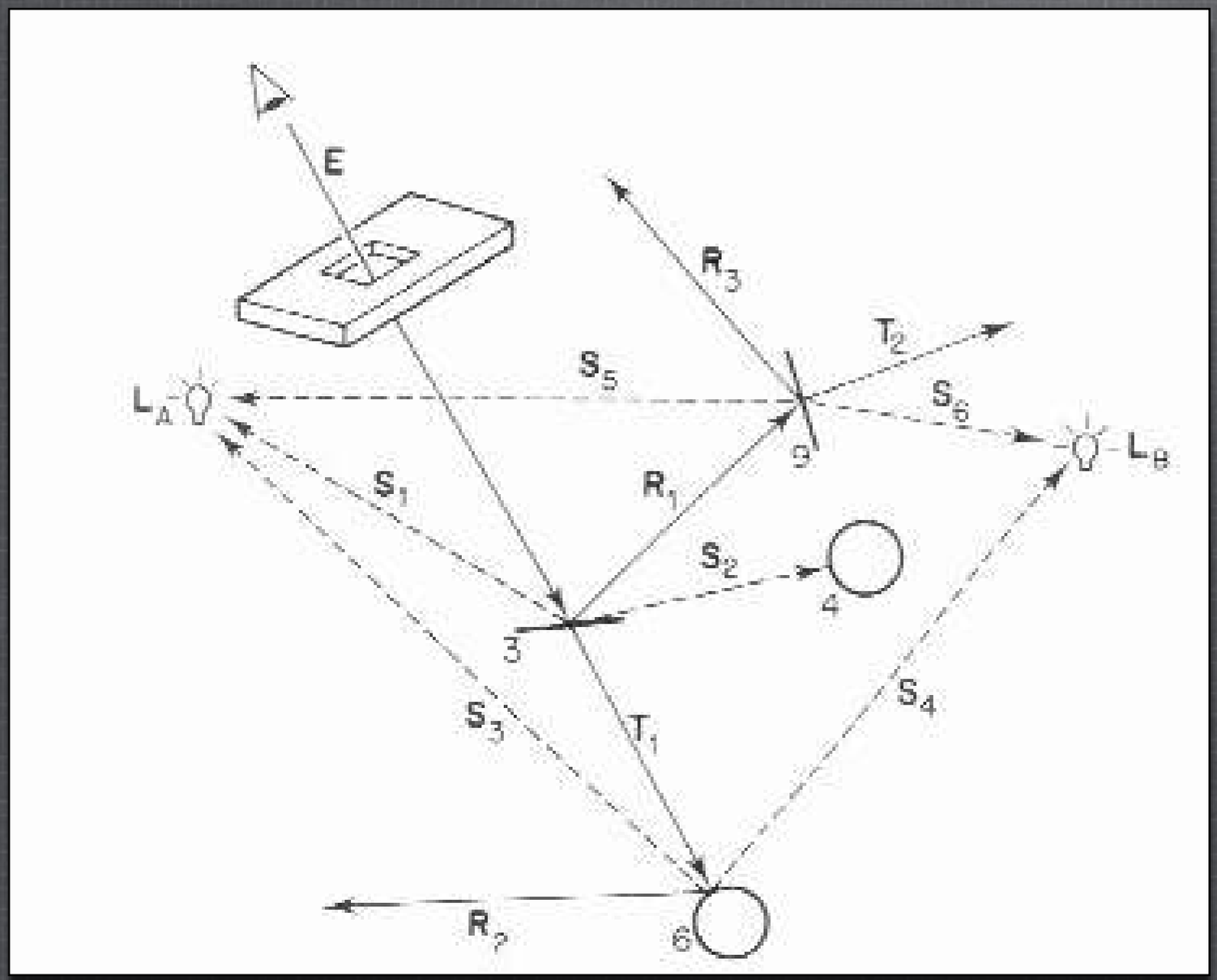
Raytracing (Whitted 1980)

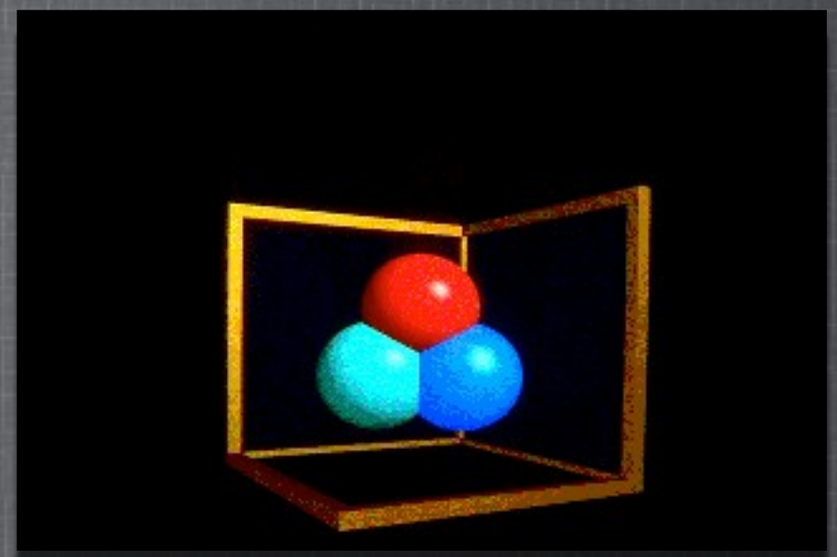


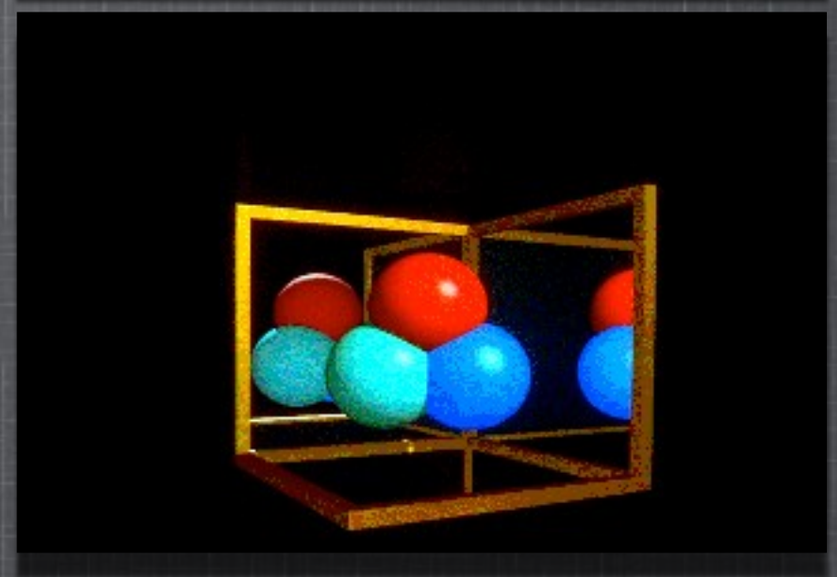


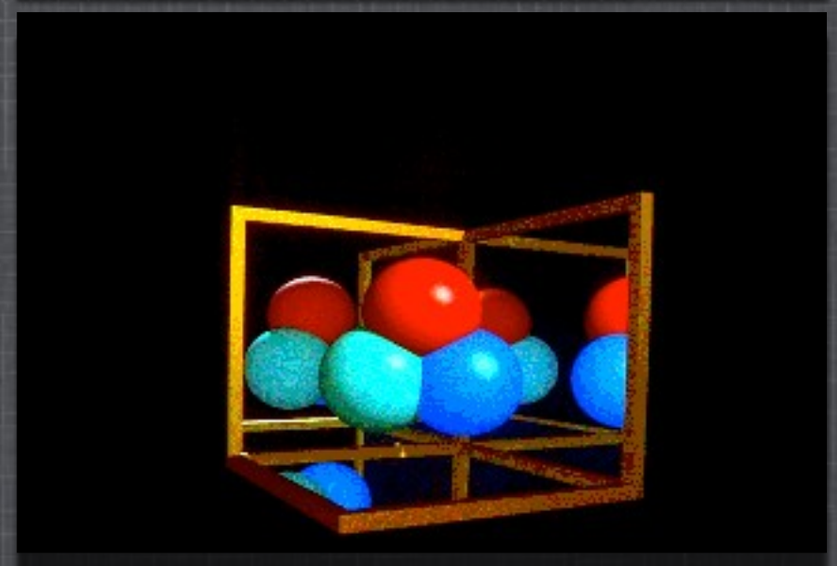
RENDERING

Raytracing (Whitted 1980)

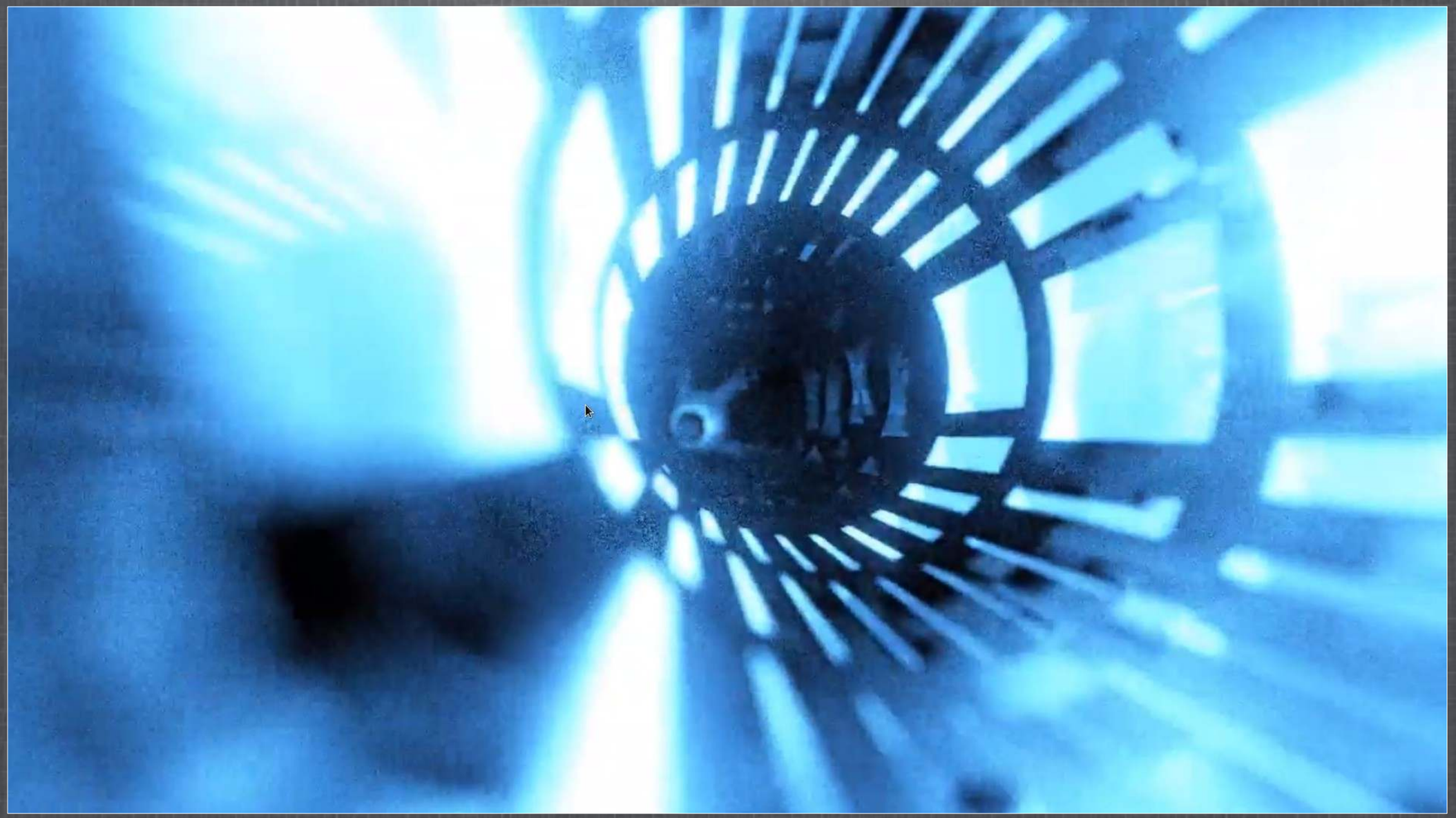








DEMO SCENE



Sync Cord by NuSan & Valden (PC 4k intro, 2020)

REFLECTIONS

Real-time Ray-Tracing in UE4 (2018)



RENDERING

Global Illumination (1980's and 1990's)

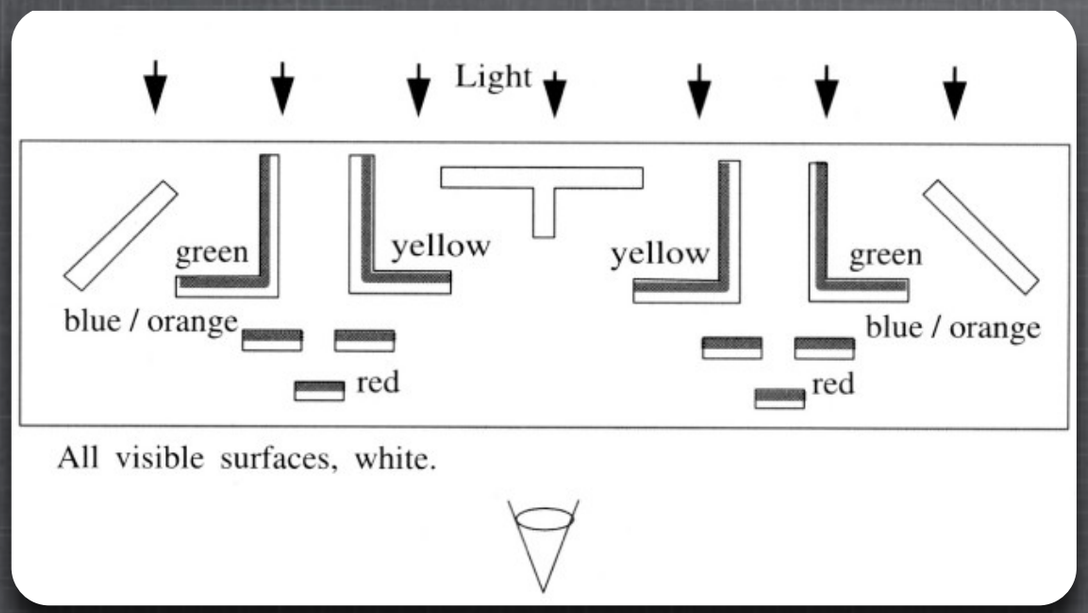


Raytracing
(Whitted 1980)

Radiosity
(Goral, Torrance et al. 1984)

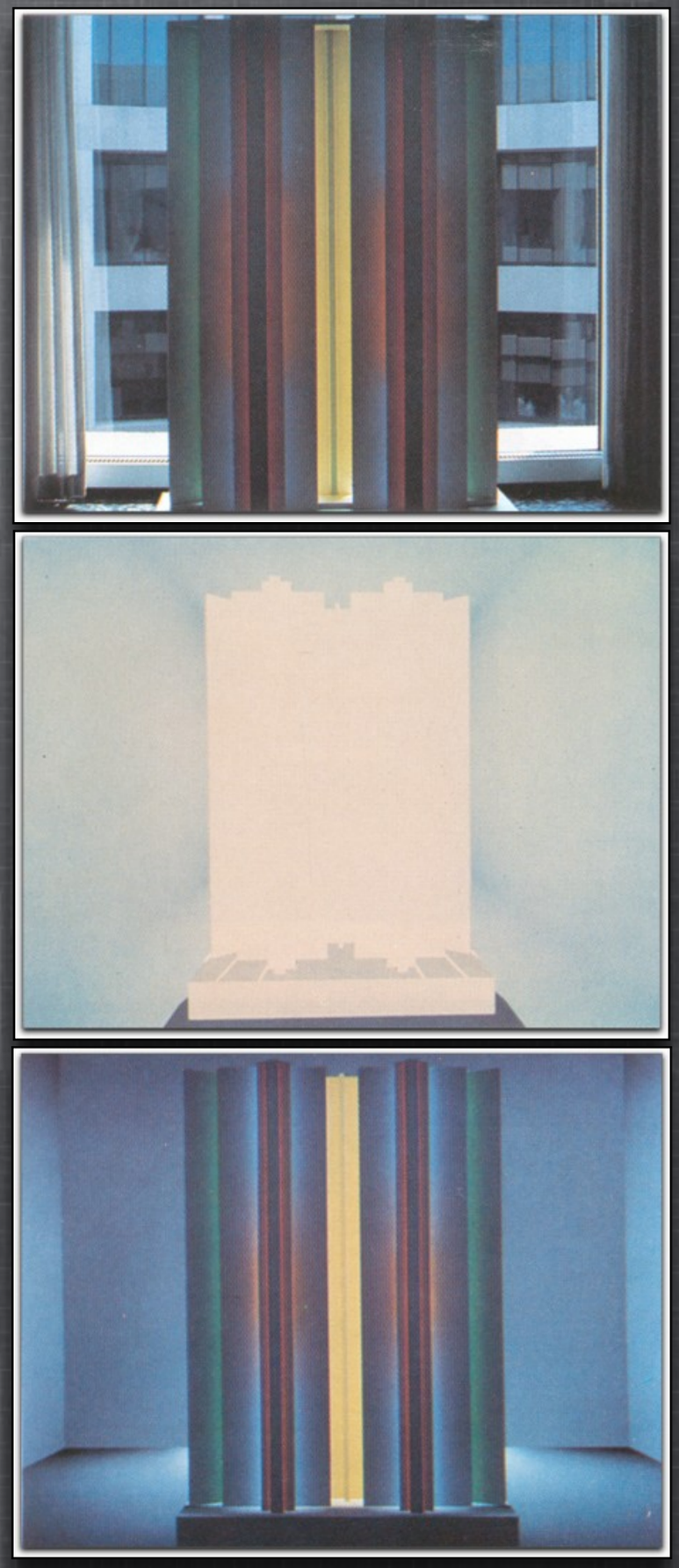
RENDERING

Radiosity (Goral, Torrance et al. 1984)



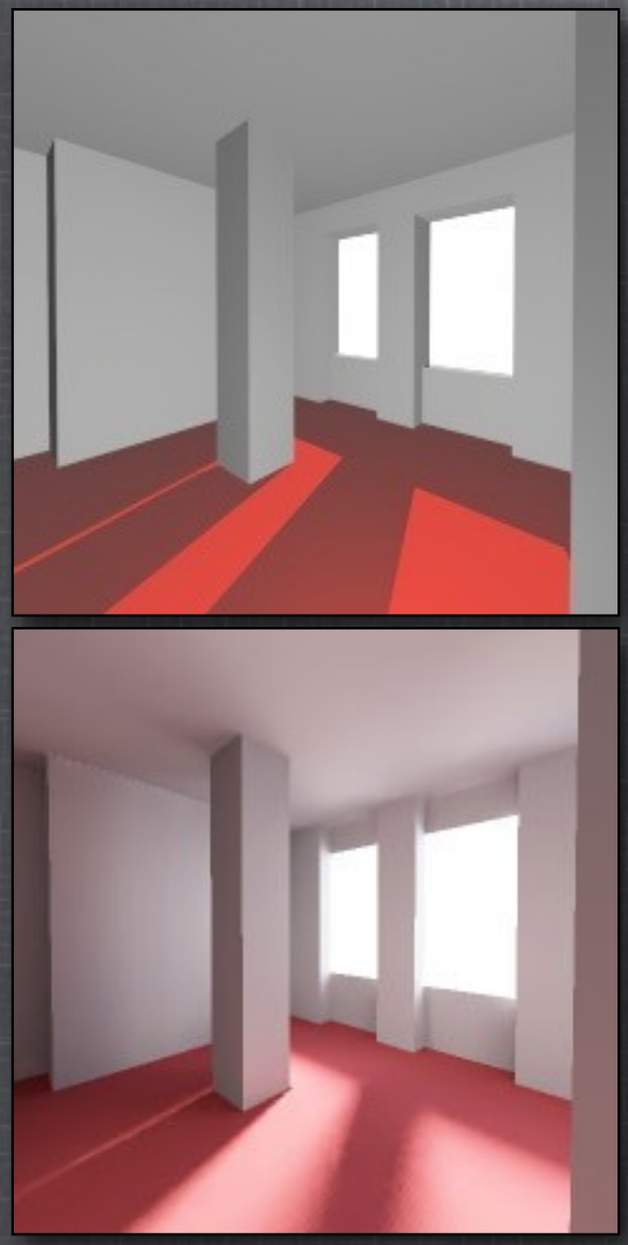
Sculpture by John Ferren

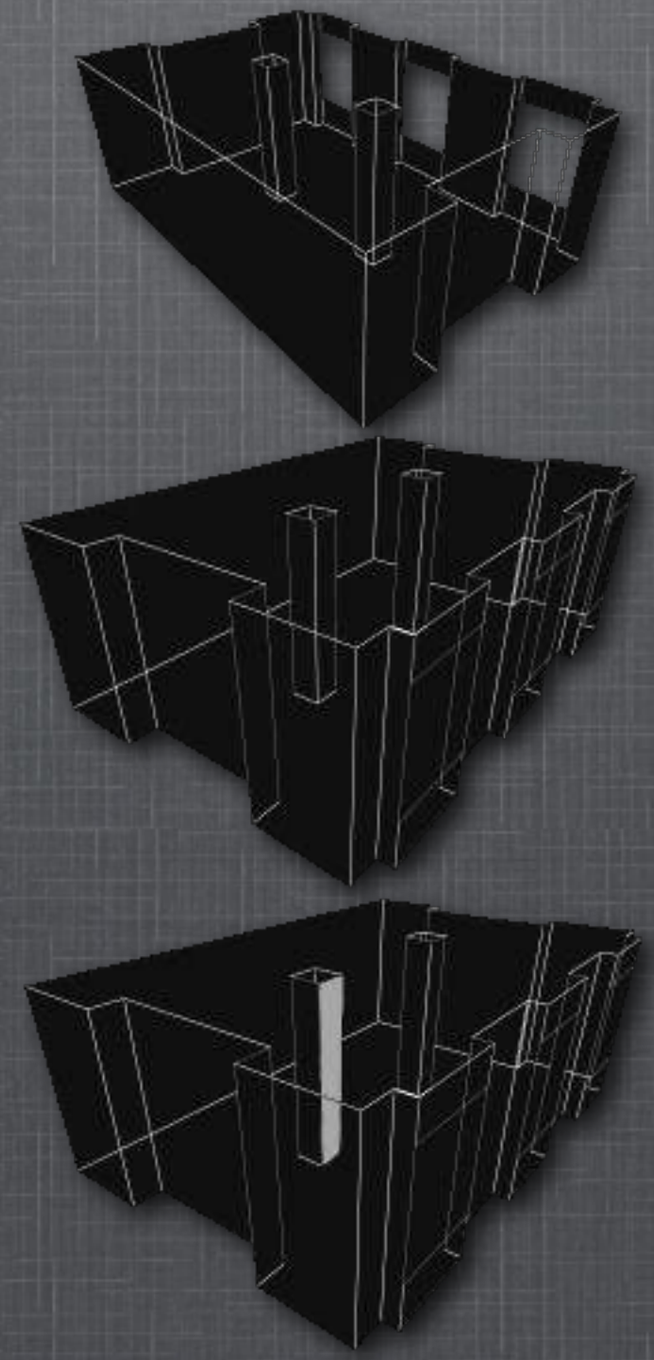
Goral et al. 1984

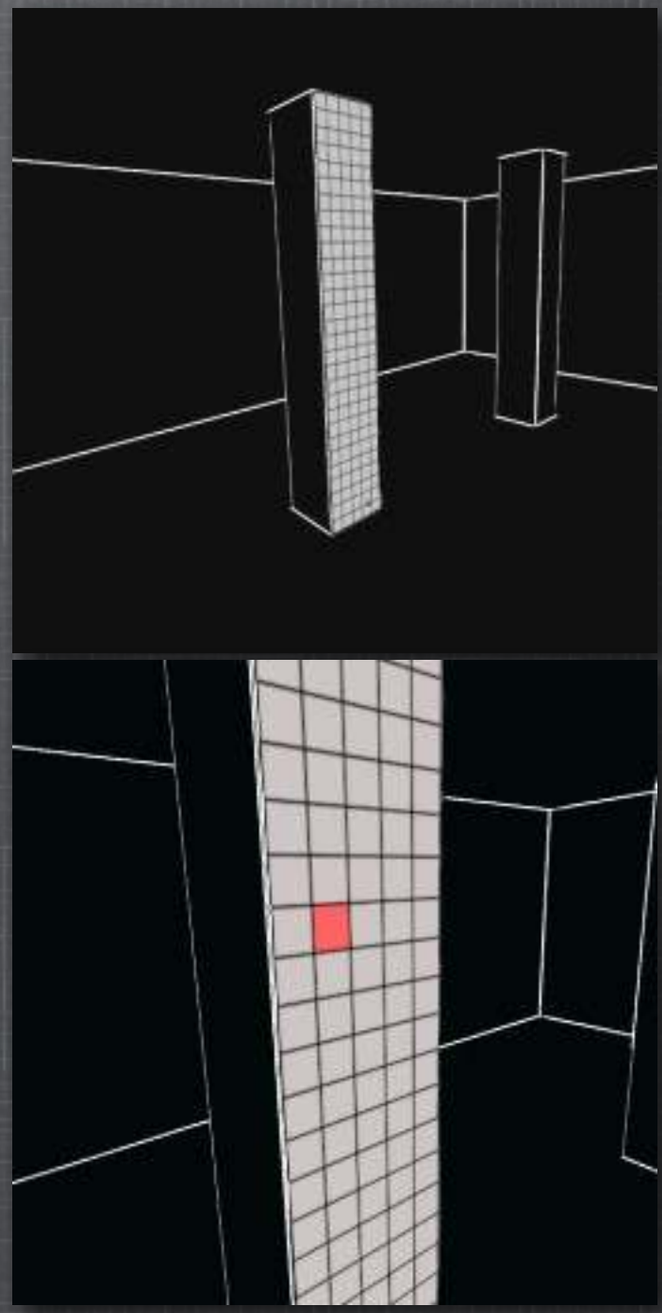


RENDERING

Radiosity (Goral, Torrance et al. 1984)

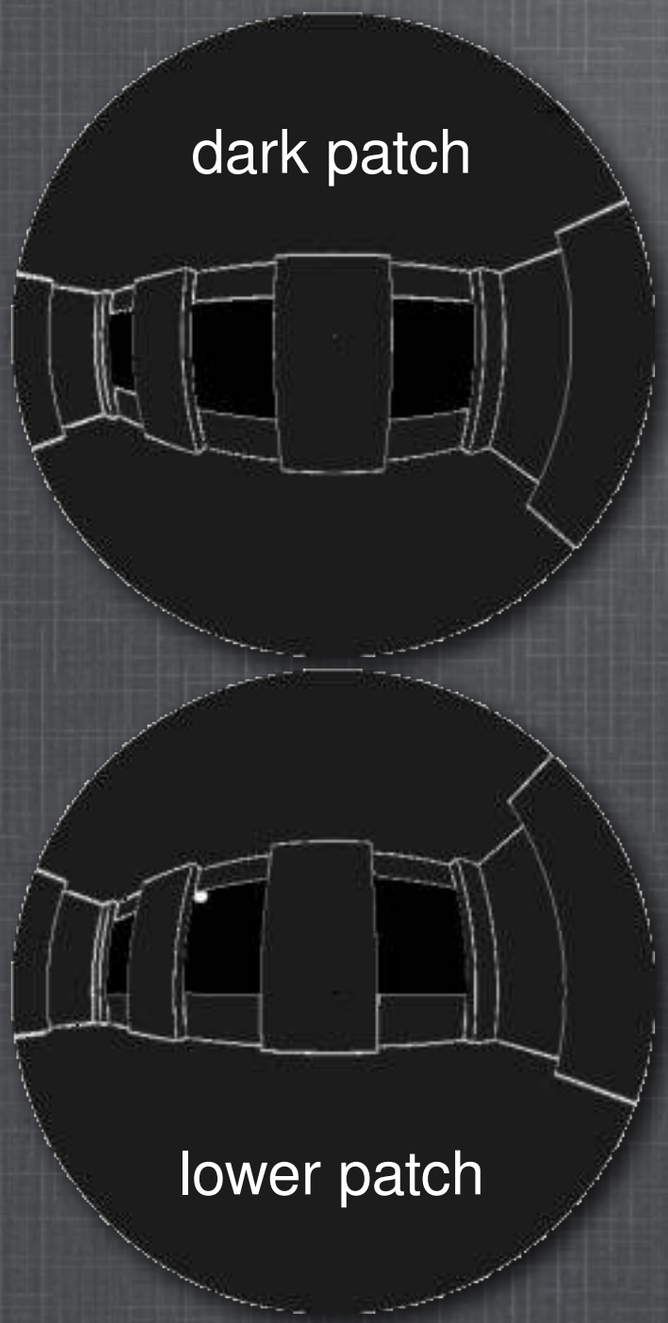


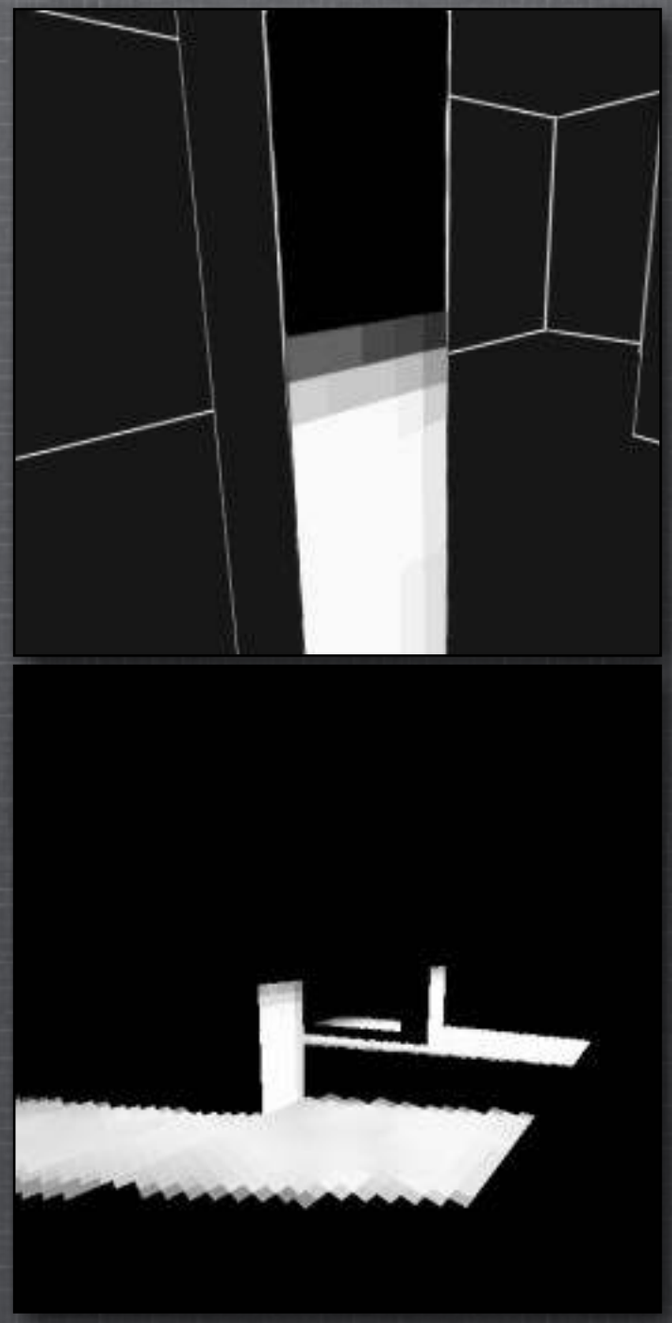




RENDERING

Radiosity (Goral, Torrance et al. 1984)

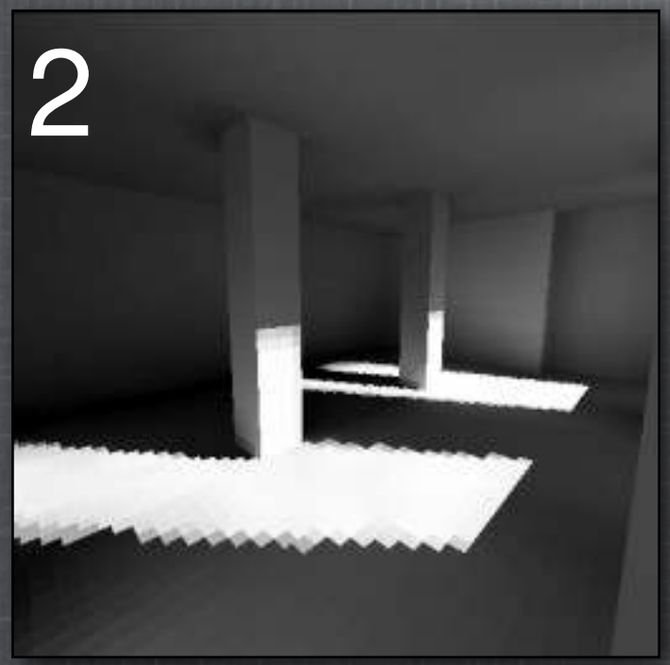


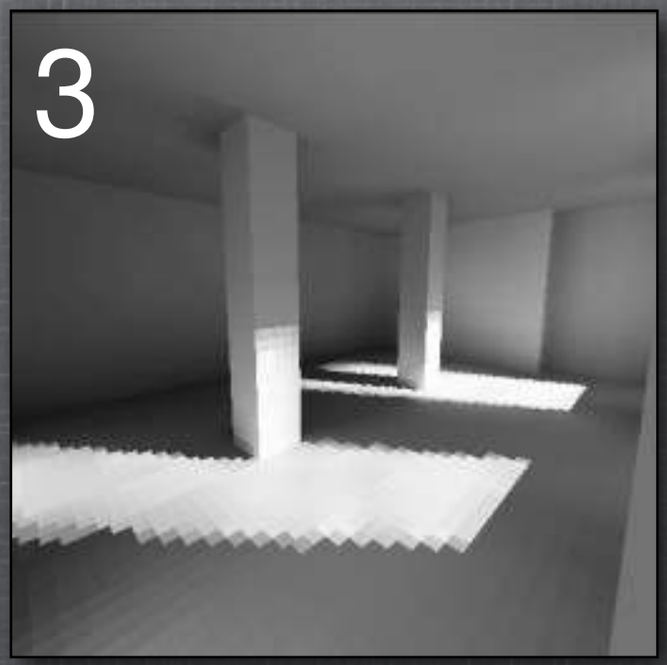


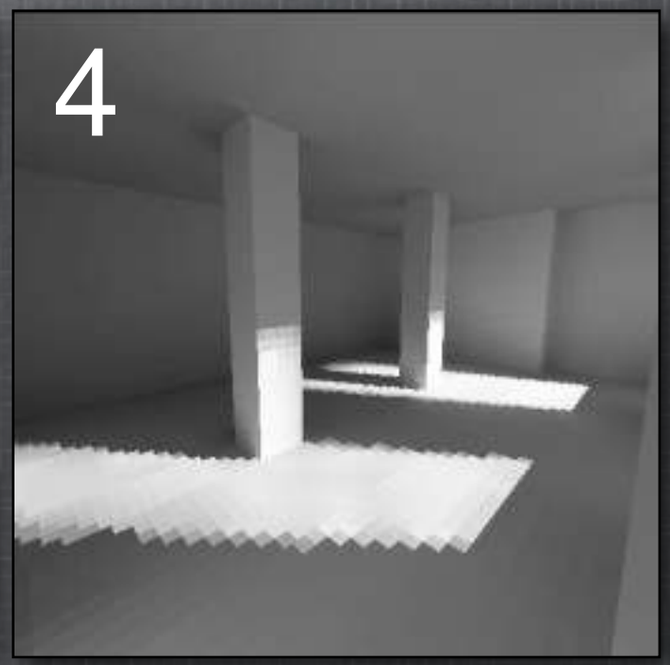
dark patch
not dark
anymore

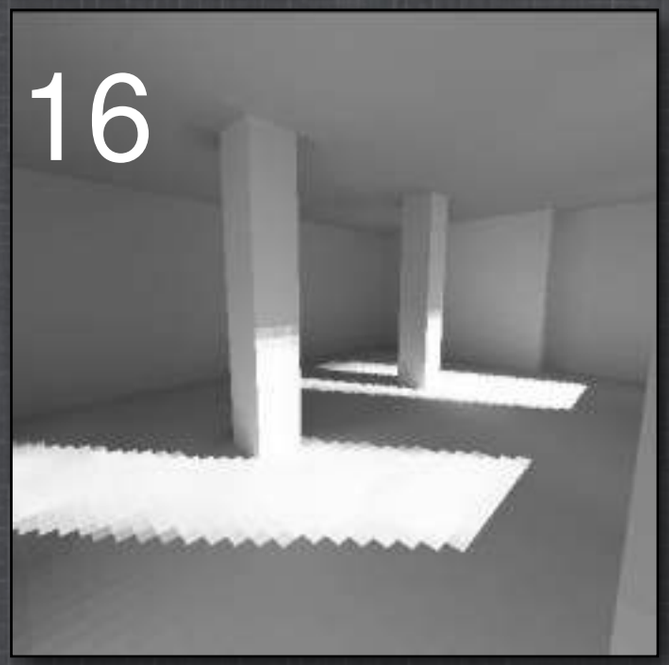
RENDERING

Radiosity (Goral, Torrance et al. 1984)





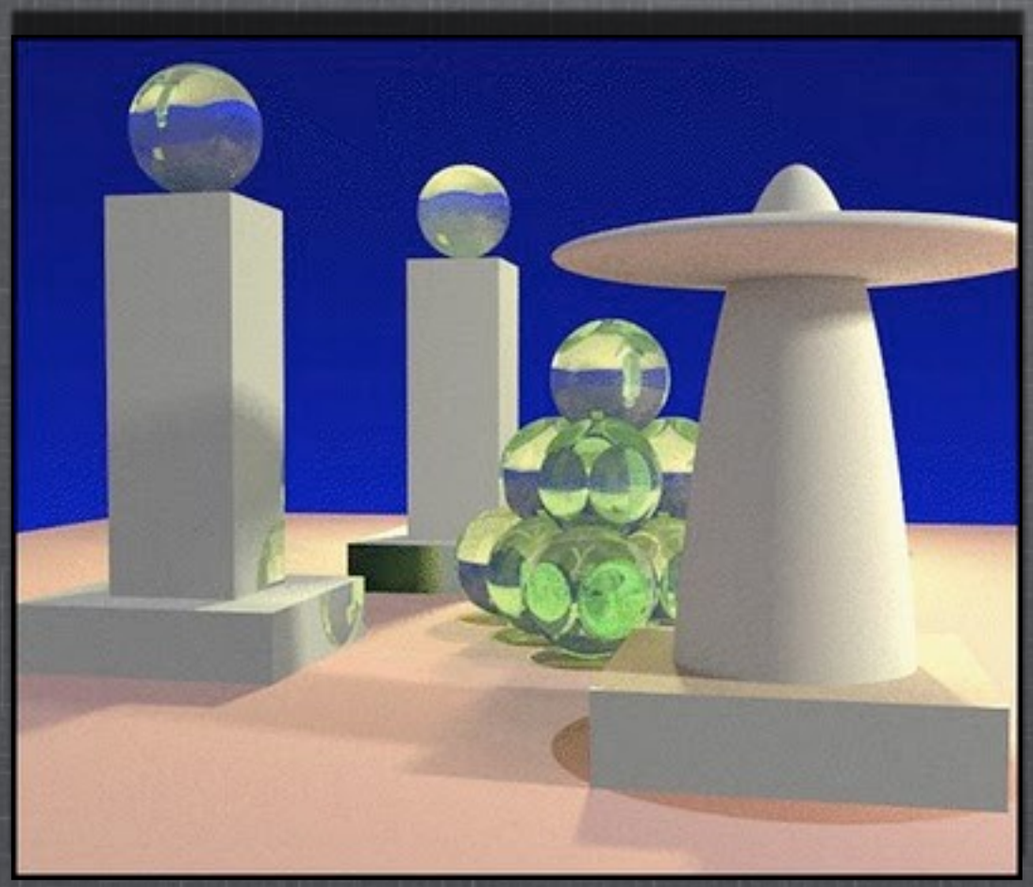






RENDERING

Global Illumination (1980's and 1990's)



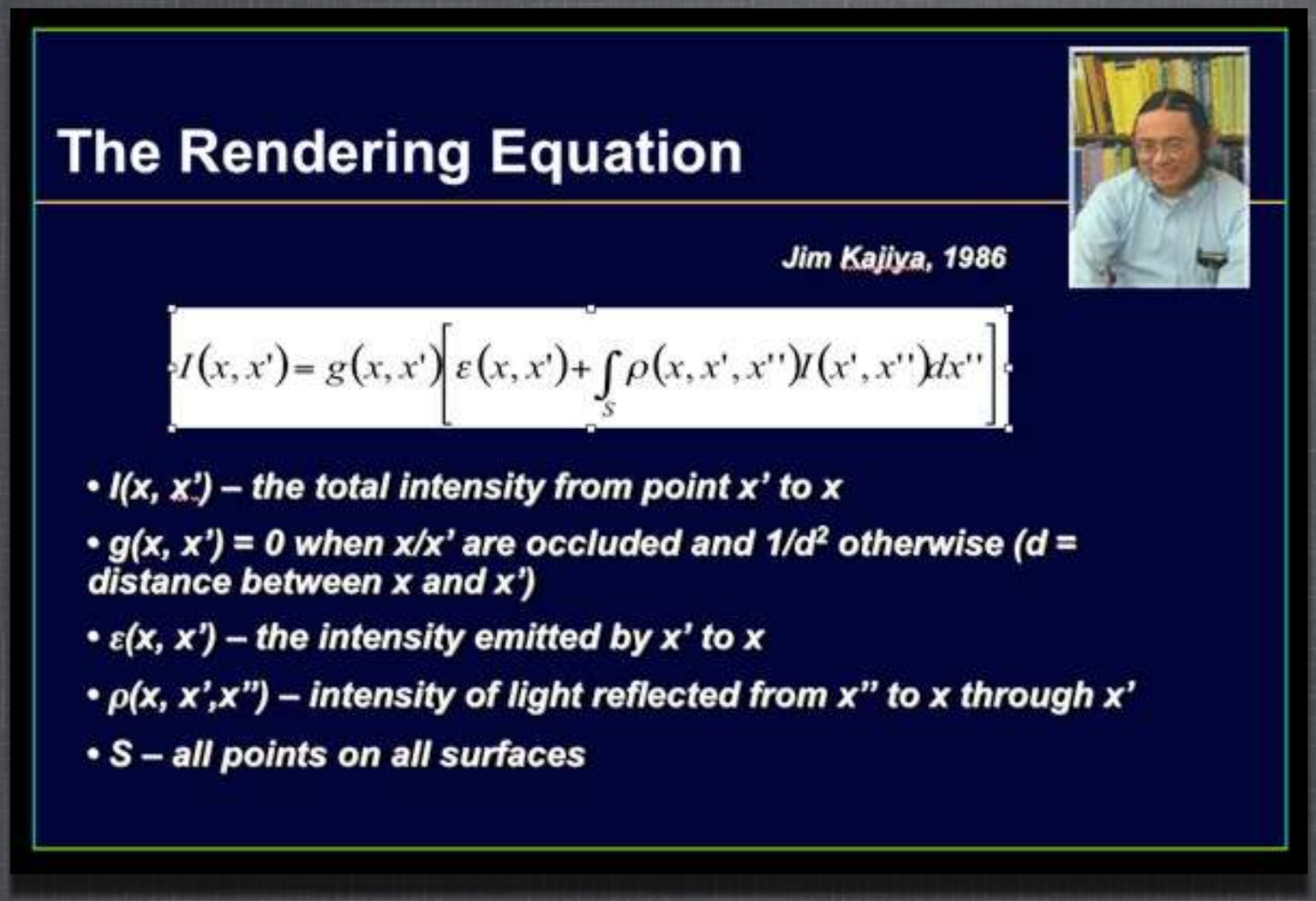
Raytracing
(Whitted 1980)

Radiosity
(Goral, Torrance et al. 1984)

Rendering equation
(Kajiya, 1986)

RENDERING

Global Illumination (1980's and 1990's)



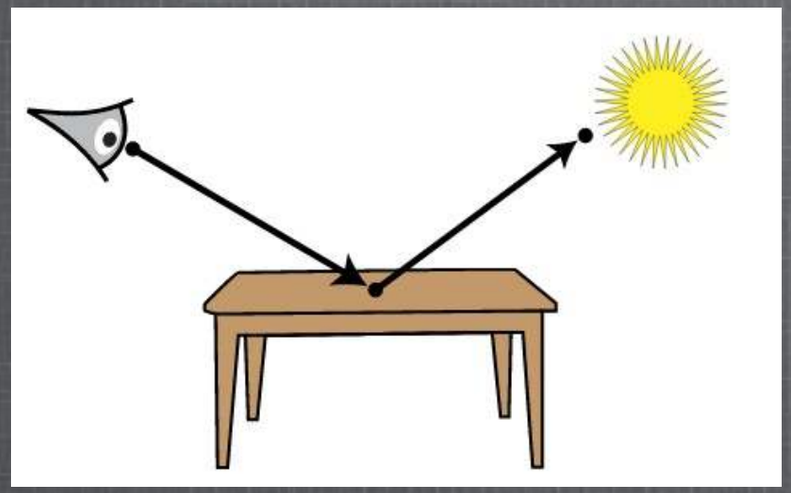
Raytracing
(Whitted 1980)

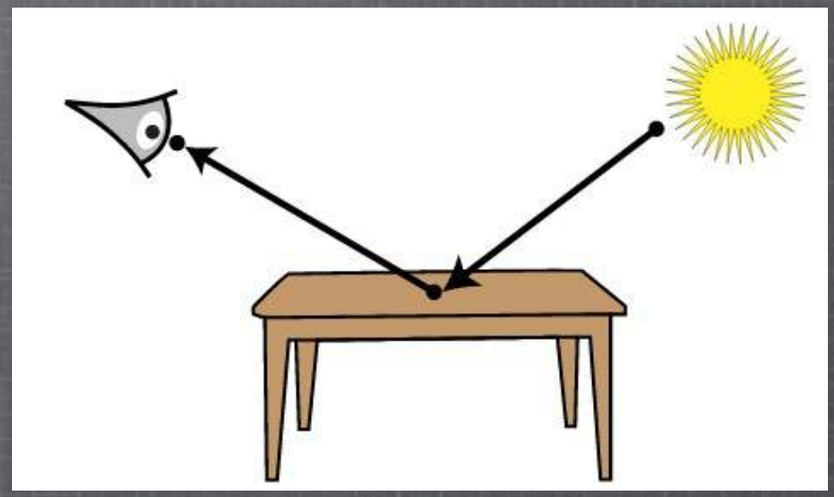
Radiosity
(Goral, Torrance et al. 1984)

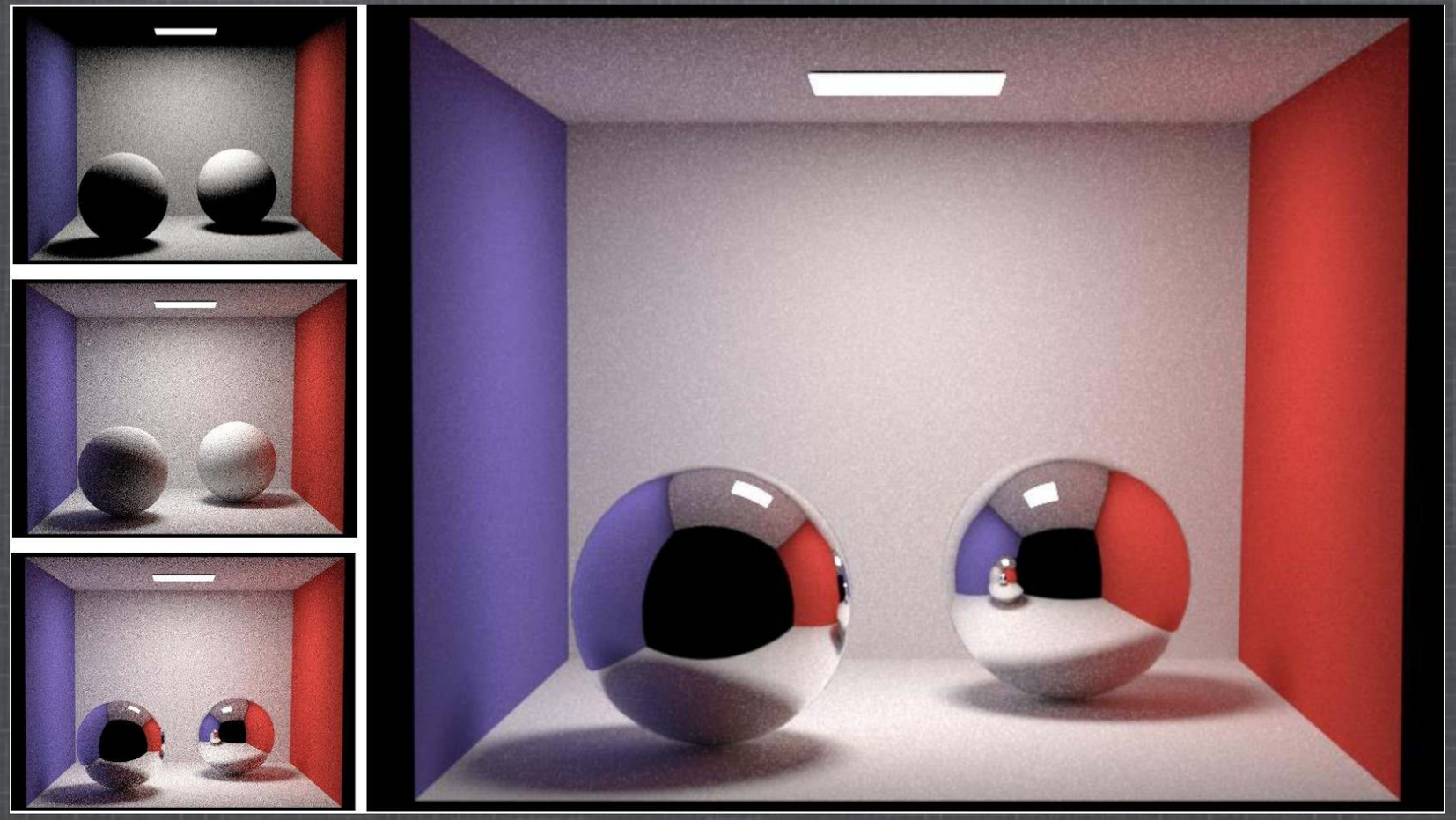
Rendering equation
Path Tracing
(Kajiya, 1986)

RENDERING

Path Tracing

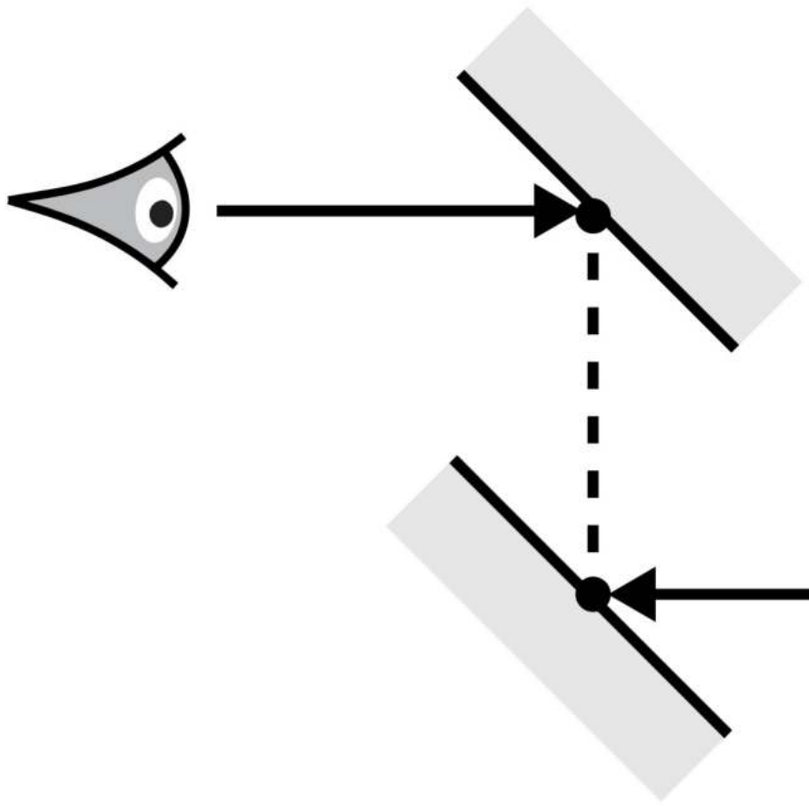


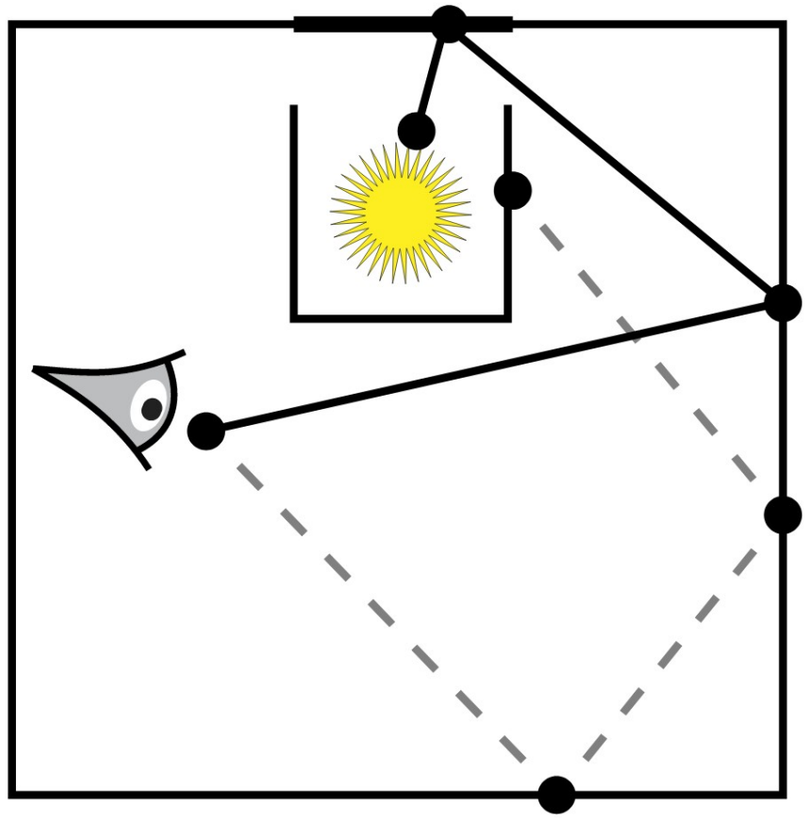




MODERN
PATH TRACING

Bidirectional Path Tracing





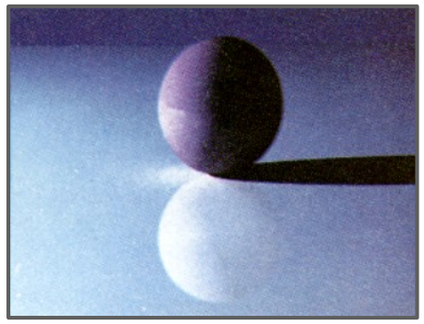
RENDERING

(early) Realtime Path Tracing

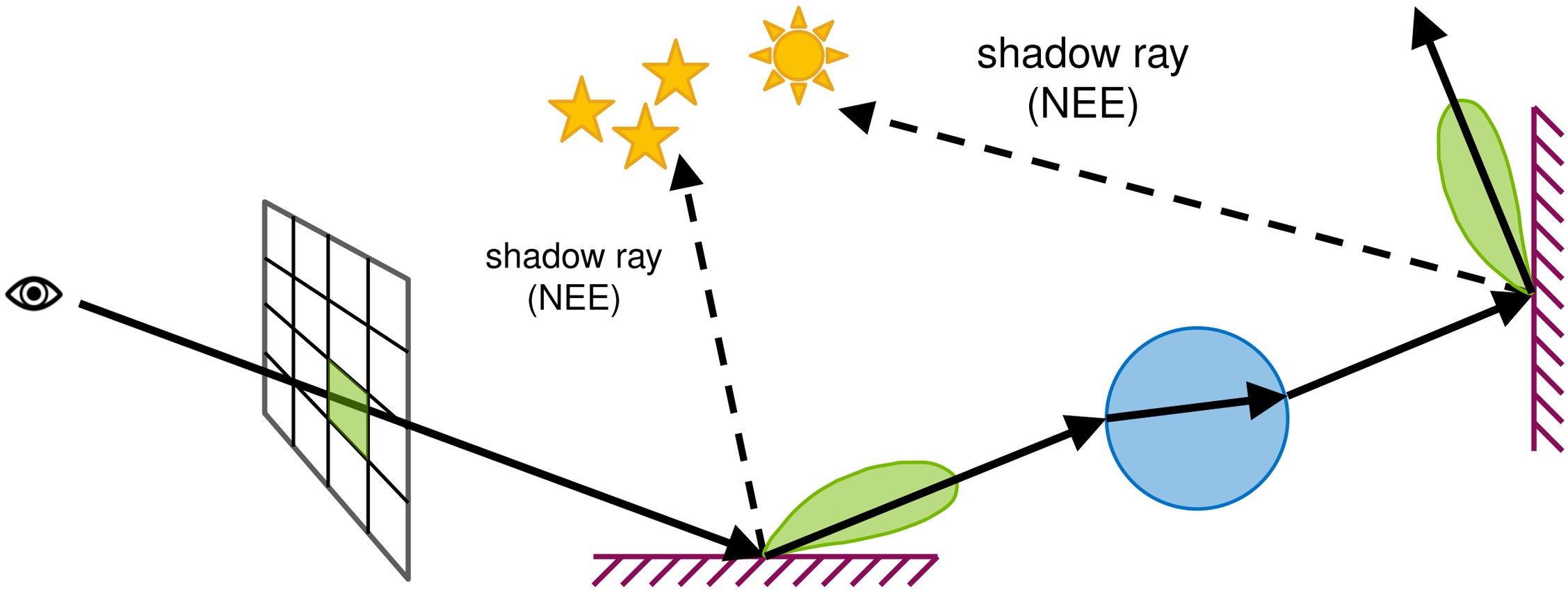


RENDERING EQUATION

Kajiya @ SIGGRAPH 1986



MODERN PATH TRACING



RENDERING WITH DENOISING



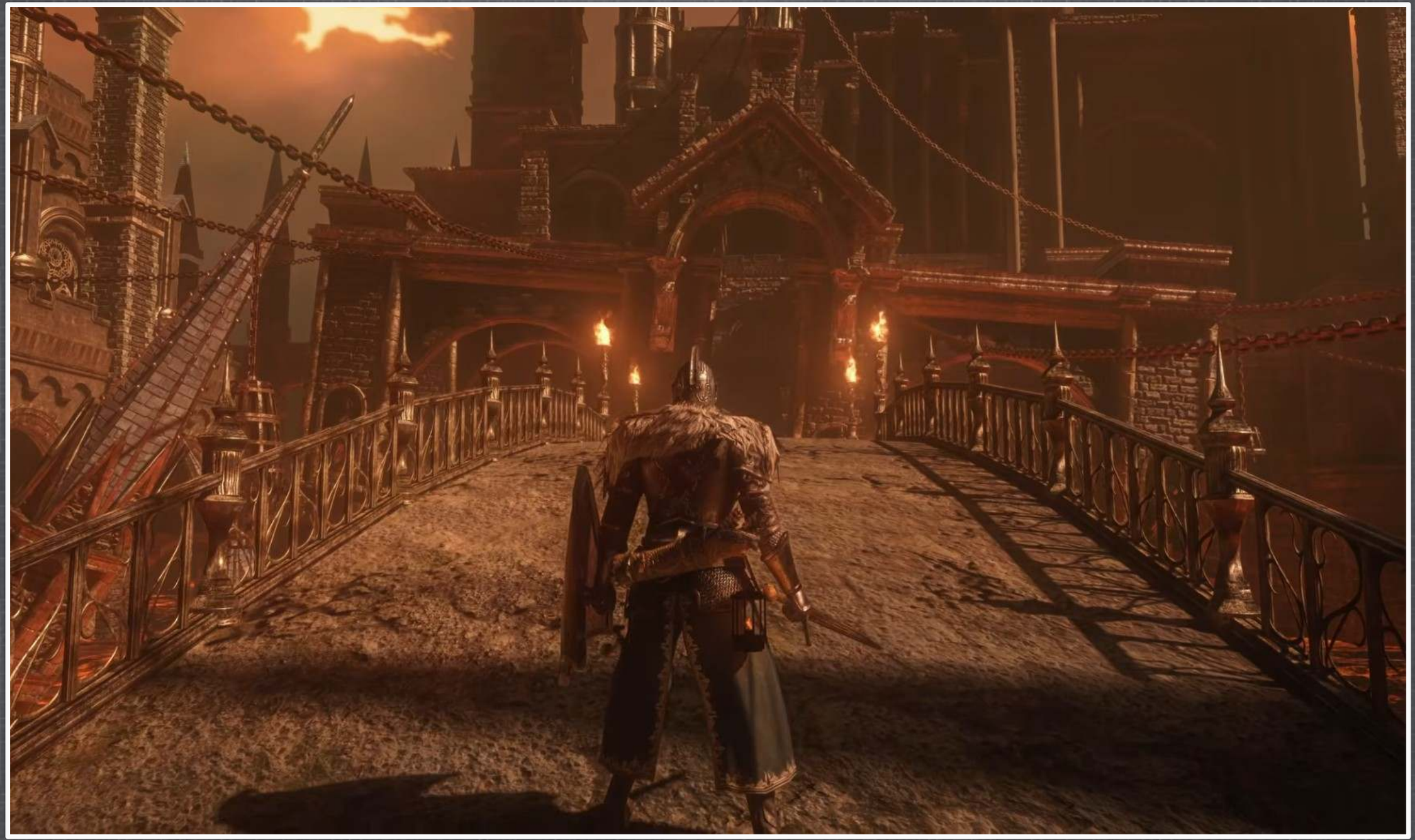
Realtime Path Tracing NVIDIA 2022 (30FPS, 1080p, 30 bounces, 30B triangles)



Realtime Path Tracing NVIDIA 2022 (30FPS, 1080p, 30 bounces, 30B triangles)

RENDERING

Dark Souls II Path Tracing + DLSS 5



QUARTER PLANNING

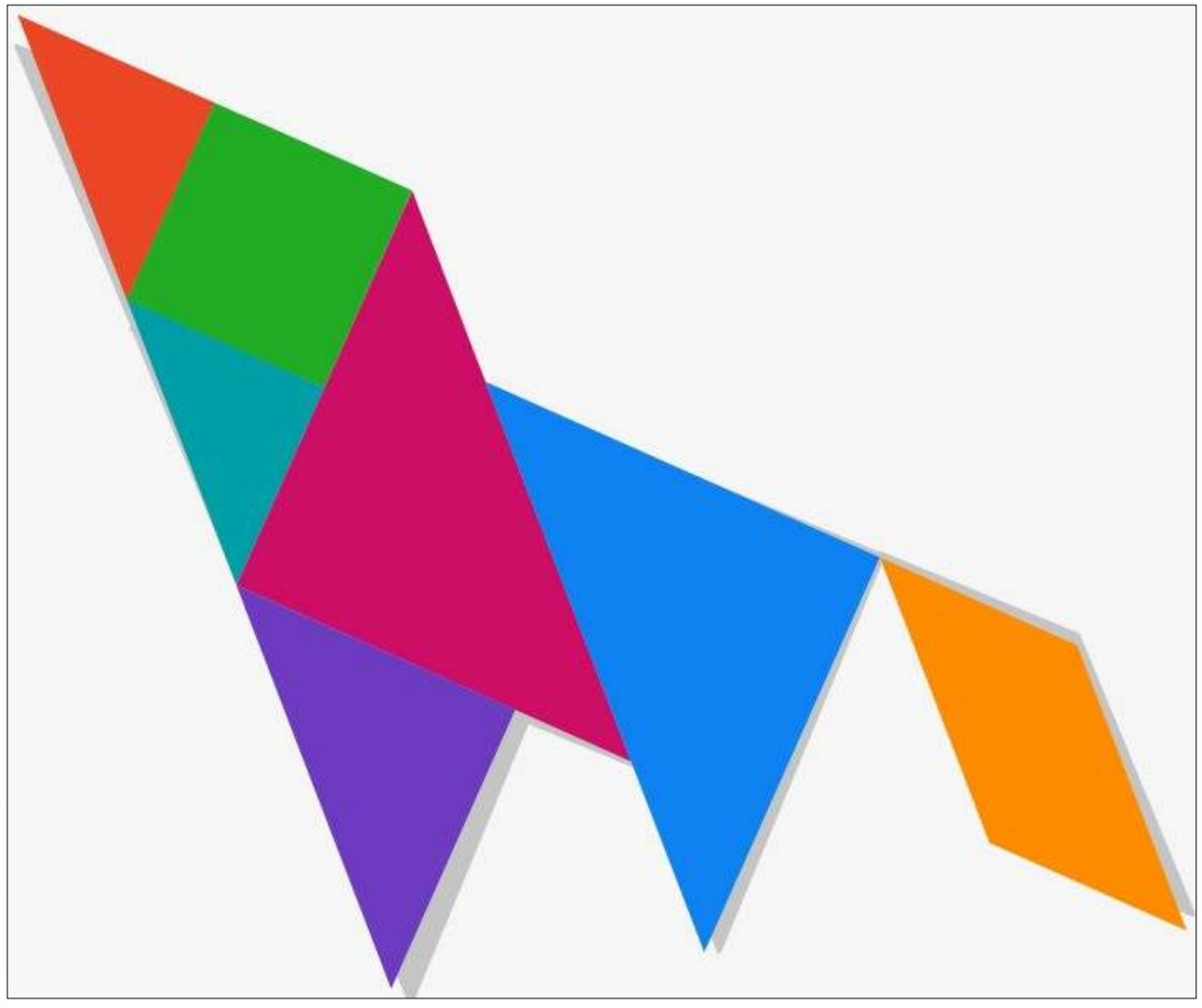


November 2026

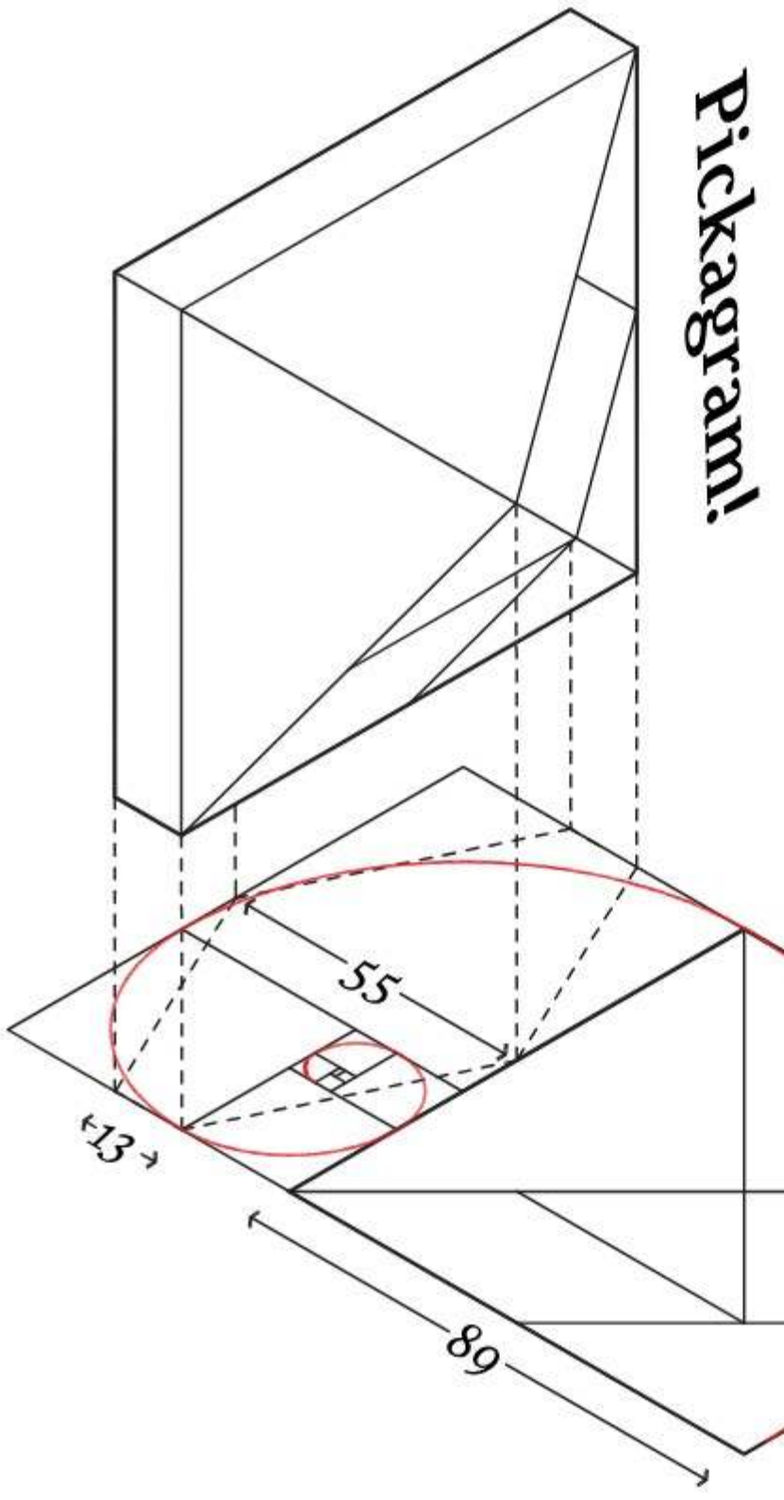
W	Date	Lectures	Labs	Assignments due	Grade
1	Nov 3 (Tue)	Introduction, Review	Team forming C++ IDE		
1	Nov 6 (Fri)	Mathematics for CG (part 1): dot, cross, matrices	GLM / GLM setup		
2	Nov 10 (Tue)	Mathematics for CG (part 2): matrices, quaternions	GLM library	**A1 : GLM (50% in class)** Nov 12 (Thu)	**1.0**
2	Nov 13 (Fri)	Programmable Graphics Pipeline, Drawing in OpenGL	OpenGL / GLFW / GLEW Setup	Tan/Picka-gram Shape (in class)	
3	Nov 17 (Tue)	Viewing Pipeline Viewing in OpenGL	Drawing + Transformations	**A2 : 2D Tangram** Nov 19 (Thu)	**3.0**
3	Nov 20 (Fri)	Geometry Management Meshes and Scene Graphs	Viewing + Quaternions		

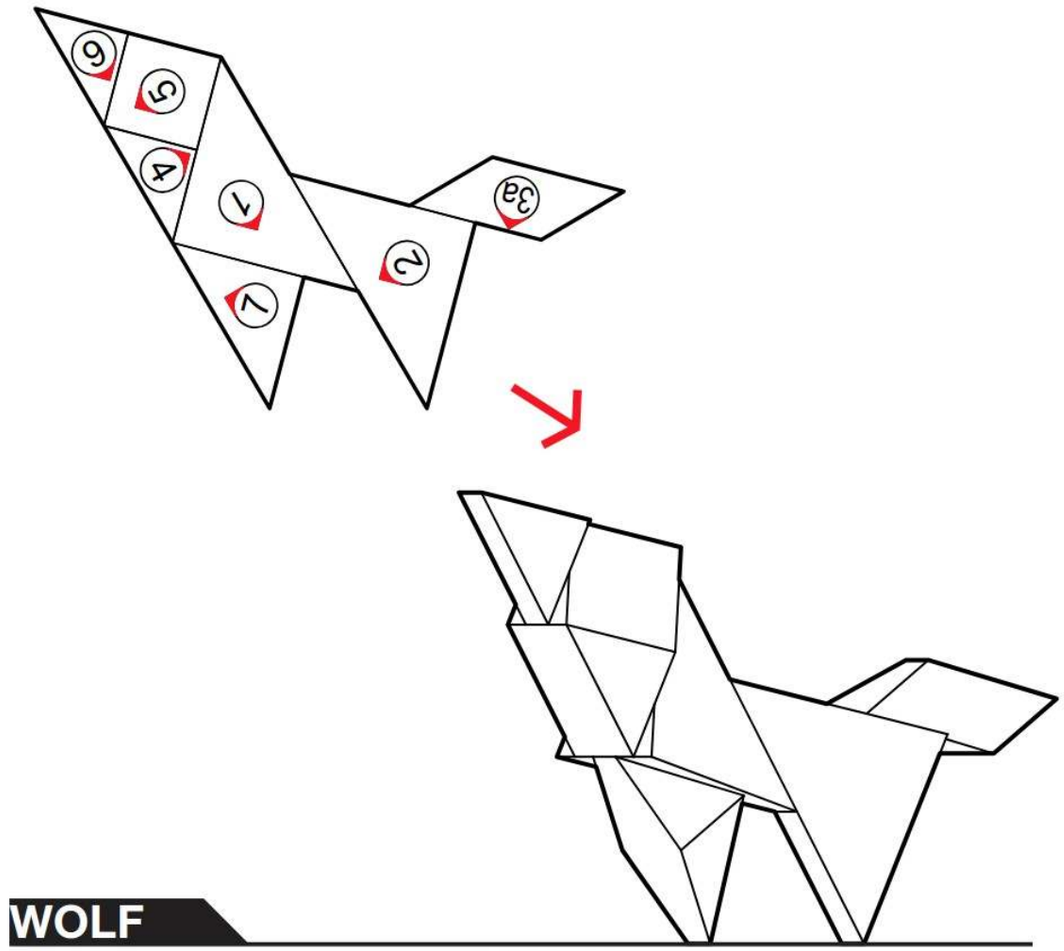
Tangram



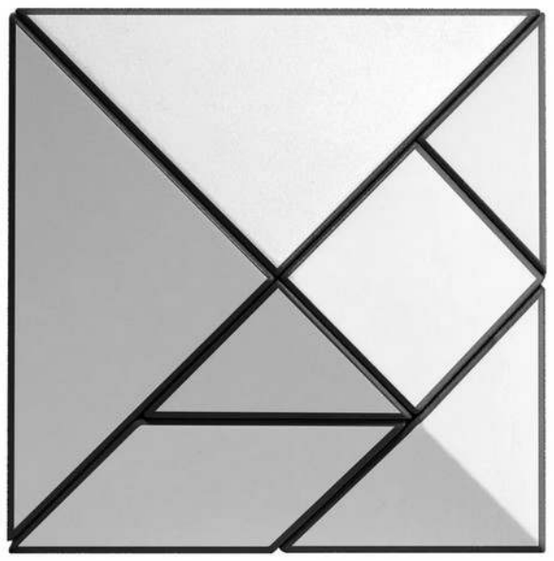


Pickagram



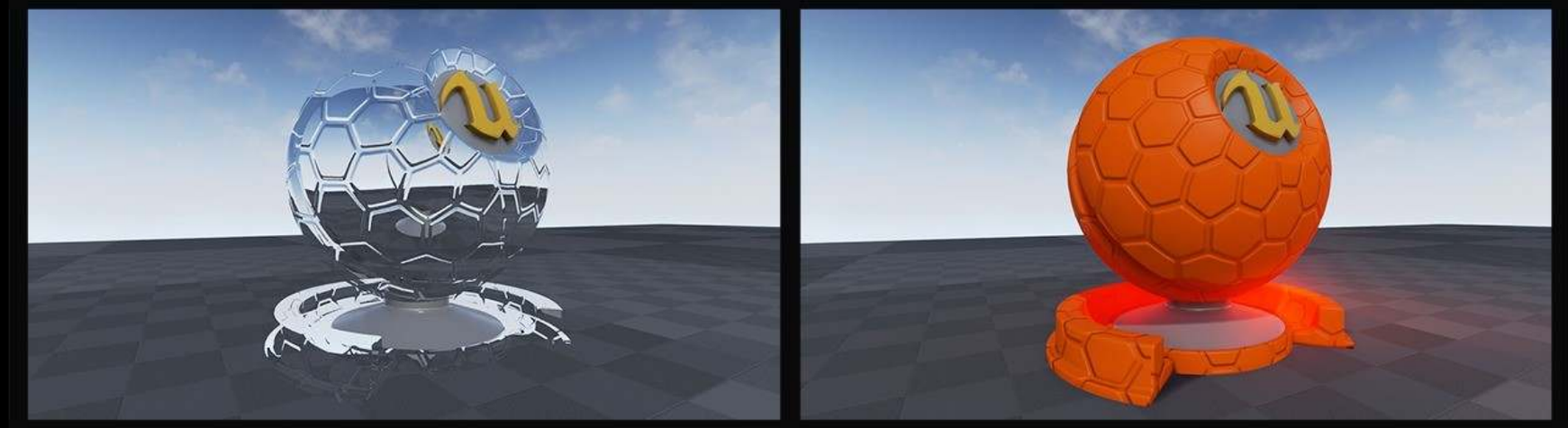


WOLF



November - December 2026

W	Date	Lectures	Labs	Assignment due	Grade
4	Nov 24 (Tue)	GLSL Shaders Lighting (1): Cel Shading	Meshes + ASSIMP Library		
4	Nov 27 (Fri)	Lighting (2): Gouraud and Phong Shading	Project Support	**A3 : 3D Pickagram** Nov 30 (Mon)	**4.0**
5	Dec 1 * (Tue feriado)	Mapping (1): Texturing, Sampling and Filtering	Shader Project Individual Project Definition	**A4 : Project Proposal** Dec 3 (Thu)	**1.0**
5	Dec 4 (Fri)	Mapping (2): Procedural Textures and Noise	Goal Definition and Execution Planning		
6	Dec 8 * (Tue feriado)	Mapping (3): Environment Mapping and Fresnel Reflectance	Project Support		
6	Dec 11 (Fri)	Mapping (4): Render Target Textures, Tangent Space, Normal and Displacement Mapping	Project Support		



December 2026 - January 2027

W	Date	Lectures	Labs	Assignments Due	Grade
7	Dec 15 (Tue)	Mapping (5): Ambient Occlusion and Physically Based Rendering	Project Support		
7	Dec 18 (Fri)	(project report Q&A)	Project Support	**A5 : Shader Project** Jan 4 (Mon)	**8.0**
8				**A5: Project Video Report** Jan 4 (Mon) **Project Discussions** Jan 6-7 (Wed-Thu)	**3.0**

IN BRIEF

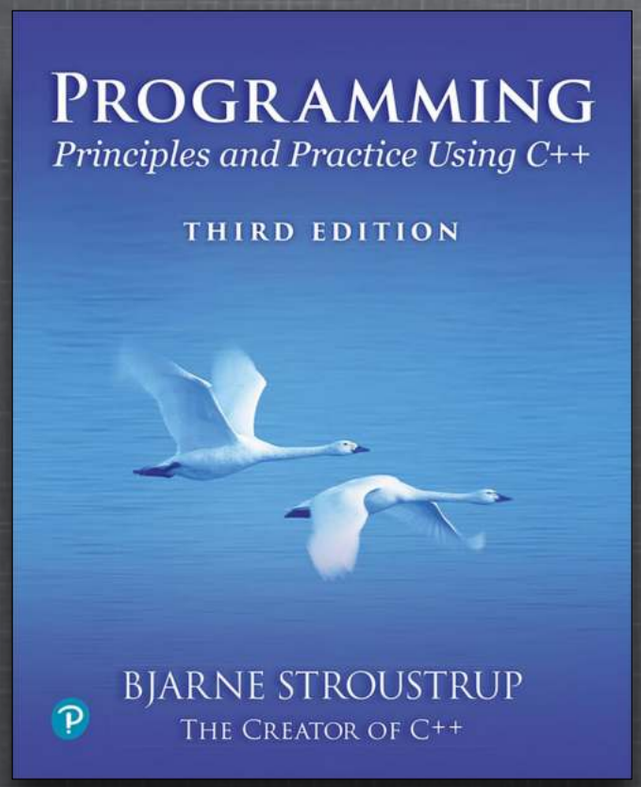
- 40% - Base Project (teams of 2, 3 checkpoints)
- 60% - Shader Project (individual, 2 checkpoints)

IMPORTANT

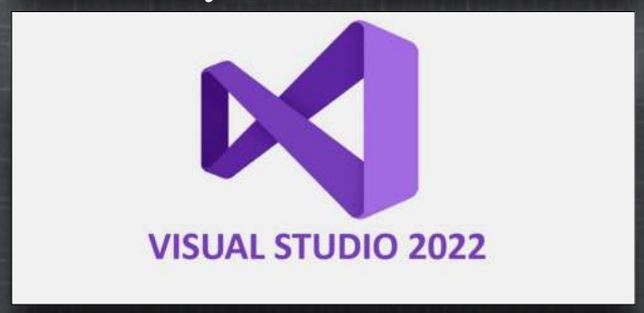
Reports must include:

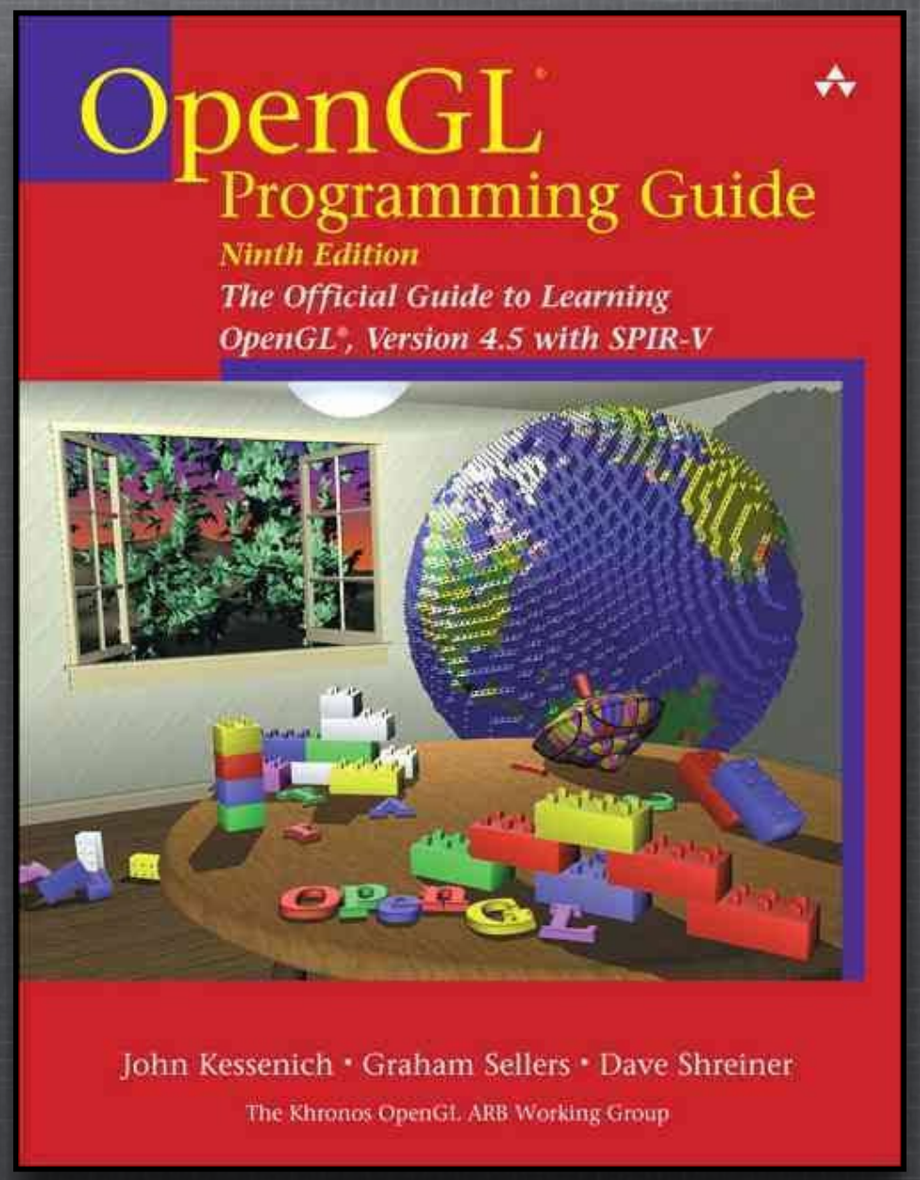
- Declaration of use of AI in assessment components
- Declaration of honour for group work

Check the *evaluation methods* section of the course website for more information.



Choose your C++/GLSL IDE





CONTACT

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CGJ Office hours on Fénix.